

Pelvic girdle, gluteal region and thigh

The pelvic girdle consists of the paired hip bones (each composed of the ilium, ischium and pubis) and the sacrum. The two pubic bones articulate anteriorly at the pubic symphysis and the sacrum articulates posteriorly with the two iliac bones at the sacroiliac joint; the bones are virtually incapable of independent movement, except in the female during parturition or as a result of pathological change. The pelvic girdle is massively constructed and serves as a weight-bearing and protective structure, as an attachment for trunk and lower limb muscles, and as the skeletal framework of a birth canal capable of accommodating passage of the fetus.

The gluteal region or buttock is an area demarcated by the gluteal fold inferiorly, a line joining the greater trochanter and the anterior superior iliac spine laterally, the iliac crest superiorly and the midline medially. It contains a large bulk of skeletal muscle that covers several vulnerable neurovascular structures, and incorporates junctional zones between the lower limb, pelvis and perineum at the sciatic foramina. Direct and indirect musculoskeletal injuries in this region may damage the sciatic nerve and gluteal vessels.

The thigh consists of a cylinder of compact bone, the femoral shaft, surrounded by muscle groups traversed by important neurovascular structures. The muscles are grouped according to function and lie within osteofascial compartments that are defined by fascial septa running between the femur and an enveloping tube of thick fascia, the fascia lata. The femoral artery gives off its major branch, the profunda femoris artery (deep artery of the thigh), in the anterior compartment, and the sciatic nerve usually divides into its main branches, the tibial and common fibular nerves, as it passes through the posterior compartment of the thigh. The femoral nerve divides soon after entering the anterior compartment of the thigh beneath the inguinal ligament; the obturator nerve enters the medial thigh proximally and medially from the pelvis and divides into its main branches, which run anterior and posterior to adductor brevis.

SKIN AND SOFT TISSUES

SKIN

See also [Chapter 78](#) (p. 1316).

Cutaneous vascular supply and lymphatic drainage

Buttock

Most of the skin of the buttock is supplied by musculocutaneous perforating vessels from the superior and inferior gluteal arteries. There are also small peripheral contributions from similar branches of the internal pudendal, iliolumbar and lateral sacral arteries.

Cutaneous veins are tributaries of vessels that correspond to the named arteries. Cutaneous lymphatic drainage is to the superficial inguinal nodes.

Thigh

The skin of the thigh distal to the inguinal ligament and gluteal fold is supplied mainly by branches of the femoral and profunda femoris arteries. There is some contribution from the obturator, inferior gluteal and popliteal arteries, and from direct cutaneous, musculocutaneous and fasciocutaneous vessels. For further details, consult [Cormack and Lamberty \(1994\)](#).

Cutaneous veins are tributaries of vessels that correspond to the named arteries. Cutaneous lymphatic drainage is to the superficial inguinal nodes, mainly via collecting trunks accompanying the long saphenous vein.

Innervation

For the dermatomes and cutaneous nerves, see [Figures 78.12B, 78.13, 78.14](#).

SOFT TISSUES

Subcutaneous tissue

The subcutaneous tissue (tela subcutanea) of the thigh and buttock consists, as elsewhere in the limbs, of loose areolar tissue containing a variable quantity of fat. In some regions, particularly near the inguinal ligament, it splits into recognizable layers, between which may be found the branches of superficial vessels and nerves. It is thick in the inguinal region, where its two layers enclose the superficial inguinal lymph nodes, long saphenous vein and other smaller vessels. Here, the superficial layer is continuous with that of the abdominal fascia. The deep layer, a thin fibroelastic stratum, is most marked medial to the long saphenous vein and inferior to the inguinal ligament, and is interposed between the subcutaneous vessels and nerves and the deep fascia, fusing with the latter a little below the ligament. This membranous layer of subcutaneous tissue overlies the saphenous opening, blending with its circumference and with the femoral sheath. Over the opening, it is perforated by the long saphenous vein, by the superficial branches of the femoral artery other than the superficial circumflex iliac branch (which perforates the fascia lata separately), and lymphatic vessels; hence the term cribriform fascia. The subcutaneous tissue of the buttock is continuous superiorly with that over the low back and contains a variable quantity of fat.

Deep fascia

The deep fascia (fascia musculorum) covering the gluteal muscles varies in thickness. Over gluteus maximus it is thin, but over the anterior two-thirds of gluteus medius it forms the thick, strong gluteal aponeurosis. This is attached to the lateral border of the iliac crest superiorly, and splits anteriorly to enclose tensor fasciae latae and posteriorly to enclose gluteus maximus.

Fascia lata

The fascia lata, the wide, deep fascia of the thigh, is thicker in the proximal and lateral parts of the thigh where tensor fasciae latae and an expansion from gluteus maximus are attached to it. It is thin posteriorly and over the adductor muscles, but thicker around the knee, where it is strengthened by expansions from the tendon of biceps femoris laterally, sartorius medially and quadriceps femoris anteriorly. The fascia lata is attached superiorly and posteriorly to the back of the sacrum and coccyx, laterally to the outer margin of the iliac crest, anteriorly to the inguinal ligament and superior ramus of the pubis, and medially to the inferior ramus of the pubis, the ramus and tuberosity of the ischium, and the lower border of the sacrotuberous ligament. From the iliac crest, it descends as a dense layer over gluteus medius to the upper border of gluteus maximus, where it splits into two layers, one passing superficial and the other deep to the muscle, the layers reuniting at the lower border of the muscle.

Iliotibial tract

Over the flattened lateral surface of the thigh, the fascia lata thickens to form a strong band, the iliotibial tract. The upper end of the tract splits into two layers, where it encloses and anchors tensor fasciae latae and receives, posteriorly, most of the tendon of gluteus maximus. The superficial layer ascends lateral to tensor fasciae latae to the iliac crest; the deeper layer passes up and medially, deep to the muscle, and blends with the lateral part of the capsule of the hip joint. Distally, the iliotibial

tract is attached to a smooth, triangular facet (Gerdy's tubercle) on the anterolateral aspect of the lateral condyle of the tibia, where it is superficial to, and blends with, an aponeurotic expansion from vastus lateralis. When the knee is extended against resistance, it stands out as a strong, visible ridge on the anterolateral aspect of the thigh and knee.

Distally, the fascia lata is attached to all exposed bony points around the knee joint, such as the condyles of the femur and tibia, and the head of the fibula. On each side of the patella, the deep fascia is reinforced by transverse fibres, which receive contributions from the lateral and medial vasti. The stronger lateral fibres are continuous with the iliotibial tract.

Intermuscular septa

The deep surface of the fascia lata yields two intermuscular septa, which are attached to the whole of the linea aspera and to its proximal and distal prolongations. The lateral septum, thicker and stronger than the medial one, extends from the attachment of gluteus maximus to the lateral femoral condyle; it lies between vastus lateralis in front and the short head of biceps femoris behind, and provides partial attachment for them. The medial, thinner and weaker septum lies between vastus medialis and the adductors and pectineus. Numerous smaller septa, such as that separating the thigh adductors and flexors, pass between the individual muscles, ensheathing them and sometimes providing partial attachment for their fibres.

Saphenous opening

The saphenous opening is an aperture in the deep fascia, inferolateral to the medial end of the inguinal ligament, which allows passage for the long saphenous vein and other smaller vessels (Fig. 80.1). The cribriform fascia, which is pierced by these structures, fills in the aperture and must be removed to reveal it. Adjacent subsidiary openings may exist to transmit venous tributaries. In the adult, the approximate centre of the saphenous opening is 3 cm lateral to a point just distal to the pubic tubercle. The length and width of the opening vary considerably. The fascia lata in this part of the thigh displays superficial and deep strata (not to be confused with the superficial and deep layers of the subcutaneous tissue described above). They lie, respectively, anterior and posterior to the femoral sheath, with the saphenous opening situated where the two layers are in continuity. This serves to explain the somewhat oblique and spiral configuration of the saphenous opening.

The superficial layer, lateral and superior to the saphenous opening, is attached, in continuity, to the crest and anterior superior spine of the ilium, to the whole length of the inguinal ligament, and to the pecten pubis and lacunar ligament. It is reflected inferolaterally from the pubic tubercle as the arched falciform margin, which forms the superior, lateral and inferior boundaries of the saphenous opening; this margin adheres to the anterior layer of the femoral sheath, and the cribriform fascia is attached to it. The falciform margin is considered to have superior and inferior horns. The inferior horn is well defined, and is continuous behind the long saphenous vein with the deep stratum of the fascia lata.

The deep layer is medial to the saphenous opening and is continuous with the superficial stratum at its lower margin. Traced upwards, it covers pectineus, adductor longus and gracilis, passes behind the femoral sheath, with which it blends, and continues to the pecten pubis.

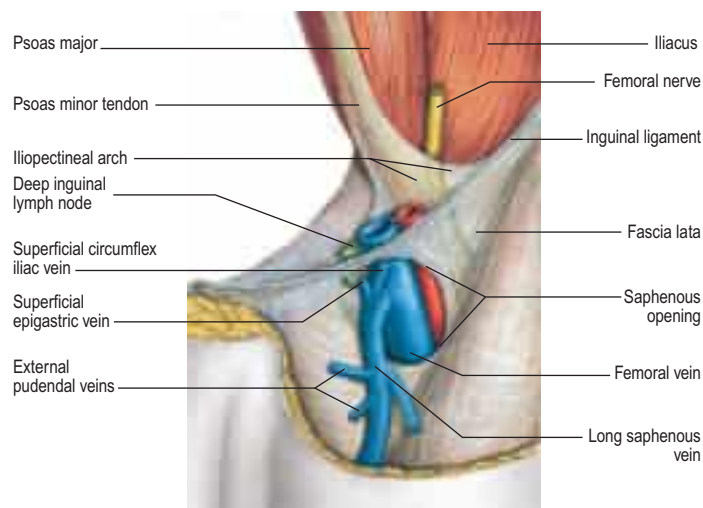


Fig. 80.1 The saphenous opening following removal of the cribriform fascia. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Fascial compartments

There are three functional groups of muscle in the thigh: namely, anterior (extensor), posterior (flexor) and medial (adductor). The anterior and posterior groups occupy separate osteofascial compartments that are limited peripherally by the fascia lata and separated from each other by the femur and the medial and lateral intermuscular septa (Fig. 80.2). The adductor muscles do not possess a separate compartment limited by fascial planes. Nevertheless, it is customary to speak of three compartments: anterior, posterior and medial. The muscles of the three compartments are described below. Adductor magnus, adductor longus and pectineus could each be considered to be constituents of two compartments, i.e. adductor magnus in the posterior and the medial compartments, and adductor longus and pectineus in the anterior and the medial compartments.

The nerve supply to the compartments of the thigh mainly follows the 'one compartment – one nerve' principle. Thus, the femoral nerve supplies the anterior compartment muscles, the obturator nerve supplies the medial compartment muscles, and the sciatic nerve supplies those in the posterior compartment. The dual functional and compartmental attribution of adductor magnus, adductor longus and pectineus are reflected in their dual nerve supplies.

In contrast to the motor innervation, the arterial supply to the compartmental muscle groups does not exhibit such a direct relationship. All groups receive a supply from the femoral system, particularly from the profunda femoris artery and its branches. The adductors receive a contribution from the obturator artery, and the hamstrings receive a proximal supply from the inferior gluteal artery. Further details are given in the descriptions of the individual muscles.

Femoral sheath

The femoral sheath is a funnel-shaped distal prolongation of extraperitoneal fascia, formed of transversalis fascia anterior to the femoral vessels, and of the iliac fascia posteriorly. It is wider proximally and its tapered distal end fuses with the vascular adventitia 3 or 4 cm distal to the inguinal ligament. At birth the sheath is shorter; it elongates when extension at the hips becomes habitual. The femoral branch of the genitofemoral nerve perforates its lateral wall. The medial wall slopes laterally and is pierced by the long saphenous vein and lymphatic vessels. Like the carotid sheath, the femoral sheath encloses a mass of connective tissue in which the vessels are embedded. Three compartments are described: a lateral one containing the femoral artery; an intermediate one for the femoral vein; and a medial compartment, the femoral canal, which contains lymph vessels and an occasional lymph node embedded in areolar tissue. The presence of this canal allows the femoral vein to distend. The canal is conical and approximately 1.25 cm in length. Its proximal (wider) end, termed the femoral ring, is bounded in front by the inguinal ligament, behind by pectineus and its fascia and the pectineal ligament, medially by the crescentic, lateral edge of the lacunar ligament and laterally by the femoral vein. The spermatic cord, or the round ligament of the uterus, is just above its anterior margin, while the inferior epigastric vessels are near its anterolateral rim. It is larger in women than in men: this is due partly to the relatively greater width of the female pelvis and partly to the smaller size of the femoral vessels in women. The ring is filled by condensed extraperitoneal tissue, the femoral septum, which is covered on its proximal aspect by the parietal peritoneum. Numerous lymph vessels that connect the deep inguinal to the external iliac lymph nodes traverse the femoral septum (Fig. 80.3).

Femoral hernia

Femoral hernia is described with other groin hernias on [page 1081](#).

Iliac fascia

The iliac fascia covers psoas and iliacus. It is thin above but thickens progressively towards the inguinal ligament. The part covering psoas is thickened above as the medial arcuate ligament. Medially, the fascia over psoas is attached by a series of fibrous arches to the intervertebral discs, the margins of vertebral bodies, and the upper part of the sacrum. Laterally, it blends with the fascia anterior to quadratus lumborum above the iliac crest, and with the fascia covering iliacus below the crest.

The iliac part is connected laterally to the whole of the inner lip of the iliac crest and medially to the pelvic brim, where it blends with the periosteum. It is attached to the iliopubic ramus, where it receives a slip from the tendon of psoas minor, when this muscle is present. The external iliac vessels are anterior to the fascia but the branches of the lumbar plexus are posterior to it. The fascia is separated from the peritoneum by loose extraperitoneal tissue. Lateral to the femoral vessels, the iliac fascia is continuous with the posterior margin of the inguinal

A

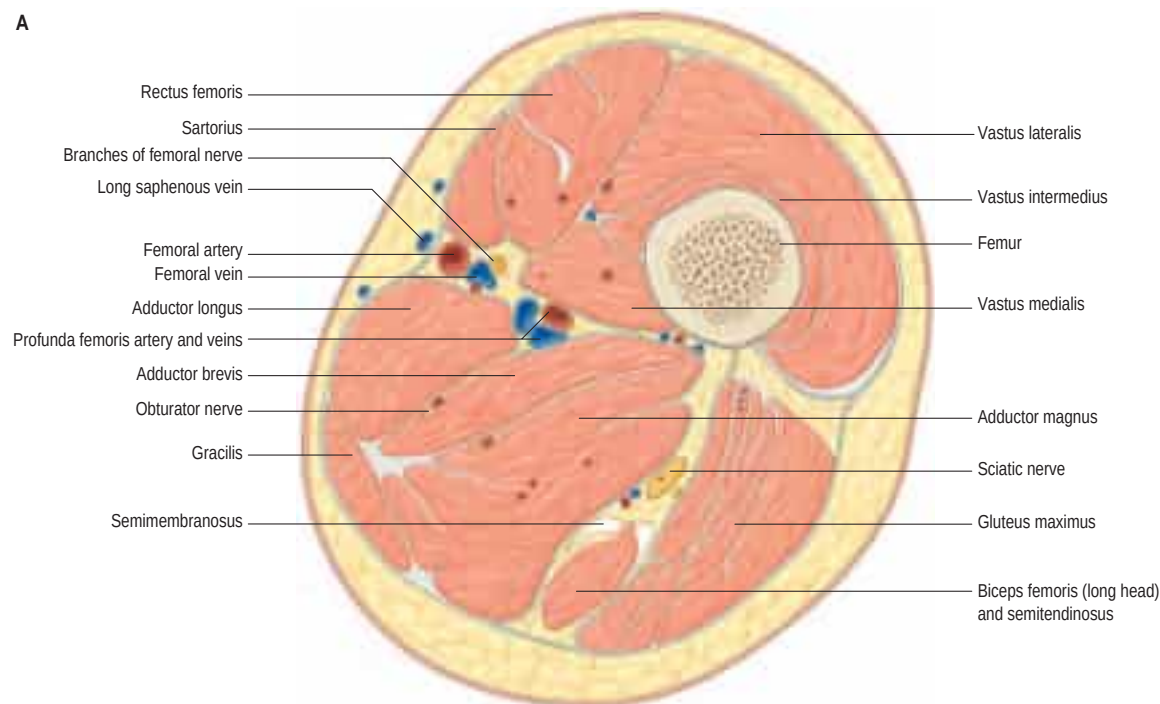
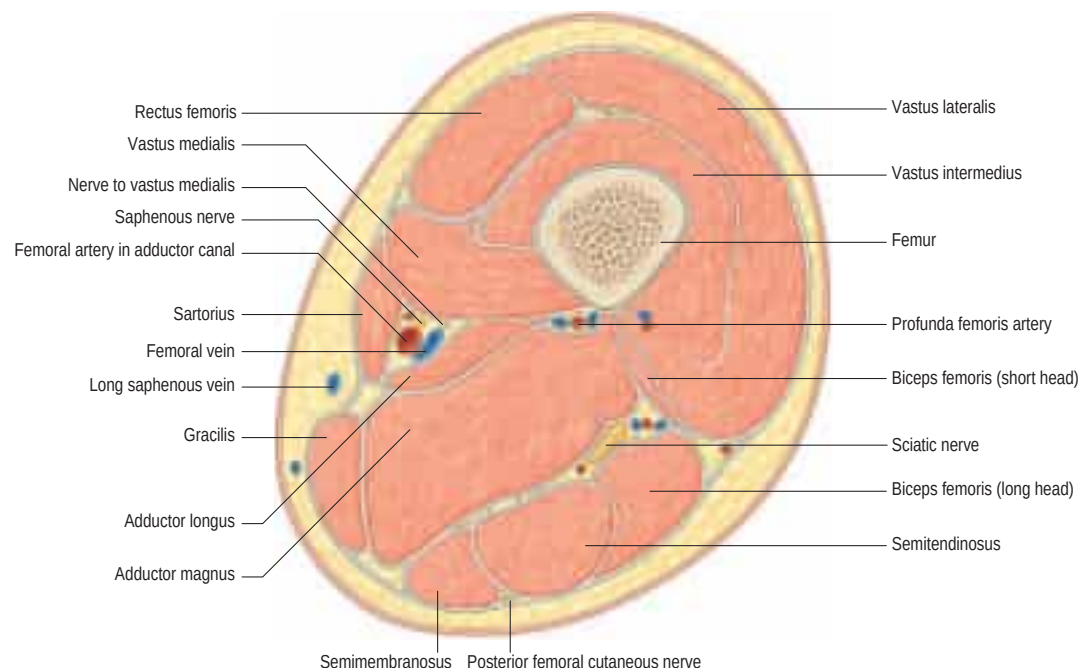


Fig. 80.2 A transverse (axial) section through the thigh. **A**, At the level of the apex of the femoral triangle. **B**, At the level of the mid-thigh.

B



ligament and the transversalis fascia. Medially, it passes behind the femoral vessels to become the pectineal ligament, attached to the pecten pubis. At the junction of its lateral and medial parts, it is attached to the iliopubic ramus and the capsule of the hip joint. It thus forms a septum between the inguinal ligament and the hip bone, dividing the space here into a lateral part, the muscular space, containing psoas major, iliacus and the femoral nerve, and a medial part, the vascular space, transmitting the femoral vessels (see Fig. 80.3). The iliac fascia continues downwards to form the posterior wall of the femoral sheath.

Obturator membrane

The obturator membrane (Fig. 80.4) is a thin aponeurosis that closes (obturates) most of the obturator foramen, leaving a superolateral aperture, the obturator canal, through which the obturator vessels and nerve leave the pelvis and enter the thigh. The membrane is attached to the sharp margin of the obturator foramen except at its inferolateral angle, where it is fixed to the pelvic surface of the ischial ramus, i.e. internal to the foramen. Its fibres are arranged mainly transversely in interlacing bundles; the uppermost bundle, which is attached to the obturator tubercles, completes the obturator canal. The outer and inner surfaces of the obturator membrane provide attachment for the obturator externus and internus, respectively. Some fibres of the pubofemoral ligament of the hip joint are attached to the outer surface.

BONES

The pelvic girdle is an entity consisting of the two hip bones and the sacrum (strictly speaking, the sacrum is part of the vertebral column). The pelvic girdle is massively constructed and serves as a weight-bearing and protective structure, an attachment for trunk and limb muscles, and as the skeletal framework of the birth canal.

HIP BONE

The hip bone is large, irregular, constricted centrally and expanded above and below (Fig. 80.5). Its lateral surface has a deep, cup-shaped acetabulum, articulating with the femoral head, anteroinferior to which is the large, oval or triangular obturator foramen. Above the acetabulum, the bone widens into an undulant plate surmounted by a sinuously curved iliac crest.

The bone articulates in front with its fellow and posteriorly with the side of the sacrum to form the pelvic girdle. Each hip bone has three parts – ilium, ischium and pubis, connected to each other by cartilage in youth but united as one bone in adults. The principal union is in the acetabulum. The ilium includes the upper acetabulum and expanded

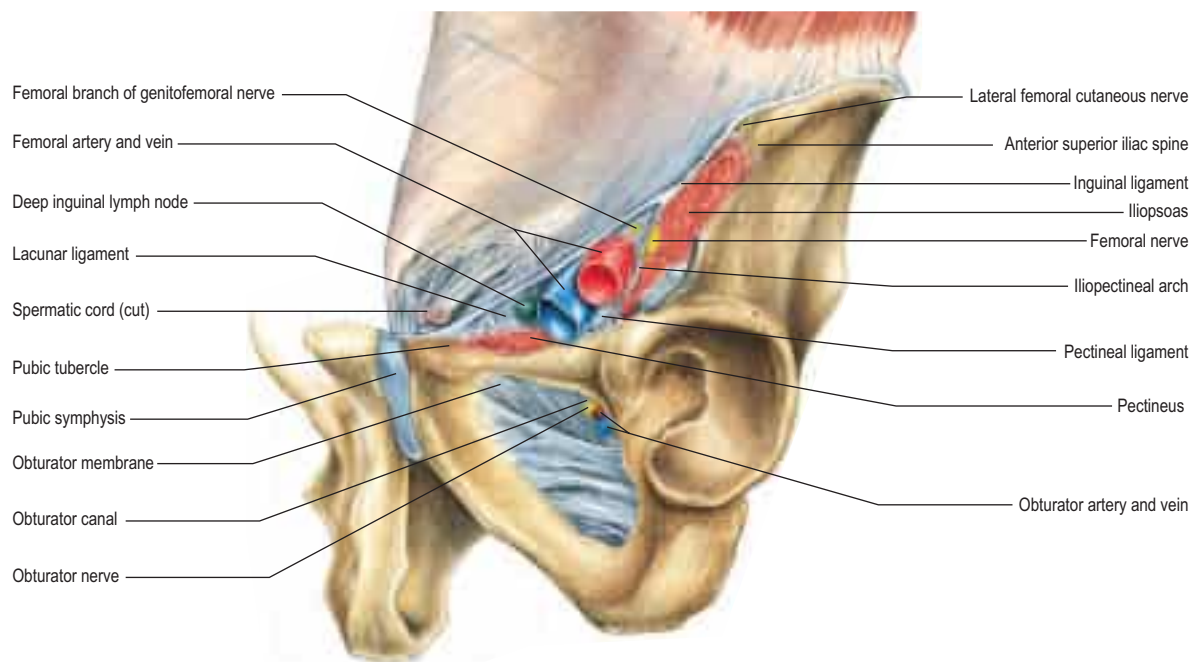


Fig. 80.3 Structures passing beneath the inguinal ligament. (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

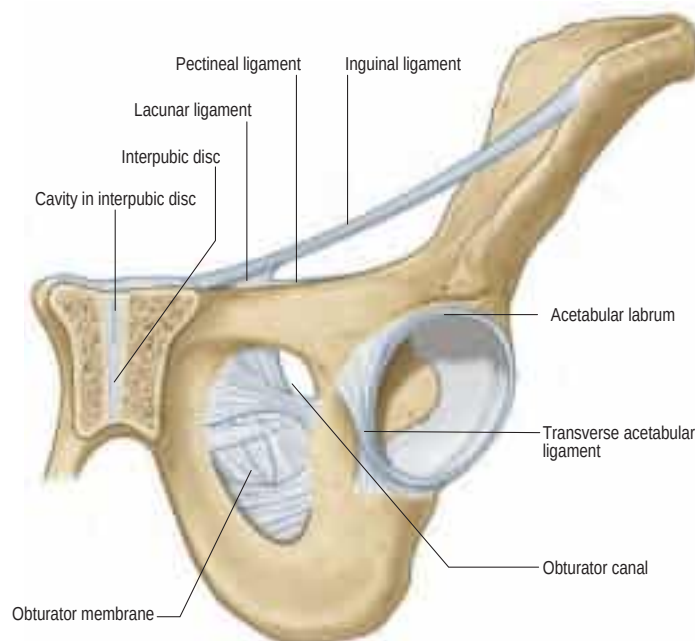


Fig. 80.4 An anterolateral view of left bony pelvis, showing associated ligaments and the obturator membrane.

area above it; the ischium includes the posteroinferior acetabulum and bone posteroinferior to it; the pubis forms the anteroinferior acetabulum.

Acetabulum

The acetabulum (**Fig. 80.5A,C**) is an approximately hemispherical cavity situated about the centre of the lateral aspect of the hip bone. It faces anteroinferiorly and is circumscribed by an irregular margin deficient inferiorly at the acetabular notch. The acetabular fossa forms the central floor and is rough and non-articular. The articular lunate surface is widest above (the 'dome'), where weight is transmitted to the femur. Consequently, fractures through this region tend to be associated with unsatisfactory outcomes. All three components of the hip bone contribute to the acetabulum, although unequally. The pubis forms the antero-superior fifth of the articular surface, the ischium forms the floor of the fossa and rather more than the posteroinferior two-fifths of the articular surface, and the ilium forms the remainder. Occasionally, a linear defect may be seen to cross the acetabular surface from the superior border to the acetabular fossa. This does not correspond to any junction between the main morphological parts of the hip bone.

Obturator foramen

The obturator foramen lies below and slightly anterior to the acetabulum, between the pubis and ischium. It is bordered above by the grooved obturator surface of the superior pubic ramus, medially by the pubic body and its inferior ramus, below by the ischial ramus, and laterally by the anterior border of the ischial body, including the margin of the acetabulum. The foramen is almost closed by the obturator membrane (see above), which is attached to its margins, except superolaterally, where a communication remains between the pelvis and thigh. This free edge of the membrane is attached to an anterior obturator tubercle at the anterior end of the inferior border of the superior pubic ramus, and a posterior obturator tubercle on the anterior border of the acetabular notch; these tubercles are sometimes indistinct. Since the tubercles lie in different planes and the obturator groove crosses the upper border of the foramen, the acetabular margin is in fact a spiral. The foramen is large and oval in males, but smaller and nearly triangular in females.

Structure

The thicker parts of the hip bone are trabecular, encased by two layers of compact bone, while the thinner parts, as in the acetabulum and central iliac fossa, are often translucent and consist of a single lamina of compact bone. In the upper acetabulum and along the arcuate line, i.e. the route of weight transmission from the sacrum to the femur, the amount of compact bone is increased and the subjacent trabecular bone displays two sets of pressure lamellae. These start together near the upper auricular surface and diverge to meet two strong buttresses of compact bone, from which two similar sets of lamellar arches start and converge on the acetabulum. The anterior part of the iliac crest has been much studied with regard to distribution of cortical and trabecular bone. [Whitehouse \(1977\)](#) described the cortical bone as very porous, being only 75% bone, decreasing to 35% near the anterior superior iliac spine.

Studies of the internal stresses within the hip bone have revealed a pattern of trabeculae that corresponds well with the theoretically expected patterns of stress trajectories ([Holm 1980](#)). These patterns are considerably more complex than in any other major bone. Stresses are higher in the acetabulum than in the iliac region. In the ilium, the pelvic surface is subjected to considerably less stress than is the gluteal surface.

Muscle attachments

See individual bones.

Vascular supply

In the infant, nutrient arteries are clearly demonstrable for each component of the hip bone. Each nutrient artery branches in a fan-like fashion within its bone of supply ([Crock 1996](#)). Later, a periosteal arterial network develops, with contributions from numerous local arteries (see under individual bones).

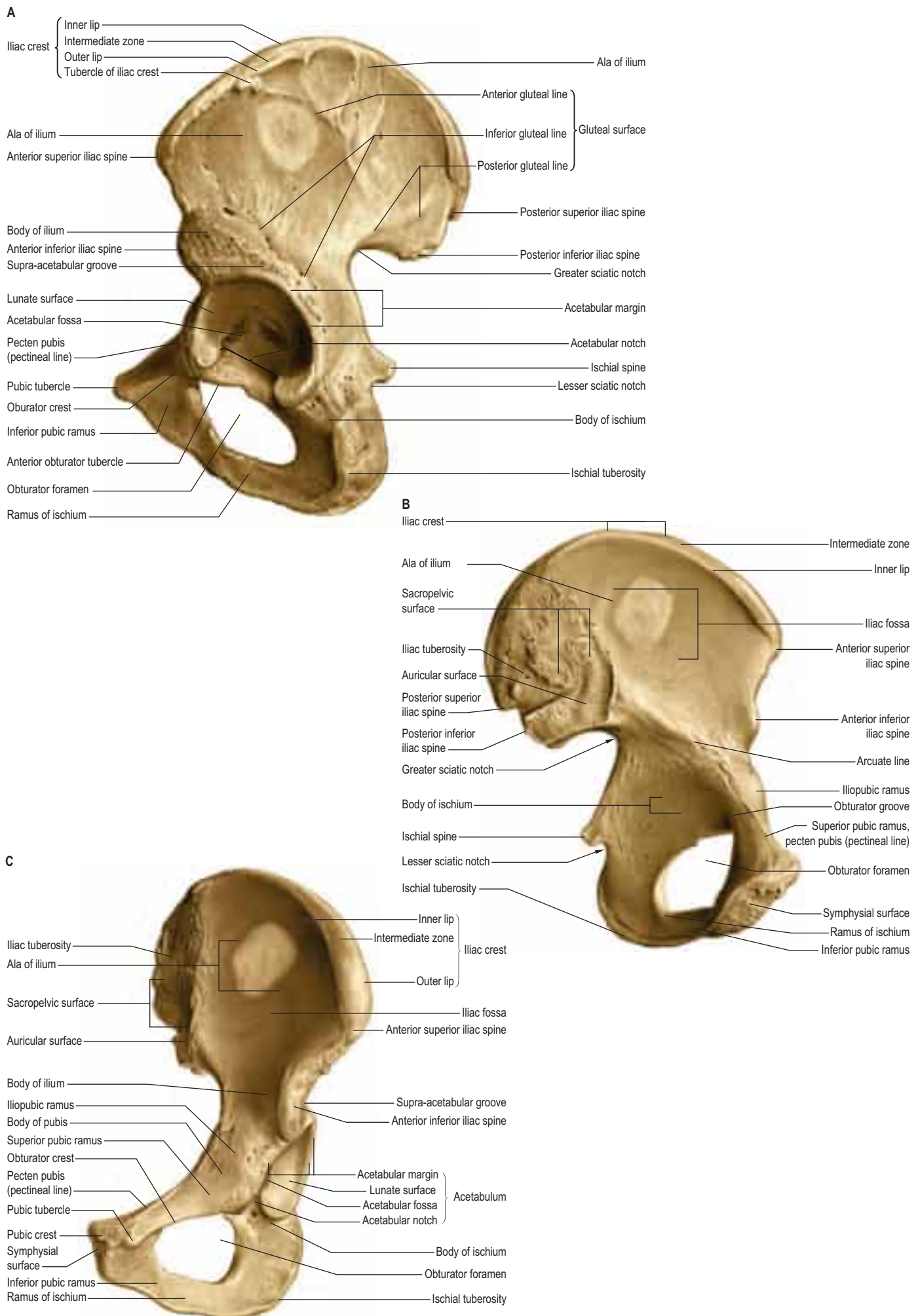


Fig. 80.5 The left hip bone. **A**, Outer aspect. **B**, Inner aspect. **C**, Anterosuperior view. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Innervation

Periosteal innervation is by a network of nerves derived from branches of local nerves. These nerves also supply muscles attaching to the periosteum and the joints involving the hip bone. Autonomic nerves accompany nutrient arteries and branch within the bone.

Ossification

Ossification (Figs 80.6–80.7) is by three primary centres: one each for the ilium, ischium and pubis. The iliac centre appears above the greater sciatic notch prenatally at about the ninth week; the ischial centre in its body in the fourth month; and the pubic centre in its superior ramus between the fourth and fifth months. The pubis is often not recovered from prenatal remains due to its size and fragility and because it is the last of the hip bones to begin ossification (Scheuer and Black 2004). At birth the whole iliac crest, the acetabular floor and the inferior margin are cartilaginous. Gradual ossification of the three components of the acetabulum results in a triradiate cartilaginous stem extending medially to the pelvic surface as a Y-shaped epiphyseal plate between the ilium, ischium and pubis, and including the anterior inferior iliac spine. Cartilage along the inferior margin also covers the ischial tuberosity, forms

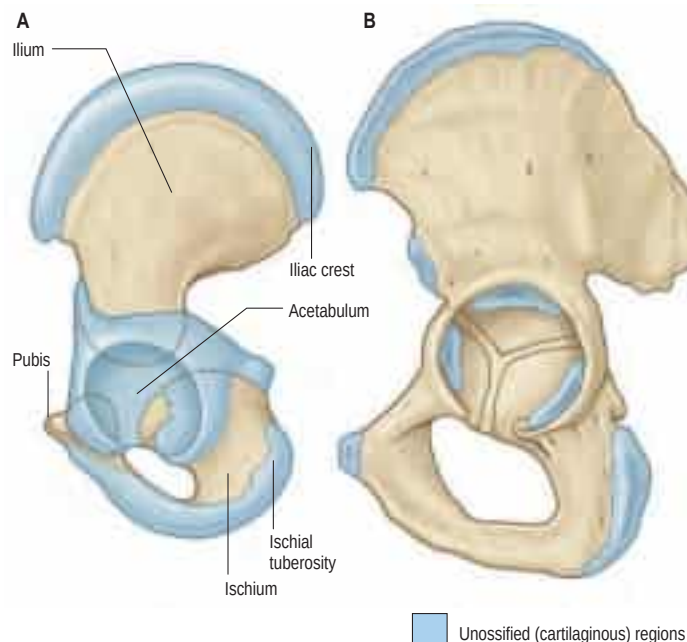


Fig. 80.6 The hip bone. **A**, At birth. **B**, In adolescence; more heavily stippled areas indicate the secondary centres of ossification.



Fig. 80.7 An anteroposterior radiograph of the pelvis of a boy aged 7 years. Key: 1, ilium; 2, part of triradiate growth cartilage; 3, superior femoral epiphysis; 4, cartilaginous growth plates; 5, ossifying greater trochanter; 6, ischium; 7, pubis; 8, cartilage between inferior pubic and ischial rami.

conjoined ischial and pubic rami, and continues to the pubic symphyseal surface and along the pubic crest to the pubic tubercle.

The ossifying ischium and pubis fuse to form a continuous ischio-pubic ramus at the seventh or eighth year. Secondary centres, other than for the acetabulum, appear at about puberty and fuse between the fifteenth and twenty-fifth years. There are usually two for the iliac crest (which fuse early), and single centres for the ischial tuberosity (in cartilage close to the inferior acetabular margin and spreading forwards), anterior inferior iliac spine (although it may ossify from the triradiate cartilage) and symphyseal surface of the pubis (the pubic tubercle and crest may have separate centres). Progression of ossification of the iliac crest in girls is an index of skeletal maturity and is useful in determining the optimal timing of surgery for spinal deformity.

Between the ages of 8 and 9 years, three major centres of ossification appear in the acetabular cartilage. The largest appears in the anterior wall of the acetabulum and fuses with the pubis, the second in the iliac acetabular cartilage superiorly, fusing with the ilium, and the third in the ischial acetabular cartilage posteriorly, fusing with the ischium. At puberty, these epiphyses expand towards the periphery of the acetabulum and contribute to its depth (Ponseti 1978). Fusion between the three bones within the acetabulum occurs between the sixteenth and eighteenth years. Delaere et al (1992) have suggested that ossification of the ilium is similar to that of a long bone, possessing three cartilaginous epiphyses and one cartilaginous process, although it tends to undergo osteoclastic resorption comparable with that of cranial bones. During development, the acetabulum increases in breadth at a faster rate than it does in depth.

Avulsion fractures of pelvic apophyses may occur from excessive pull on tendons, usually in athletic adolescents. The most frequent examples of such injuries are those to the ischial tuberosity (hamstrings) and anterior inferior iliac spine (rectus femoris).

Pubis

The pubis (see Figs 80.5–80.6) is the ventral part of the hip bone and forms a median cartilaginous pubic symphysis with its fellow. The body of the pubis occupies the anteromedial part of the bone, and from the body a superior ramus passes up and back to the acetabulum and an inferior ramus passes back, down and laterally to join the ischial ramus inferomedial to the obturator foramen.

Body

The body, anteroposteriorly compressed, has anterior, posterior and symphyseal (medial) surfaces and an upper border, the pubic crest. The anterior surface also faces inferolaterally; it is rough superomedially and smooth elsewhere, giving attachment to medial thigh muscles. The smooth posterior surface faces upwards and backwards as the oblique anterior wall of the lesser pelvis and is related to the urinary bladder. The symphyseal surface is elongated and oval, united by cartilage to its fellow at the pubic symphysis. Denuded of cartilage, it has an irregular surface of small ridges and furrows or nodular elevations, varying considerably with age, features that are of forensic interest. The pubic crest is the rounded upper border of the body and overhangs the anterior surface; the pubic tubercle is a small rounded eminence on its lateral end. Both crest and tubercle are palpable; the latter partly is obscured in males by the spermatic cord that crosses above it from the scrotum to the anterior abdominal wall. The pubic rami diverge posterolaterally from the superolateral corners of the body.

The anterior surface of the pubic body faces the adductor region. The anterior pubic ligament attaches to its medial part along a rough strip, which is wider in females. The posterior surface is separated from the urinary bladder by retropubic fat. The puboprostatic ligaments are attached to this surface medial to levator ani.

Superior pubic ramus

The superior pubic ramus passes upwards, backwards and laterally from the body, superolateral to the obturator foramen, to reach the acetabulum. It is triangular in section and has three surfaces and borders. Its anterior, pectineal surface, tilted slightly up, is triangular in outline and extends from the pubic tubercle to the iliopubic ramus. It is bounded in front by the rounded obturator crest and behind by the sharp pecten pubis (pectineal line), which, with the crest, is the pubic part of the linea terminalis (i.e. anterior part of the pelvic brim). The posterosuperior, pelvic surface, medially inclined, is smooth and narrows into the posterior surface of the body, which is bounded above by the pecten pubis and below by a sharp inferior border. The obturator surface, directed down and back, is crossed by the obturator groove sloping down and forwards. Its anterior limit is the obturator crest and its posterior limit is the inferior border.

Inferior pubic ramus

The inferior pubic ramus, an inferolateral process of the body, descends inferolaterally to join the ischial ramus medial to, and below, the obturator foramen. The union may be locally thickened, but not obviously so in adults. The ramus has two surfaces and borders. The anterolateral surface, continuous above with that of the pubic body, faces the thigh and is marked by muscles. It is limited laterally by the margin of the obturator foramen, and medially by the rough anterior border. The posteromedial surface is continuous above with that of the body and is transversely convex; its medial part is often everted in males and gives attachment to the crus of the penis. This surface faces the perineum medially, its smooth lateral part tilted up towards the pelvic cavity.

The internal surface is indistinctly divided into medial, intermediate and lateral areas. The medial area faces inferomedially in direct contact with the crus of the penis or clitoris and is limited above and behind by an indistinct ridge for attachment of the fascia overlying the superficial perineal muscles. The medial margin of the ramus, strongly everted in males, provides attachment for the fascia lata and the stratum membranousum perinei.

Pubic tubercle

The pubic tubercle provides attachment to the medial end of the inguinal ligament. It forms part of the floor of the superficial inguinal ring and is crossed by the spermatic cord.

Pecten pubis

The pecten pubis is the sharp, superior edge of the pectineal surface. The conjoint tendon (inguinal falx) and lacunar ligament are attached at its medial end and a strong, fibrous pectineal ligament is attached along the rest of its surface. The smooth pelvic surface is separated from parietal peritoneum only by areolar tissue, in which the lateral umbilical ligament descends forwards across the ramus and, laterally, the vas deferens (ductus deferens) passes backwards. The obturator groove, which is converted to a canal by the upper borders of the obturator membrane and obturator muscles, transmits the obturator vessels and nerve from the pelvis to the thigh. Some fibres of the pubofemoral ligament are attached to the lateral end of the obturator crest.

Muscle attachments

The tendon of adductor longus is attached on the anterior (external) surface of the body, below the pubic crest. Below adductor longus, gracilis is attached to a line near the lower margin extending down on to the inferior ramus. Above gracilis, adductor brevis is attached to the body and inferior ramus. Above again, obturator externus is attached to the anterior surface, spreading on to inferior pubic and ischial rami. Adductor magnus usually extends from the ischial ramus on to the lower part of the inferior pubic ramus between adductor brevis and obturator externus. Pectineus is attached to the pectineal surface of the superior ramus along its upper part. Ascending loops of cremaster are attached to the pubic tubercle. The lateral part of rectus abdominis and, inferiorly, pyramidalis, are attached lateral to the tubercle, on the pubic crest. Medially, the crest is crossed by the medial part of rectus abdominis, ascending from ligamentous fibres that interlace in front of the pubic symphysis. Anterior fibres of levator ani are attached on the posterior (internal) surface of the body near its centre. More laterally, obturator internus is attached on this surface, extending on to both rami. Psoas minor, when present, is attached near the centre of the pecten pubis.

Vascular supply

The pubis is supplied by a periosteal anastomosis of branches from the obturator, inferior epigastric and medial circumflex femoral arteries. The superficial and deep external pudendal arteries may also contribute. Multiple vascular foramina are present, mainly at the lateral (acetabular) end of the bone, but there is no consistently placed nutrient foramen.

Innervation

The pubic periosteum is innervated by branches of the nerves that supply muscles attached to the bone, the hip joint and the symphysis pubis.

Ossification

Ossification of the pubis is described on [page 1342](#).

Ilium

The ilium has upper and lower parts and three surfaces (see [Figs 80.5, 80.6](#)). The smaller, lower part forms a little less than the upper two-fifths of the acetabulum. The upper part is much expanded, and has gluteal,

sacropelvic and iliac (internal) surfaces. The posterolateral gluteal surface is an extensive rough area; the anteromedial iliac fossa is smooth and concave; and the sacropelvic surface is medial and posteroinferior to the fossa, from which it is separated by the medial border.

Iliac crest

The iliac crest is the superior border of the ilium. It is convex upwards but sinuous from side to side, being internally concave in front and convex behind. Its ends project as anterior and posterior superior iliac spines. The anterior superior iliac spine is palpable at the lateral end of the inguinal fold; the lateral end of the inguinal ligament is attached to the anterior superior iliac spine. The posterior superior iliac spine is not palpable but is often indicated by a dimple, approximately 4 cm lateral to the second sacral spinous process, above the medial gluteal region.

The iliac crest has ventral and dorsal segments. The ventral segment occupies slightly more than the anterior two-thirds of the iliac crest and its prominence is associated with changes in iliac form as a result of the emergence of the upright posture. It has internal and external lips and a rough intermediate zone that is narrowest centrally. The dorsal segment, which occupies approximately the posterior one-third in humans, is a feature of all land vertebrates. It has two sloping surfaces separated by a longitudinal ridge ending at the posterior superior spine. The interosseous and posterior sacroiliac ligaments arise from the medial margin of the dorsal segment. The tubercle of the iliac crest projects outwards from the outer lip approximately 5 cm posterosuperior to the anterior superior spine. The summit of the iliac crest, a little behind its midpoint, is level with the fourth lumbar vertebral body in adults and with the fifth lumbar vertebral body in children aged 10 years or less ([Tame and Burstall 2003](#)).

Anterior border

The anterior border descends to the acetabulum from the anterior superior spine. Superiorly it is concave forwards. Inferiorly, immediately above the acetabulum, is a rough anterior inferior iliac spine, which is divided indistinctly into an upper area for the straight head of rectus femoris and a lower area extending laterally along the upper acetabular margin to form a triangular impression for the proximal end of the iliofemoral ligament.

Posterior border

The posterior border is irregularly curved and descends from the posterior superior spine, at first forwards, with a posterior concavity forming a small notch. At the lower end of the notch is a wide, low projection: the posterior inferior iliac spine. Here the border turns almost horizontally forwards for approximately 3 cm, then down and back to join the posterior ischial border. Together these borders form a deep notch, the greater sciatic notch, which is bounded above by the ilium and below by the ilium and ischium. The upper fibres of the sacrotuberous ligament are attached to the upper part of the posterior border. The superior rim of the notch is related to the superior gluteal vessels and nerve. The lower margin of the greater sciatic notch is covered by piriformis and is related to the sciatic nerve.

Medial border

The medial border separates the iliac fossa and the sacropelvic surface. It is indistinct near the crest, rough in its upper part, then sharp where it bounds an articular surface for the sacrum, and finally rounded. The latter part is the arcuate line, which inferiorly reaches the posterior part of the iliopubic ramus, marking the union of the ilium and pubis.

Gluteal surface

The gluteal surface, facing inferiorly in its posterior part and laterally and slightly downwards in front, is bounded above by the iliac crest, and below by the upper acetabular border and by the anterior and posterior borders. It is rough and curved, convex in front, concave behind, and marked by three gluteal lines. The posterior gluteal line is shortest, descending from the external lip of the crest approximately 5 cm in front of its posterior limit and ending in front of the posterior inferior iliac spine. Above, it is usually distinct, but inferiorly it is poorly defined and frequently absent. The anterior gluteal line, the longest, begins near the midpoint of the superior margin of the greater sciatic notch and ascends forwards into the outer lip of the crest, a little anterior to its tubercle. The inferior gluteal line, seldom well marked, begins posterosuperior to the anterior inferior iliac spine, curving posteroinferiorly to end near the apex of the greater sciatic notch. Between the inferior gluteal line and the acetabular margin is a rough, shallow groove. Behind the acetabulum, the lower gluteal surface is continuous

with the posterior ischial surface, the conjunction marked by a low elevation.

The articular capsule is attached to an area adjoining the acetabular margin, most of which is covered by gluteus minimus. Posteroinferiorly, near the union of the ilium and ischium, the bone is related to piriformis.

Iliac fossa

The iliac fossa, the internal concavity of the ilium, faces anterosuperiorly. It is limited above by the iliac crest, in front by the anterior border and behind by the medial border, separating it from the sacropelvic surface. It forms the smooth and gently concave posterolateral wall of the greater pelvis. Below it is continuous with a wide shallow groove, bounded laterally by the anterior inferior iliac spine and medially by the iliopectineal ramus.

The converging fibres of iliacus occupy the wide groove between the anterior inferior iliac spine and the iliopectineal ramus laterally and the tendon of psoas major medially; the tendon is separated from the underlying bone by a bursa. The right iliac fossa contains the caecum, and often the vermiform appendix and terminal ileum. The left iliac fossa houses the terminal part of the descending colon and the proximal sigmoid colon.

Sacropelvic surface

The sacropelvic surface, the posteroinferior part of the medial iliac surface, is bounded posteroinferiorly by the posterior border, anterosuperiorly by the medial border, posterosuperiorly by the iliac crest and anteroinferiorly by the line of fusion of the ilium and ischium. It is divided into iliac tuberosity and auricular and pelvic surfaces. The iliac tuberosity, a large, rough area below the dorsal segment of the iliac crest, shows cranial and caudal areas separated by an oblique ridge and connected to the sacrum by the interosseous sacroiliac ligament. The sacropelvic surface gives attachment to the posterior sacroiliac ligaments and, behind the auricular surface, to the interosseous sacroiliac ligament. The iliolumbar ligament is attached to its anterior part. The auricular surface, immediately anteroinferior to the tuberosity, articulates with the lateral sacral mass. Shaped like an ear, its widest part is anterosuperior, and its 'lobule' posteroinferior and on the medial aspect of the posterior inferior spine. Its edges are well defined but the surface, though articular, is rough and irregular. It articulates with the sacrum and is reciprocally shaped. The anterior sacroiliac ligament is attached to its sharp anterior and inferior borders. The narrow part of the pelvic surface, between the auricular surface and the upper rim of the greater sciatic notch, often shows a rough pre-auricular sulcus (that is usually better defined in females) for the lower fibres of the anterior sacroiliac ligament. For the reliability of this feature as a sex discriminant, refer to [Finnegan \(1978\)](#) and [Brothwell and Pollard \(2001\)](#). The pelvic surface is anteroinferior to the acutely curved part of the auricular surface, and contributes to the lateral wall of the lesser pelvis. Its upper part, facing down, is between the auricular surface and the upper limb of the greater sciatic notch. Its lower part faces medially and is separated from the iliac fossa by the arcuate line. Anteroinferiorly, it extends to the line of union between the ilium and ischium. Though usually obliterated, it passes from the depth of the acetabulum to approximately the middle of the inferior limb of the greater sciatic notch.

Muscle attachments

The attachment of sartorius extends down the anterior border below the anterior superior iliac spine. The iliac crest gives attachment to the anterolateral and dorsal abdominal muscles, and to the fasciae and muscles of the lower limb. The fascia lata and iliotibial tract are attached to the outer lip and tubercle of its ventral segment. Tensor fasciae latae is attached anterior to the tubercle. The lower fibres of external oblique and, just behind the summit of the crest, the lowest fibres of latissimus dorsi are attached to its anterior two-thirds. A variable interval exists between the most posterior attachment of external oblique and the most anterior attachment of latissimus dorsi, and here the crest forms the base of the lumbar triangle through which herniation of abdominal contents may rarely occur. Internal oblique is attached to the intermediate area of the crest. Transversus abdominis is attached to the anterior two-thirds of the inner lip of the crest, and behind this to the thoracolumbar fascia and quadratus lumborum. The highest fibres of gluteus maximus are attached to the dorsal segment of the crest on its lateral slope. Erector spinae arises from the medial slope of the dorsal segment. The straight head of rectus femoris is attached to the upper area of the anterior inferior spine. Some fibres of piriformis are attached in front of the posterior inferior spine on the upper border of the greater sciatic foramen.

The gluteal surface is divided by three gluteal lines into four areas. Behind the posterior line, the upper rough part gives attachment to the

upper fibres of gluteus maximus and the lower, smooth region to part of the sacrotuberous ligament and iliac head of piriformis. Gluteus medius is attached between the posterior and anterior lines, below the iliac crest, and gluteus minimus is attached between the anterior and inferior lines. The fourth area, below the inferior line, contains vascular foramina.

The reflected head of rectus femoris attaches to a curved groove above the acetabulum. Iliacus is attached to the upper two-thirds of the iliac fossa and is related to its lower one-third. The medial part of quadratus lumborum is attached to the anterior part of the sacropelvic surface, above the iliolumbar ligament. Piriformis is sometimes partly attached lateral to the pre-auricular sulcus, and part of obturator internus is attached to the more extensive remainder of the pelvic surface.

Vascular supply

Branches of the iliolumbar artery run between iliacus and the ilium; one or more enter large nutrient foramina lying posteroinferiorly in the iliac fossa. The superior gluteal, obturator and superficial circumflex iliac arteries contribute to the periosteal supply. The obturator artery may supply a nutrient branch. Vascular foramina on the ilium underlying the gluteal muscles may lead into large vascular canals in the bone.

Innervation

The periosteum is innervated by branches of nerves that supply muscles attached to the bone, the hip joint and the sacroiliac joint.

Ossification

Ossification of the ilium is described on [page 1342](#).

Ischium

The ischium, the inferoposterior part of the hip bone, has a body and ramus. The body has upper and lower ends and femoral, posterior and pelvic surfaces (see [Figs 80.5–80.7](#)). Above, it forms the posteroinferior part of the acetabulum; below, its ramus ascends anteromedially at an acute angle to meet the inferior pubic ramus, thereby completing the boundary of the obturator foramen. The ischiofemoral ligament is attached to the lateral border below the acetabulum ([Fuss and Bacher 1991](#)).

The femoral surface faces downwards, forwards and laterally towards the thigh. It is bounded in front by the margin of the obturator foramen. The lateral border, indistinct above but well defined below, forms the lateral limit of the ischial tuberosity. At a higher level, the femoral surface is covered by piriformis, from which it is partially separated by the sciatic nerve and the nerve to quadratus femoris. The posterior surface, facing superolaterally, is continuous above with the iliac gluteal surface, and here a low convexity follows the acetabular curvature. Inferiorly, this surface forms the upper part of the ischial tuberosity, above which is a wide, shallow groove on its lateral and medial aspects. Above the ischial tuberosity, the posterior surface is crossed by the tendon of obturator internus and the gemelli. The nerve to quadratus femoris lies between these structures and the ischium. The ischial tuberosity is a large, rough area on the lower posterior surface and inferior extremity of the ischium. Though obscured by gluteus maximus in hip extension, it is palpable in hip flexion. It is 5 cm from the midline and about the same distance above the gluteal fold. It is elongated and widest above, and tapers inferiorly. The ischial posterior aspect lies between the lateral and posterior borders. The posterior border blends above with that of the ilium, helping to complete the inferior rim of the greater sciatic foramen, the posterior end of which has a conspicuous ischial spine. Below this, the rounded border forms the floor of the lesser sciatic foramen, between the ischial spine and tuberosity. The pelvic surface is smooth and faces the pelvic cavity; inferiorly, it forms part of the lateral wall of the ischio-anal fossa.

Ischial ramus

The ischial ramus has anteroinferior and posterior surfaces continuous with the corresponding surfaces of the inferior pubic ramus. The anteroinferior surface is roughened by the attachment of the medial thigh muscles. The smooth posterior surface is partly divided into perineal and pelvic areas, like the inferior pubic ramus. The upper border completes the obturator foramen; the rough lower border, together with the medial border of the inferior pubic ramus, contributes to the pubic arch. The fascia overlying the superficial muscles of the perineum is attached below the ridge between the perineal and pelvic areas of the posterior surface of the ischial ramus. Above the ridge, areas give attachment to the crus of the penis or clitoris and the external urethral sphincter. The lower border of the ramus provides an attachment for the fascia lata and the stratum membranosum perinei.

Ischial tuberosity

The ischial tuberosity is divided nearly transversely into upper and lower areas. The upper area is subdivided by an oblique line into superolateral and inferomedial parts. The lower area, narrowing as it curves on to the inferior ischial aspect, is subdivided by an irregular vertical ridge into lateral and medial areas. The medial area is covered by fibroadipose tissue that usually contains the sciatic bursa of gluteus maximus, which supports the body in sitting. Medially, the tuberosity is limited by a curved ridge that passes on to the ramus and which gives attachment to the sacrotuberous ligament and its falciform process.

Ischial spine

The ischial spine projects downwards and a little medially (see Fig. 80.5A,B). The sacrospinous ligament is attached to its margins, separating the greater from the lesser sciatic foramen. The ligament is crossed posteriorly by the internal pudendal vessels, pudendal nerve and the nerve to obturator internus.

Muscle attachments

Part of obturator externus is attached to the lower femoral surface of the ischial body. Part of obturator externus, the anterior fibres of adductor magnus and, near the lower border, gracilis are all attached to the anterior surface of the ischial ramus. Between adductor magnus and gracilis, the attachment of adductor brevis may descend from the inferior pubic ramus. The posterior surface is divided into pelvic and perineal areas. The pelvic area, facing back, has part of obturator internus attached to it. The perineal area faces medially; its upper part is related to the crus of the penis or clitoris, and its lower part gives attachment to sphincter urethrae, ischiocavernosus and the superficial transverse perineal muscle.

The ischial tuberosity gives attachment to the posterior thigh muscles. Quadratus femoris is attached along the upper part of its lateral border. The upper area of the tuberosity is subdivided by an oblique line into a superolateral part for semimembranosus and an inferomedial part for the long head of biceps femoris and semitendinosus. The lower area is subdivided by an irregular vertical ridge into lateral and medial areas. The larger lateral area is for part of adductor magnus. Superomedial to the tuberosity, the posterior surface has a wide, shallow groove, usually covered by hyaline cartilage, with a bursa between it and the tendon of obturator internus. Gemellus inferior is attached to the lower margin of the groove, near the tuberosity. Gemellus superior is attached to the upper margin, near the ischial spine. Forceful contraction of the hamstrings can result in an avulsion fracture of the ischial tuberosity.

The pelvic surface of the ischial spine gives attachment to coccygeus and to the most posterior fibres of levator ani. Obturator internus is attached to the upper part of the smooth pelvic ischial surface and converges on the bony part of the lesser sciatic foramen, covering the rest of this surface other than the pelvic aspect of the ischial spine; the muscle and its fascia separate the bone from the ischio-anal fossa.

Vascular supply

There are multiple vascular foramina at the acetabular margins and a few are usually present on the pelvic surface. Branches of the obturator, medial circumflex femoral and inferior gluteal arteries supply the ischium.

Innervation

The periosteum is innervated by branches of nerves that supply the hip joint and muscles attached to the bone.

Ossification

Ossification of the ischium is described on [page 1342](#).

SACRUM

See [page 726](#).

COCYX

See [page 729](#).

BONY PELVIS AS A WHOLE

The term pelvis ('basin') is applied variously to the skeletal ring formed by the hip bones and the sacrum, the cavity therein, and even the entire

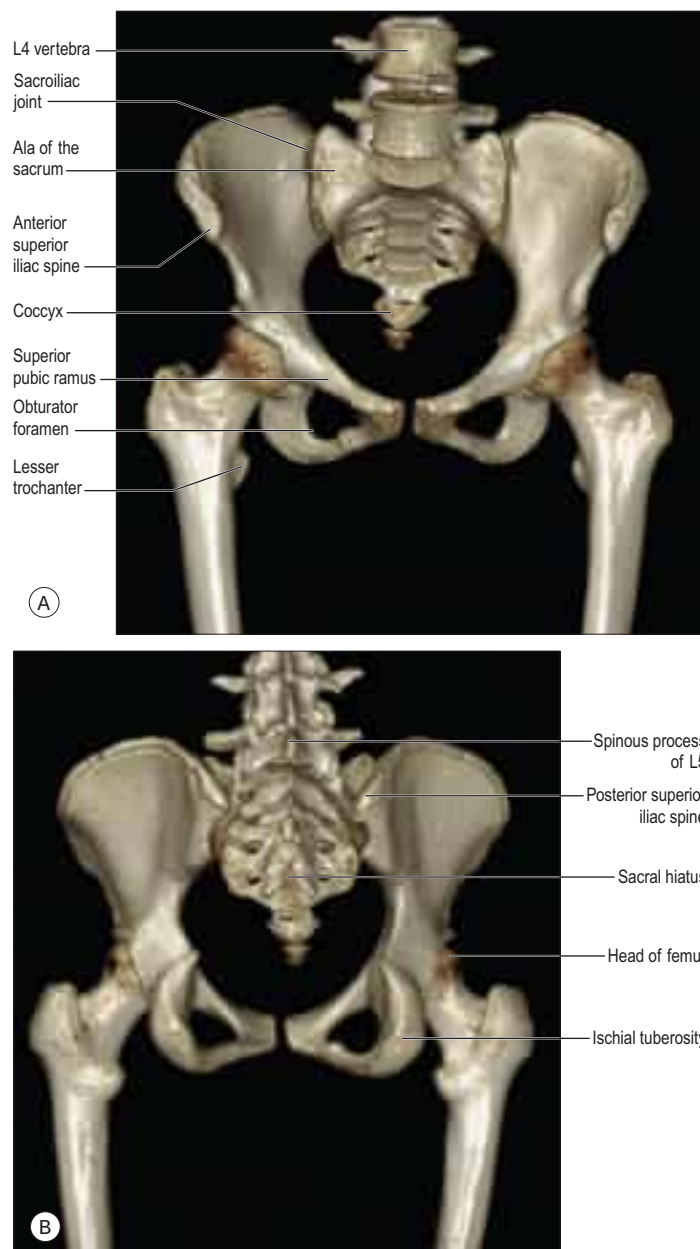


Fig. 80.8 Anterior (A) and posterior (B) views of the female pelvis, upper femur and lower lumbar spine, using three-dimensional computed tomography. (Courtesy of Dr Yoginder Vaid and Mr Jon C Betts, Jr.)

region where the trunk and lower limbs meet (Fig. 80.8). It is used here in the skeletal sense, to describe the irregular osseous girdle between the femoral heads and fifth lumbar vertebra. It is large because its primary function is to withstand the forces of body weight and musculature. In this section, its obstetric, forensic and anthropological significance will be considered.

The pelvis can be regarded as having greater and lesser segments, the true and false pelves. The segments are arbitrarily divided by an oblique plane passing through the sacral promontory posteriorly and the lineae terminales elsewhere. Each linea terminalis includes the iliac arcuate line, pectineal line (pecten pubis) and pubic crest.

Greater pelvis

The greater pelvis consists of the ilium and pubis above the lineae terminales and the base of the sacrum. This junctional zone is structurally massive and forms powerful arches from the acetabular fossae to the vertebral column around the visceral cavity, which is part of the abdomen. It has little anterior wall because of the pelvic inclination.

Pelvic inlet (superior pelvic aperture)

The pelvic inlet or brim may be round or oval in shape, and is indented posteriorly by the sacral promontory. The pelvic brim is obstetrically important and has also long been measured for anthropological reasons, as has the pelvic cavity.

By convention, the pelvic inlet is described in three dimensions. The anteroposterior diameter (true conjugate) is measured between the

midpoints of the sacral promontory and upper border of the pubic symphysis, and on average is 10 cm in the adult male and 11.2 cm in the adult female. The transverse diameter is the maximum distance between similar points (assessed by eye) on opposite sides of the pelvic brim, and is on average 12.5 cm in the adult male and 13.1 cm in the adult female. The oblique diameter is measured from the iliopubic ramus to the opposite sacroiliac joint, and is on average 12 cm in the adult male and 12.5 cm in the adult female. These measurements vary with the individual and with racial group. In children, dimensions of the thorax and spine are significantly correlated with the width of the pelvic inlet, which are age-independent predictors of paediatric chest width and may be useful in assessing growth of the thorax and spine in children with early-onset spinal deformity (Emans et al 2005).

Articulated bony pelvis

The lesser pelvis encloses a true basin when soft tissues of the pelvic floor are in place. Skeletally, it is a narrower continuation of the greater pelvis, with irregular but more complete walls around its cavity. Of obstetric importance, it has a curved median axis, and superior and inferior openings. The superior opening is occupied by viscera. The pelvic floor, viscera and subjacent perineal sphincters close the inferior opening.

Cavity of the lesser pelvis

The cavity of the lesser pelvis is short, curved, and markedly longer in its posterior wall. Anteroinferiorly, it is bounded by pubic bones, their rami and symphysis. Posteriorly, it is bounded by the concave anterior sacral surface and coccyx. Laterally on each side, its margins are the smooth quadrangular pelvic aspect of the fused ilium and ischium. The region so enclosed is the pelvic cavity proper, through which pass

the rectum, urinary bladder and parts of the reproductive organs. The cavity in females must also permit passage of the fetus.

The pelvic cavity diameters are measured at approximately the mid-level. The anteroposterior diameter is measured between the midpoints of the third sacral segment and posterior surface of the pubic symphysis, and is about 10.5 cm in the adult male and 13 cm in the adult female. The transverse diameter is the widest transverse distance between the side walls of the cavity, and often the greatest transverse dimension in the whole cavity. It measures about 12 cm in the adult male and 12.5 cm in the adult female. The oblique diameter is the distance from the lowest point of one sacroiliac joint to the midpoint of the contralateral obturator membrane, and measures about 11 cm in the adult male and 13.1 cm in the adult female. All measurements vary with the individual and with racial group.

Pelvic outlet (inferior pelvic aperture)

Less regular in outline than the pelvic inlet, the pelvic outlet is indented behind by the coccyx and sacrum, and bilaterally by the ischial tuberosities. Its perimeter thus consists of three wide arcs. Anteriorly is the pubic arch, between the converging ischiopubic rami. Posteriorly and laterally on both sides are the sciatic notches between the sacrum and ischial tuberosities. The sciatic notches are divided by the sacrotuberous and sacrospinous ligaments into greater and lesser sciatic foramina (Fig. 80.9).

With ligaments included, the pelvic outlet is rhomboidal. Its anterior limbs are the ischiopubic rami (joined by the inferior pubic ligament) and its posterior margins are the sacrotuberous ligaments, with the coccyx in the midline. The outlet is thus not rigid in its posterior half, being limited by ligaments and the coccyx, all slightly yielding. Even with the sacrum taken as the posterior midline limit (more reliable for

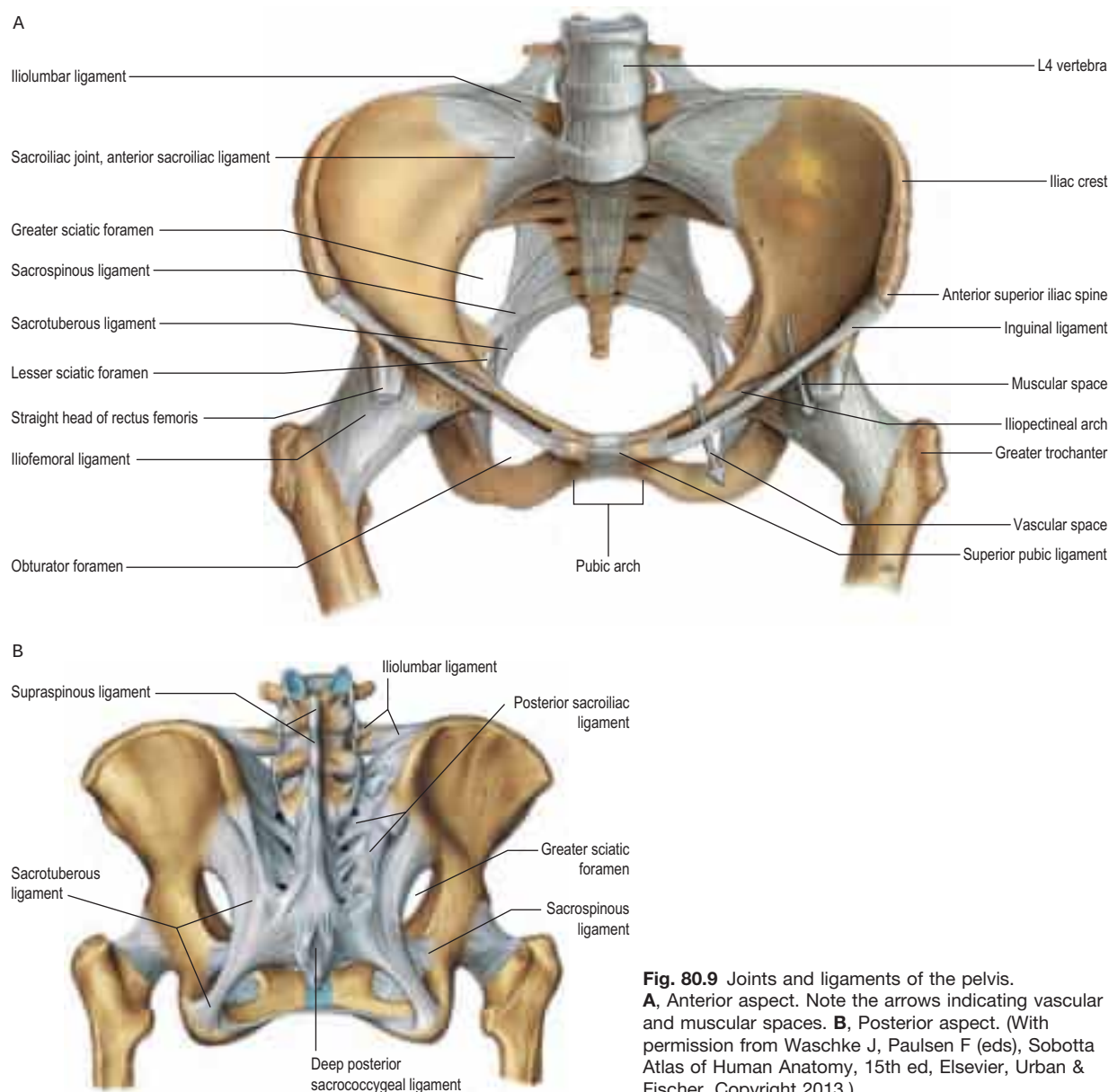


Fig. 80.9 Joints and ligaments of the pelvis. **A**, Anterior aspect. Note the arrows indicating vascular and muscular spaces. **B**, Posterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

measurement), there may be slight mobility at the sacroiliac joints. Note also that a plane of the pelvic outlet is merely conceptual. The anterior, ischiopubic part has a plane that is inclined down and back to a transverse line between the lower limits of the ischial tuberosities, and the posterior half has a plane approximating to the sacrotuberous ligaments, sloping down and forwards to the same line.

Three measurements are made for the pelvic outlet. The anteroposterior diameter is usually measured from the apex of the coccyx to the midpoint of the lower rim of the pubic symphysis. The lowest sacral point may also be used (on average: male 8 cm, female 12.5 cm). The transverse (bituberous) diameter is measured between the ischial tuberosities at the lower borders of their medial surfaces (on average: male 8.5 cm, female 11.8 cm). The oblique diameter extends from the midpoint of the sacrotuberous ligament on one side to the contralateral ischiopubic junction (on average: male 10 cm, female 11.8 cm). All measurements vary among individuals and racial groups.

Other measurements

Apart from these main measurements, by consensus the basis of pelvic osteometry, other planes and measurements are used in obstetric practice. The plane of greatest pelvic dimensions is an obstetric concept. It represents the most capacious pelvic level, between the pelvic brim and midlevel plane, and corresponds with the latter anteriorly at the middle part of the pubic symphysis and posteriorly at the level of the second and third sacral segments.

The plane of least dimensions is said to be at about mid-pelvic level. Its transverse diameter is between the apices of the ischial spines. This measurement is about 9.5 cm in an adult female and is just wide enough to allow passage of the biparietal diameter of a fetal head (about 9 cm). Not surprisingly, most difficulty in parturition occurs here.

The above measurements are sometimes made in clinical practice using radiographs or magnetic resonance imaging (MRI) pelvimetry. Precise measurement is not possible without radiological techniques, and even these do not take into account the adjacent soft tissues. In the past, measurements were made at physical (e.g. vaginal) examinations. However, these manual measurements have proved to be of little clinical value and are now more or less obsolete.

Morphological classification of pelves



Available with the *Gray's Anatomy* e-book

Pelvic axes and inclination

The axis of the superior pelvic aperture traverses its centre at right angles to its plane, directed down and backwards (Fig. 80.10). When prolonged (projected), it passes through the umbilicus and mid-coccyx. An axis is similarly established for the inferior aperture: projected upwards, it impinges on the sacral promontory. Axes can likewise be constructed for any plane, and one for the whole cavity is a concatenation of an infinite series of such lines (see Fig. 80.10). The fetal head, however,

descends in the axis of the inlet as far as the level of the ischial spines; it is then directed forwards into the axis of the vagina at right angles to that axis. The form of this pelvic axis and the disparity in depth between the anterior and posterior contours of the cavity are prime factors in the mechanism of fetal transit in the pelvic canal.

In the standing position, the pelvic canal curves obliquely backwards relative to the trunk and abdominal cavity. The whole pelvis is tilted forwards, the plane of the pelvic brim making an angle of 50–60° with the horizontal. The plane of the pelvic outlet is tilted to about 15°. Strictly, the pelvic outlet has two planes: an anterior passing backwards from the pubic symphysis and a posterior passing forwards from the coccyx, both descending to meet at the intertuberous line. In standing, the pelvic aspect of the pubic symphysis faces nearly as much upwards as backwards and the sacral concavity is directed anteroinferiorly. The front of the pubic symphysis and anterior superior iliac spines are in the same vertical plane. While sitting, body weight is transmitted through the inferomedial parts of the ischial tuberosities, with variable soft tissues intervening. The anterior superior iliac spines are in a vertical plane through the acetabular centres, and the whole pelvis is tilted back with the lumbosacral angle somewhat diminished at the sacral promontory.

Pelvic mechanism

The skeletal pelvis supports and protects the contained viscera but is primarily part of the lower limbs, affording wide attachment for muscles of the thigh, leg and trunk. It constitutes the major mechanism for transmitting the weight of the head, trunk and upper limbs to the lower limbs. It may be considered as two arches divided by a coronal trans-acetabular plane. The posterior arch, chiefly concerned in transmitting weight, consists of the upper three sacral vertebrae and strong pillars of bone from the sacroiliac joints to the acetabular fossae. The anterior arch, formed by the pubic bones and their superior rami, connects these lateral pillars as a tie beam to prevent separation; it also acts as a compression strut against medial femoral thrust. The sacrum, as the summit of the posterior arch, is loaded at the lumbosacral joint. Theoretically, this force has two components: one thrusting the sacrum downwards and backwards between the iliac bones, the other thrusting its upper end downwards and forwards. Sacral movements are regulated by osseous shape and massive ligaments. The first component therefore acts against the wedge, its tendency to separate iliac bones resisted by the sacroiliac and iliolumbar ligaments and pubic symphysis.

Vertical coronal sections through the sacroiliac joints suggest division of the (synovial) articular region of the sacrum into three segments. In the anterosuperior segment, involving the first sacral vertebra, the articular surfaces are slightly sinuous and almost parallel. In the middle segment, the posterior width between the articular markings is greater than the anterior, and centrally a sacral concavity fits a corresponding iliac convexity, an interlocking mechanism relieving the strain on the ligaments produced by body weight. In the posteroinferior segment, the anterior sacral width is greater than the posterior and here its sacral surfaces are slightly concave. Anteroinferior sacral dislocation by the second component (of force) is prevented, therefore, mainly by the middle segment, owing to its cuneiform shape and interlocking mechanism. However, some rotation occurs, in which the anterosuperior segment tilts down and the posteroinferior segment up. 'Superior' segmental movement is limited to a small degree by wedging but primarily by tension in the sacrotuberous and sacrospinous ligaments. In all movements, the sacroiliac and iliolumbar ligaments and pubic symphysis resist separation of the iliac bones.

SEXUAL DIFFERENCES IN THE PELVIS

The pelvis provides the most marked skeletal differences between male and female. Distinction can be made even during fetal life, particularly in the subpubic arch. In infancy, dimensions of the whole pelvis are greater in males than in females, but the size of the pelvic cavity is usually greater in females. This distinction prevails in childhood but the difference is maximal at about 22 months. Sexual differences in adults are divisible into metrical and non-metrical features; the range of most features overlaps between the sexes.

Differences are inevitably linked to function. While the primary pelvic function in both sexes is locomotor, the pelvis, particularly the lesser pelvis, is adapted to parturition in females, and these changes variably affect the proportions and dimensions of the greater pelvis. Since males are distinctly more muscular and therefore more heavily built, overall pelvic dimensions, such as the intercrural distance (distance between the iliac crests), are greater, markings for muscles and ligaments more pronounced, and general architecture more robust. The

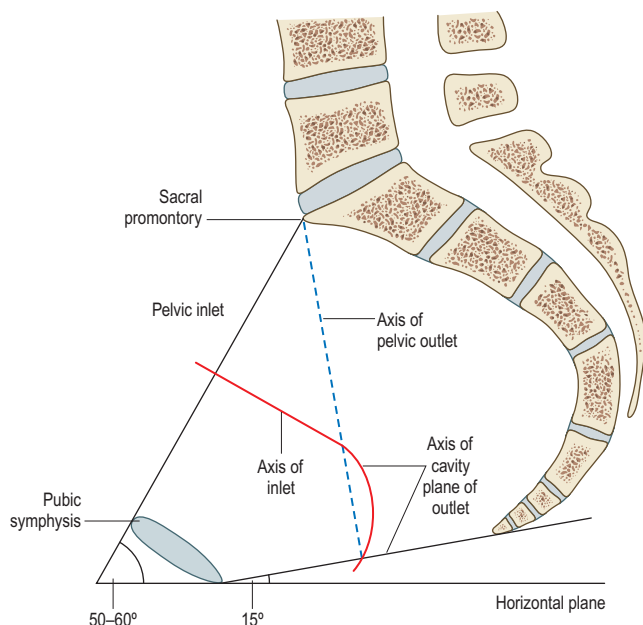


Fig. 80.10 A median sagittal section through the female pelvis, showing the planes of the inlet and outlet and the axis of the pelvic cavity.

Interest in the dimensions described above is primarily obstetric and, less frequently, forensic. All pelvic measurements display individual variation and the values quoted are means from limited surveys. These measurements have been analysed by many anatomists, anthropologists, obstetricians and radiologists in attempts to classify human pelves, especially female. The four most common terms used today are gynaecoid, anthropoid, platypelloid and android. The gynaecoid pelvis is the traditional Western female pelvis with a heart-shaped brim and the range of measurements quoted above. An anthropoid pelvis has a larger mid-cavity and a wide anteroposterior inlet, which is oval in shape; it is more common in women of African origin and may be associated with a 'high-assimilation' pelvis where there is an additional lumbar vertebra. A platypelloid pelvis is flat and oval from side to side at the brim; it is a contracted pelvis that is rarely seen nowadays, having previously been associated with rickets. An android pelvis has a triangular brim and is the shape of a male pelvis.

male iliac crest is more rugged and more medially inclined at its anterior end; in females the iliac crests are less curved in all parts. The iliac alae are more vertical in females but do not ascend so far; the iliac fossae are therefore shallower. These iliac peculiarities probably account for the greater prominence of female hips.

The male is relatively and absolutely more heavily built above the pelvis, with consequent differences at the lumbosacral and hip joints. The sacral basal articular facet for the fifth lumbar vertebra and intervening intervertebral disc is more than one-third of the total sacral basal width in males but less than one-third in females, in whom the sacrum is also relatively broader, accentuating this difference. The female has relatively broader sacral alae. The male acetabulum is absolutely larger, and its diameter is approximately equal to the distance between its anterior rim and pubic symphysis. In females, acetabular diameter is usually less than this distance, not only because it is absolutely smaller but also because the anterolateral wall of the cavity is comparatively and often absolutely wider. The height of the female pubic symphysis and adjoining parts of the pubis and ischium, which form the anterior pelvic wall, are also absolutely less, producing a somewhat triangular obturator foramen, which is more ovoid in males. Differing pubic growth is also expressed in the pubic arch below the pubic symphysis and between the inferior pubic rami. It is more angular in males, being 50–60°; in females it is rounded, less easy to measure and usually 80–85°. A greater separation of the pubic tubercles in females contributes to the pubic width. The ischiopubic rami are also much more lightly built and narrowed near the symphysis; in males they bear a distinctly rough, everted area for attachment of the penile crura, the corresponding attachment for the clitoris being poorly developed. The ischial spines are closer in males and are more turned. The greater sciatic notch is usually wider in females: mean values for males and females are 50.4° and 74.4°, respectively. The greater female values for angle and width are associated with increased backward sacral tilt and greater anteroposterior pelvic diameter, especially at lower levels.

The sacrum also displays metrical sexual differences. Female sacra are less curved, the curvature being most marked between the first and second segments and the third and fifth, with an intervening flatter region. Male sacra are more evenly curved, and relatively long and narrow, and more often exceed five segments (by addition of a lumbar or coccygeal vertebra). The sacral index compares sacral breadth (between the most anterior points on the auricular surfaces) with length (between midpoints on the anterior margins of the promontory and apex): average values for males and females are 105% and 115%, respectively. Auricular surfaces are relatively smaller and more oblique in females, but extend on to the upper three sacral vertebrae in both sexes. The dorsal auricular border is more concave in females. Many differences may be summarized in the generalization that the pelvic cavity is longer and more conical in males, and shorter and more cylindrical in females; the axis is curved in both. Differences are greater at the inferior aperture than at the brim, where in absolute measurements males are not as different from females as sometimes stated.

In forensic practice, identification of human skeletal remains (which are sometimes fragmentary) usually involves determination of sex, and this is most reliably established from an examination of the pelvis. Even fragments of the pelvis may be useful in this respect. Several studies of metrical characteristics in various pelvic regions have been made, leading to the establishment of various indices. The ilium has received particular attention, e.g. one index compares the pelvic and sacroiliac parts of the bone. A line is extended back from the iliopubic ramus to the nearest point on the anterior auricular margin and thence to the iliac crest. The auricular point divides this chilotic line into anterior (pelvic) and posterior (sacral) segments, each expressed as a percentage of the other. Chilotic indices display reciprocal values in the sexes: the pelvic part of the chilotic line is predominant in females, and the sacral part in males. Detailed metrical studies of the ilium have indicated its limited reliability in 'sexing' pelvises. However, the higher incidence and definition of the female pre-auricular sulcus is recognized. The desirability of correlating all available metrical data is to be emphasized; when a range of pelvic data can be combined, especially if they are metrical, 95% accuracy should be achieved. Complete accuracy has been claimed when the rest of the skeleton is available. Assessment of sex from isolated and often incomplete human remains is less reliable. For further details, consult [Mays \(1998\)](#) and [Brothwell and Pollard \(2001\)](#).

FEMUR

The femur is the longest and strongest bone in the human body ([Figs 80.11–80.12](#)). Its length is associated with a striding gait, and its

strength with the weight and muscular forces it is required to withstand. Its shaft, almost cylindrical along most of its length, is bowed forwards. It has a proximal rounded, articular head projecting medially from its short neck, which, in turn, is a medial extension of the proximal shaft. The distal extremity is wider and more substantial, and presents a double condyle that articulates with the tibia. In standing, the femoral shafts show an inclination upwards and outwards from their tibial articulations, with the femoral heads being separated by the pelvic width. Since the tibia and fibula descend vertically from the knees, the ankles are also in the line of body weight in standing or walking. The degree of femoral obliquity varies between individuals but is generally greater in women, reflecting the relatively greater pelvic breadth and shorter femora. Proximally, the femur consists of a head, neck and greater and lesser trochanters.

Femoral head

The femoral head faces anterosuperomedially to articulate with the acetabulum ([Fig. 80.13](#)). The head, often described as rather more than half a 'sphere', is not part of a true sphere but is spheroidal. Its smoothness is interrupted posteroinferiorly to its centre by a small, rough fovea. The head is intracapsular and is encircled, distal to its equator, by the acetabular labrum. Its articular margin is distinct, except anteriorly, where the articular surface extends on to the neck. The ligamentum teres is attached to the fovea. The anterior surface of the head is separated inferomedially from the femoral artery by the tendon of *psaos major*, the iliopectineal bursa and the articular capsule.

Femoral neck

The femoral neck (see [Fig. 80.13](#)) is approximately 5 cm long, narrowest in its mid part and widest laterally, and connects the head to the shaft at an average angle of 127° ([Gilligan et al 2013](#)) (angle of inclination; neck–shaft angle): this facilitates movement at the hip joint, enabling the limb to swing clear of the pelvis. The neck also provides a lever for the action of the muscles acting about the hip joint, which are attached to the proximal femur. The neck–shaft angle is widest at birth and diminishes gradually until the age of 10 years ([Birkenmaier et al 2010](#)); it is smaller in females. The neck is laterally rotated with respect to the shaft (angle of anteversion) some 10–15°, although values of this angle vary between individuals and between populations ([Eckhoff et al 1994](#)). The contours of the neck are rounded; the upper surface is almost horizontal and slightly concave, while the lower is straighter but oblique, directed inferolaterally and backwards to the shaft near the lesser trochanter. On all aspects the neck expands as it approaches the articular surface of the head. The anterior surface of the neck is flat and marked at the junction with the shaft by a rough intertrochanteric line. The posterior surface, facing posteriorly and superiorly, is transversely convex, and concave in its long axis; its junction with the shaft is marked by a rounded intertrochanteric crest. There are numerous vascular foramina, especially anteriorly and posterosuperiorly.

The anterior surface is intracapsular, the capsule attaching laterally to the intertrochanteric line. Facets, often covered by extensions of articular cartilage, and various imprints frequently occur here. These facets may sometimes be associated with squatting. On the posterior surface the capsule does not reach the intertrochanteric crest; little more than the medial half of the neck is intracapsular. The anterior surface adjoining the head and covered by cartilage is related to the iliofemoral ligament. A groove, produced by the tendon of *obturator externus* as it approaches the trochanteric fossa, spirals across the posterior surface of the neck of the femur in a proximolateral direction.

Greater trochanter

The greater trochanter is large and quadrangular, projecting up from the junction of the neck and shaft (see [Fig. 80.13](#)). Its posterosuperior region projects superomedially to overhang the adjacent posterior surface of the neck and here its medial surface presents the rough trochanteric fossa. The proximal border of the trochanter lies approximately a hand's breadth below the iliac tuberculum, level with the centre of the femoral head. It has an anterior rough impression. Its lateral surface, continuous distally with the lateral surface of the femoral shaft, is crossed anteroinferiorly by an oblique, flat strip, which is wider above. This surface is palpable, especially when the muscles are relaxed. The trochanteric fossa occasionally presents a tubercle or exostosis.

Lesser trochanter

The lesser trochanter is a conical posteromedial projection of the shaft at the posteroinferior aspect of its junction with the neck. Its summit and anterior surface are rough, but its posterior surface, at the distal end of the intertrochanteric crest, is smooth. It is not palpable.

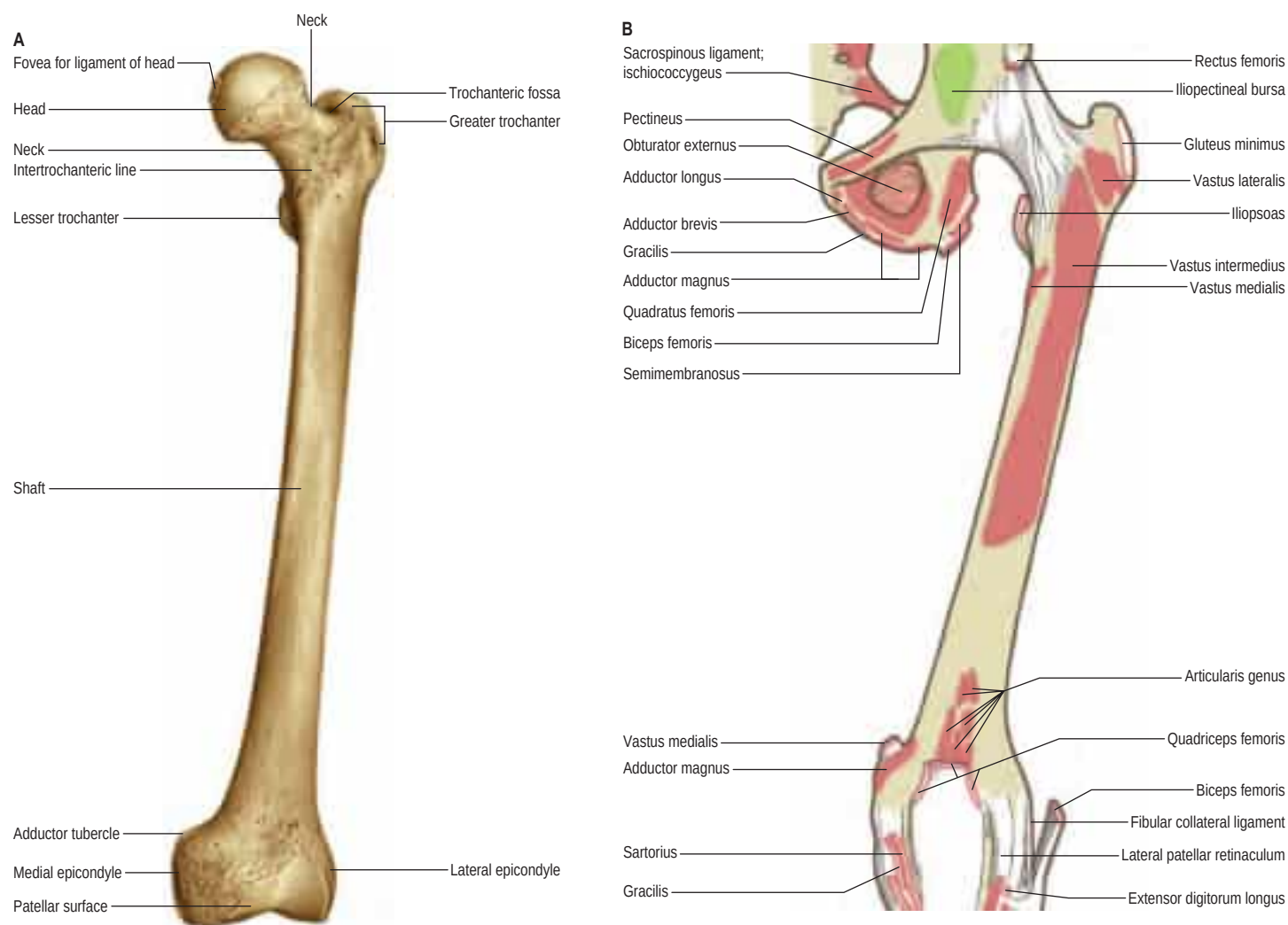


Fig. 80.11 The femur, anterior aspect. **A**, Osseous features. **B**, Muscle attachments. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Intertrochanteric line

The intertrochanteric line, a prominent ridge at the junction of the anterior surfaces of the neck and shaft, descends medially from a tubercle on the upper part of the anterior aspect of the greater trochanter to a point on the lower border of the neck, anterior to the lesser trochanter, where there may also be a tubercle. This line is the lateral limit of the hip joint capsule anteriorly. The upper and lower bands of the iliofemoral ligament are attached to its proximal and distal ends and the associated tubercles. Distally, it is continuous with the spiral line.

Intertrochanteric crest

The intertrochanteric crest, a smooth and prominent ridge at the junction of the posterior surface of the neck with the shaft, descends medially from the posterosuperior angle of the greater trochanter to the lesser trochanter. A little above its centre is a low, rounded quadrate tubercle. It is covered by gluteus maximus, from which it is separated, medial to the tubercle, by quadratus femoris and the upper border of adductor magnus.

Gluteal tuberosity

The gluteal tuberosity may be an elongated depression or a ridge. It may at times be prominent enough to merit the title of a third trochanter.

Shaft

The shaft is surrounded by muscles and is impalpable (see Figs 80.11–80.12). The distal anterior surface, for 5–6 cm above the patellar articular surface, is covered by a suprapatellar bursa, between bone and muscle. The distal lateral surface is covered by vastus intermedius. The medial surface, devoid of attachments, is covered by vastus medialis.

The shaft is narrowest centrally, expanding a little at its proximal end, and substantially more at its distal end. Its long axis makes an angle of approximately 10° with the vertical, and diverges 5–7° from the long axis of the tibia. Its middle one-third has three surfaces and borders. The extensive anterior surface, smooth and gently convex, is between the lateral and medial borders, which are both round and

indistinct. The posterolateral surface is bounded posteriorly by the broad, rough linea aspera, usually a crest with lateral and medial edges. Its subjacent compact bone is augmented to withstand compressive forces, which are concentrated here by the anterior curvature of the shaft. The linea aspera gives attachment to adductor longus, intermuscular septa and the short head of biceps femoris, all inseparably blended at their attachments. Perforating arteries cross the linea aspera laterally under tendinous arches in adductor magnus and biceps femoris. Nutrient foramina, directed proximally, appear in the linea aspera, varying in number and site, one usually near its proximal end, a second usually near its distal end. The posteromedial surface, smooth like the others, is bounded in front by the indistinct medial border and behind by the linea aspera. In its proximal third the shaft has a fourth, posterior surface, bounded medially by a narrow, rough spiral line that is continuous proximally with the intertrochanteric line and distally with the medial edge of linea aspera. Laterally, this surface is limited by the broad, rough gluteal tuberosity, ascending a little laterally to the greater trochanter and descending to the lateral edge of the linea aspera. In its distal one-third, the posterior surface of the shaft presents a further surface, the popliteal surface (see below) between the medial and lateral supracondylar lines. These lines are continuous above with the corresponding edges of the linea aspera. The lateral line is most distinct in its proximal two-thirds, where the short head of biceps femoris and lateral intermuscular septum are attached. Its distal third has a small, rough area for the attachment of plantaris, often encroaching on the popliteal surface. The medial line is indistinct in its proximal two-thirds, where vastus medialis is attached. Distally, the femoral vessels entering the popliteal fossa from the adductor canal cross the medial line obliquely. Further distally, the line is often sharp for 3 or 4 cm proximal to the adductor tubercle.

The popliteal surface, triangular in outline, lies between the medial and lateral supracondylar lines. In its distal medial part, it is rough and slightly elevated. Forming the proximal part of the floor of the popliteal fossa, the popliteal surface is covered by a variable amount of fat that separates the popliteal artery from bone. The superior medial genicular

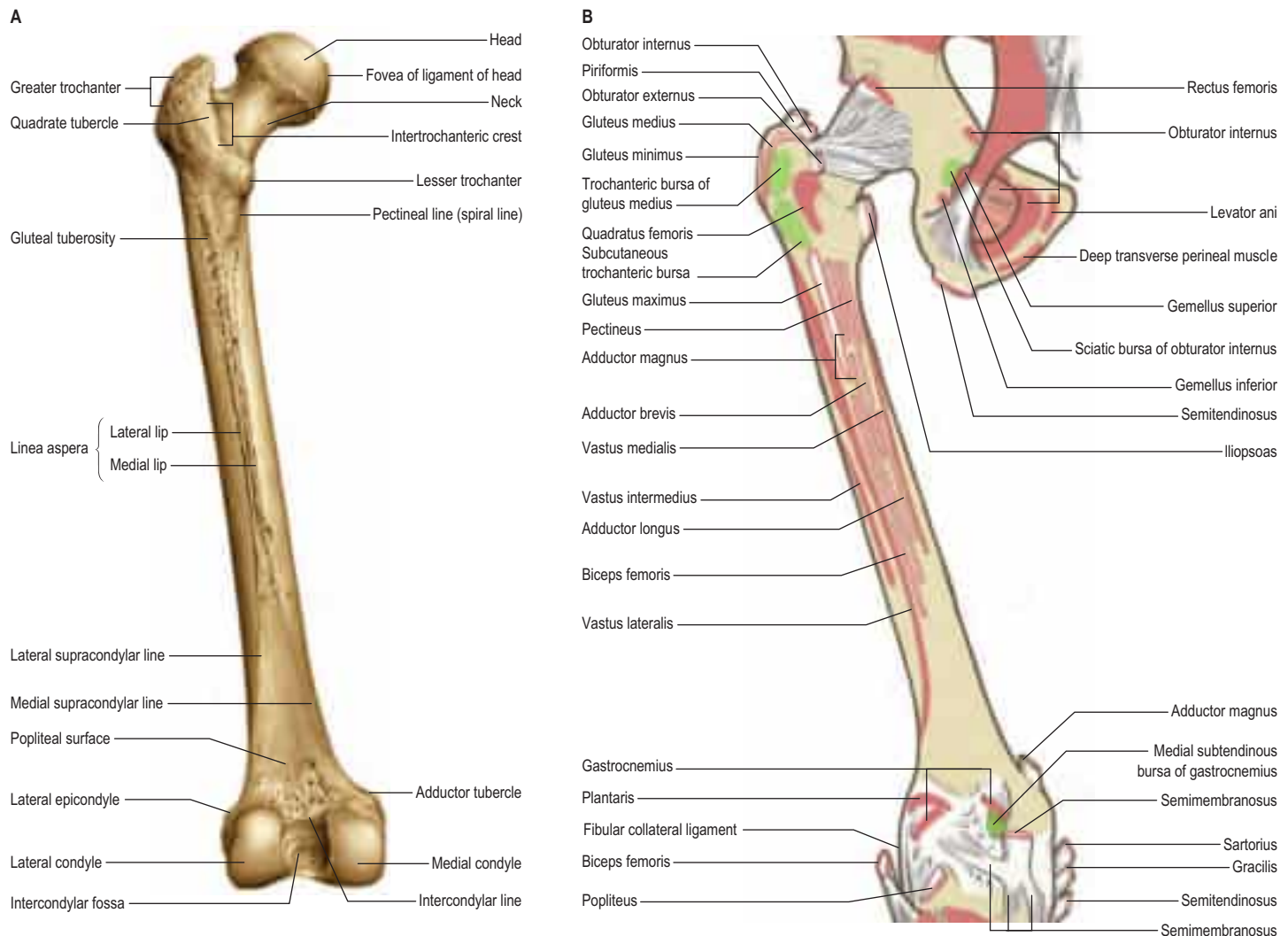


Fig. 80.12 The femur, posterior aspect. **A**, Osseous features. **B**, Muscle attachments. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

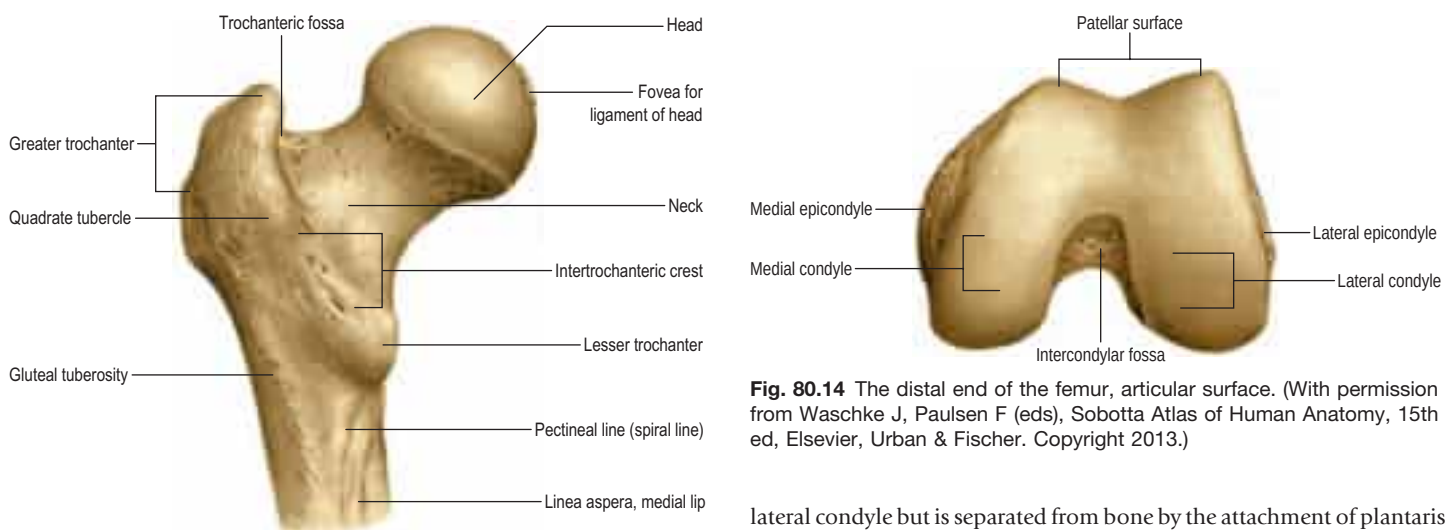


Fig. 80.13 The proximal end of the femur, posterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Fig. 80.14 The distal end of the femur, articular surface. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

lateral condyle but is separated from bone by the attachment of plantaris to the distal part of the lateral supracondylar line.

Distal end

The distal end of the femur is widely expanded as a bearing surface for transmission of weight to the tibia (Fig. 80.14). It bears two massive condyles, which are partly articular. Anteriorly, the condyles are confluent and continue into the shaft; posteriorly, they are separated by a deep intercondylar fossa and project beyond the plane of the popliteal surface. The articular surface is a broad area, like an inverted U, for the patella and the tibia. The patellar surface extends anteriorly on both condyles, especially the lateral. It is transversely concave, vertically convex and grooved for the posterior patellar surface. The tibial surface is divided by the intercondylar fossa but is anteriorly continuous with the patellar surface. Its medial part is a broad strip on the convex

inferoposterior surface of the medial condyle, and is gently curved with a medial convexity. Its lateral part covers similar aspects of the lateral condyle but is broader and passes straight back. The tibial surfaces are convex in all directions. The medial and lateral tibial surfaces have dissimilar anteroposterior curvatures. However, views differ as to the exact representation of these differences. One view holds that in both tibial portions of the femoral condyles, the sagittal radius of curvature is ever-decreasing (a 'closing helix'). More recently, it has been suggested that the medial articular surface describes arcs of two circles. The more posterior has a smaller radius. Laterally, there may only be one arc of fixed curvature with a radius similar to that of the posterior arc of the medial femoral articular surface. These differences are believed to be important determinants of knee joint motion.

Patellar surface

The patellar surface extends more proximally on the lateral side. Its proximal border is therefore oblique and runs distally and medially, separated from the tibial surfaces by two faint grooves that cross the condyles obliquely. The lateral groove is the more distinct. It runs laterally and slightly forwards from the front of the intercondylar fossa and expands to form a faint triangular depression, resting on the anterior edge of the lateral meniscus when the knee is fully extended. The medial groove is restricted to the medial part of the medial condyle and rests on the anterior edge of the medial meniscus in full extension. Where it ceases, the patellar surface continues back to the lateral part of the medial condyle as a semilunar area adjoining the anterior region of the intercondylar fossa. This area articulates with the medial vertical facet of the patella in full flexion; its outline is indistinct in most femora. In habitual squatters articular cartilage may extend to the lateral aspect of the lateral condyle under vastus lateralis.

Intercondylar fossa

The intercondylar fossa separates the two condyles distally and behind. In front, the distal border of the patellar surface limits it, and behind an intercondylar line limits it, separating it from the popliteal surface. It is intracapsular but largely extrasynovial. Its lateral wall, the medial surface of the lateral condyle, bears a flat posterosuperior impression that spreads to the floor of the fossa near the intercondylar line for the proximal attachment of the anterior cruciate ligament. The medial wall of the fossa, i.e. the lateral surface of the medial condyle, bears a similar larger area, but far more anteriorly, for the proximal attachment of the posterior cruciate ligament. Both impressions are smooth and largely devoid of vascular foramina, whereas the rest of the fossa is rough and pitted by vascular foramina. A bursal recess between the ligaments may ascend to the fossa. The capsular ligament and, laterally, the oblique popliteal ligament are attached to the intercondylar line. The infrapatellar synovial fold is attached to the anterior border of the fossa.

Lateral condyle

The lateral condyle (see Fig. 80.14) is larger anteroposteriorly than the medial. Its most prominent point is the lateral epicondyle to which the fibular collateral ligament is attached (Fig. 80.15). A short groove, deeper in front, separates the lateral epicondyle inferiorly from the articular margin. This groove allows the tendon of popliteus to run deep to the fibular collateral ligament and insert inferior and anterior to the ligament insertion. Adjoining the joint margin is a strip of condyle, 1 cm broad. It is intracapsular and covered by synovial membrane except for the attachment of popliteus.

The medial surface is the lateral wall of the intercondylar fossa. Its lateral surface projects beyond the shaft. Part of the lateral head of gastrocnemius is attached to an impression posterosuperior to the lateral epicondyle.

Medial condyle

The medial condyle has a bulging, convex medial aspect, which is easily palpable. Proximally, its adductor tubercle, which may only be a facet rather than a projection, receives the tendon of adductor magnus. The medial prominence of the condyle, the medial epicondyle, is antero-inferior to the tubercle. The lateral surface of the condyle is the medial wall of the intercondylar fossa. The condyle projects distally so that, despite the obliquity of the shaft, the profile of the distal end is almost horizontal. A curved strip, 1 cm wide and adjoining the medial articular margin, is covered by synovial membrane and is inside the joint capsule. Proximal to this, the medial epicondyle receives the tibial collateral ligament.

Structure

The femoral shaft is a cylinder of compact bone with a large medullary cavity. The wall is thick in its middle third, where the femur is narrowest



Fig. 80.15 The distal end of the femur, lateral aspect. Key: 1, attachment of plantaris; 2, attachment of lateral head of gastrocnemius; 3, lateral epicondyle; 4, attachment of fibular collateral ligament; 5, groove for popliteus in full flexion; 6, attachment of popliteus.

and the medullary cavity most capacious. Proximally and distally, the compact wall becomes progressively thinner, and the cavity gradually fills with trabecular bone. The extremities, especially where articular, consist of trabecular bone within a thin shell of compact bone, their trabeculae being disposed along lines of greatest stress. At the proximal end, the main trabeculae form a series of plates orthogonal to the articular surface, converging to a central dense wedge, which is supported by strong trabeculae passing to the sides of the neck, especially along its upper and lower profiles (Fig. 80.16). Force applied to the femoral head is therefore transmitted to the wedge and from there to the junction of the neck and shaft. This junction is strengthened by dense trabeculae extending laterally from the lesser trochanter to the end of the superior aspect of the neck, thus resisting tensile or shearing forces applied to the neck through the head (Fig. 80.17). Tensile and compressive tests indicate that axial trabeculae of the femoral head withstand much greater stresses than peripheral trabeculae. A smaller bar across the junction of the greater trochanter with the neck and shaft resists shearing produced by muscles attached to it. These two bars are proximal layers of arches between the sides of the shaft and transmit to it forces applied to the proximal end. A thin vertical plate, the so-called calcar femorale, ascends from the compact wall near the linea aspera into the trabeculae of the neck (see Fig. 80.17). Medially, it joins the posterior wall of the neck; laterally, it continues into the greater trochanter, where it disperses into general trabecular bone. It is thus in a plane anterior to the trochanteric crest and base of the lesser trochanter. Dixon (1910) demonstrated that the trabecular framework of the proximal femur was spiral, and that the 'arches' were simplified sectional profiles of this spiral. At the distal end of the femur, trabeculae spring from the entire internal surface of compact bone, descending perpendicular to the articular surface. Proximal to the condyles these are strongest and most accurately perpendicular. Horizontal planes of trabecular bone, arranged like crossed girders, form a series of cubical compartments.

Muscle attachments

The greater trochanter provides attachment for gluteus minimus and medius. Gluteus minimus is attached to its rough anterior impression and gluteus medius to its lateral oblique strip. The bone is separated from the tendon of gluteus medius by a bursa. The area behind it is covered by deep fibres of gluteus maximus, with part of its trochanteric bursa interposed. The tendon of piriformis is attached to the upper border of the trochanter and the common (tricipital) tendon of obturator internus and the gemelli is attached to its medial surface. The trochanteric fossa receives the tendon of obturator externus. Psoas major is attached to the summit and anteromedial surface of the lesser trochanter. Iliacus is attached to the medial or anterior surface of its base, descending a little behind the spiral line as its tendon fuses with that of psoas major. Adductor magnus (upper part) passes over its posterior surface, sometimes separated by an interposed bursa.



Fig. 80.16 A coronal section through the left hip joint. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

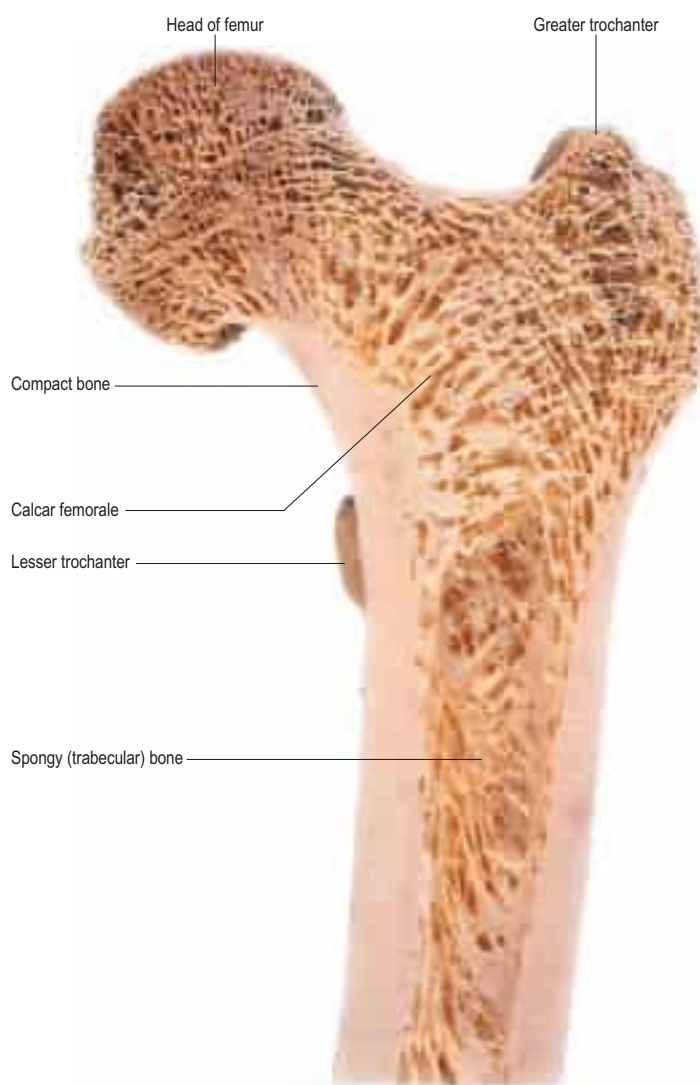


Fig. 80.17 A coronal section through the proximal end of the femur showing the trabecular architecture, calcar femorale and variations in cortical thickness. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

The most proximal fibres of vastus lateralis are attached to the proximal end of the intertrochanteric line, and those of vastus medialis to the distal end. Quadratus femoris is attached to the quadratus tubercle and the immediately distal bone. Vastus intermedius is attached to the anterior and lateral surfaces of the proximal three-quarters of the femoral shaft. Slips of articularis genu are attached distal to this.

The gluteal tuberosity receives the deeper fibres of the distal half of gluteus maximus and, at its medial edge, the uppermost fibres of adduc-

tor magnus. Distal to this, adductor magnus is attached to the linea aspera and, by an aponeurosis, to the proximal part of the medial supracondylar ridge. Its remaining fibres form a large tendon attached to the adductor tubercle, with an aponeurotic expansion to the distal part of the medial supracondylar ridge.

Pectineus and adductor brevis are attached to the posterior femoral surface between the gluteal tuberosity and spiral line. The pectineal attachment is a line, sometimes slightly rough, from the base of the lesser trochanter to the linea aspera. Adductor brevis is attached lateral to pectineus and beyond this to the proximal half of the linea aspera, medial to adductor magnus. Adductor longus, intermuscular septa and the short head of biceps femoris are attached to the linea aspera. Vastus lateralis has a linear attachment from the anterior surface of the base of the greater trochanter to the proximal end of the gluteal tuberosity, and along the lateral margin of the latter to the proximal half of the lateral edge of the linea aspera. Vastus medialis is attached from the distal end of the intertrochanteric line along the spiral line to the medial edge of the linea aspera and thence to the medial supracondylar line, which also receives many fibres from the aponeurotic attachments of adductor magnus.

The medial head of gastrocnemius is attached to the posterior surface a little above the medial condyle. The short head of biceps femoris is attached to the proximal two-thirds of the lateral supracondylar line. Plantaris attaches to the line distally. Vastus medialis is attached to the proximal two-thirds of the medial supracondylar line.

Part of the lateral head of gastrocnemius is attached posterosuperiorly to the lateral epicondyle. Popliteus is attached anteriorly in the groove on the outer aspect of the lateral epicondyle. Its tendon passes deep to the fibular collateral ligament (see Fig. 80.15). The tendon lies in the groove in full knee flexion; in extension it crosses the articular margin and may form an impression on it.

Vascular supply

The blood supply of the femoral head is derived from an arterial ring around the neck, just outside the attachment of the fibrous capsule, constituted by the medial and lateral circumflex femoral arteries with minor contributions from the superior and inferior gluteal vessels (see 'Trochanteric anastomosis') (Fig. 80.18). From this ring, ascending cervical branches pierce the capsule (under its zona orbicularis) to ascend the neck beneath the reflected synovial membrane. These vessels become the retinacular arteries and form a subsynovial intracapsular anastomosis. Here the vessels are at risk with a displaced fracture of the femoral neck. Interruption of blood supply in this way can lead to avascular necrosis of the femoral head. If the fracture is intracapsular, not only is the intraosseous blood supply damaged but the retinacular vessels are also vulnerable. If the fracture is extracapsular, the retinacular vessels will remain intact and avascular necrosis of the femoral head is much less likely. The ascending cervical vessels give off metaphyseal branches that enter the neck, while the intracapsular ring gives off lateral and inferior epiphyseal branches. A small medial epiphyseal supply, of importance in early childhood, reaches the head along the ligament of the head of femur by the acetabular branches of the obturator and medial circumflex femoral arteries, which anastomose with the other epiphyseal vessels. During growth, the epiphyseal plate separates the territories of the metaphyseal and epiphyseal vessels; these vessels anastomose freely after osseous union of the head and neck.

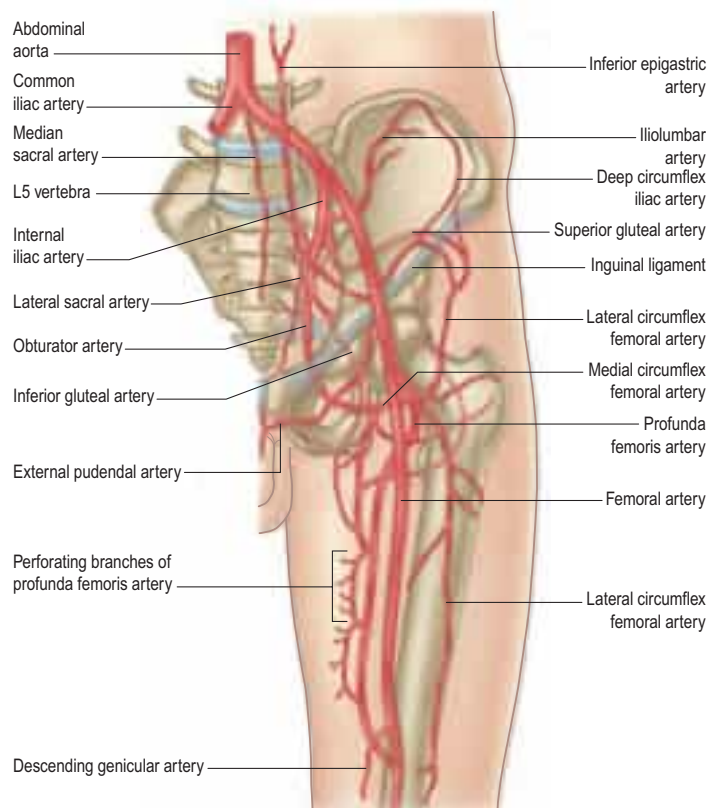


Fig. 80.18 The collateral circulation around the hip and upper thigh, anterior aspect.

Observations of developmental patterns of this supply in late fetal and early postnatal periods have revealed that although medial and lateral circumflex femoral arteries at first contribute equally, two major branches of the medial provide the final supply, both posterior to the neck. The supply from the lateral circumflex artery diminishes and the arterial ring is interrupted. As the femoral neck elongates, the extra-capsular circle becomes more distant from the epiphysal part of the head.

The trochanteric regions and subtrochanteric shaft are supplied by the trochanteric and cruciate arterial anastomoses. More distally in the shaft, nutrient foramina, directed proximally, are found in the linea aspera, varying in number and site: one is usually near its proximal end and a second usually near its distal end. The main nutrient artery is usually derived from the second perforating artery (see '[Profunda femoris artery](#)'). If two nutrient arteries occur, they may branch from the first and third perforators. Periosteal vessels arise from the perforators and from the profunda femoris artery, and run circumferentially rather than longitudinally. The distal metaphysis has many vascular foramina. Arterial supply here is from the genicular anastomosis. For further details, consult [Crook \(1980, 1996\)](#).

In general, the pattern of venous drainage of the head and neck corresponds to that of the arteries, though there may be a single large cervical vein posteroinferiorly.

Innervation

The periosteal innervation is derived proximally from nerves that supply the hip joint, distally from those supplying the knee, and in all areas from nerves that innervate muscles attached to the bone.

Ossification

The femur ossifies from five centres: in the shaft, head, greater and lesser trochanters and the distal end ([Fig. 80.19](#)). Other than the clavicle, it is the first long bone to ossify. The process starts in the mid-shaft in the seventh prenatal week and extends to produce a miniature shaft that is largely ossified at birth. Secondary centres appear in the distal end (from which the condyles and epicondyles are formed) during the ninth month, in the head during the first six months after birth ([Fig. 80.20](#)), in the greater trochanter during the fourth year and in the lesser trochanter between the twelfth and fourteenth years. In the infant, the femoral head is entirely cartilaginous; it cannot be seen on plain radiographs and is best visualized using ultrasound ([Fig. 80.21](#)). The centre in the cartilaginous head is restricted to it until the tenth year, so that the epiphysal line (see [Fig. 80.7](#)) is horizontal and the

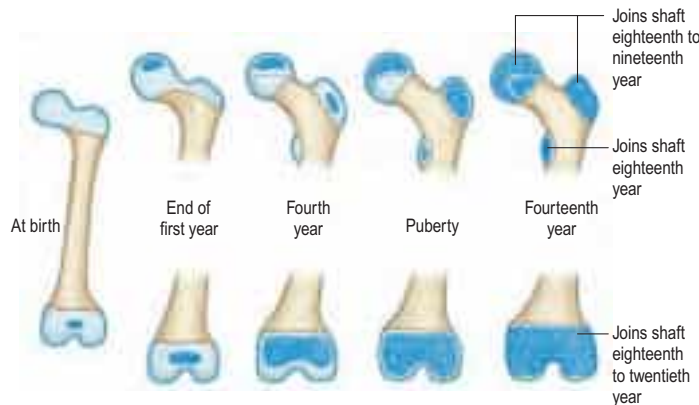


Fig. 80.19 Stages in ossification of the femur (not to scale).

inferomedial part of the articular surface is on the neck. The medial epiphysal margin later grows over this part of the articular surface. Thus, the mature epiphysis is a hollow cup on the summit of the neck. The epiphysal line follows the articular margin except where it is separated superiorly from the articular surface by a non-articular area where blood vessels enter the head ([Trueta 1957](#)). The epiphyses fuse independently: the lesser trochanter soon after puberty, followed by the greater trochanter. The capital epiphysis fuses in the fourteenth year in females and seventeenth year in males. The epiphysis at the distal end fuses in the sixteenth year in females, and eighteenth year in males. The distal epiphysal plate traverses the adductor tubercle.

Growth plate considerations

Trauma to any epiphysal plate can lead to bony union between epiphysis and metaphysis, and so cause premature cessation of growth. Any surgery in the hip region in children can injure the growth plate, resulting in abnormal proximal femoral development. In the case of fractures involving the epiphysis, expeditious restoration of normal bony alignment is essential in order to minimize the risk of subsequent abnormal growth.

The growth plate represents a line of weakness and predisposes to fracture from injury. Such acute injuries affecting the capital epiphysis are uncommon. However, a more chronic fracture through the capital epiphysis occurs in slipped capital femoral epiphysis (SCFE) (see [Fig. 81.9](#)). The condition affects pubescent adolescents, especially males. Endocrinological abnormality may be related. The femoral head epiphysis displaces posteriorly off the femoral neck. If it heals in this position, lower limb deformity and restricted hip movement occur. A classic hallmark is obligatory lateral rotation of the femur as the hip is flexed. Treatment varies according to the time taken for the 'slip' to occur. Normal anatomical restoration is not attempted because of an increased incidence of avascular necrosis. The position of the femoral head may be accepted as it is and fixed with screws in this position to stop further displacement. This treatment will deliberately cause premature growth plate fusion and so prevent future 'slippage'. Since the distal femoral growth plate accounts for most of the normal increase in longitudinal growth of the femoral shaft, an acceptable limb length difference usually results.

Infection of bone in neonates and young children tends to arise via bacteria in the blood stream that usually 'seed' in the metaphysal region, probably as a consequence of the vascular 'arcade' arrangement of arteries in this part of the bone. The proximal femoral growth plate is intra-articular. As a result, infection in the proximal femoral metaphysis can spread into the joint and result in a septic arthritis that can destroy the hip joint permanently.

The distal end of the femur is the only epiphysis in which ossification consistently starts just before birth; the phenomenon therefore serves as a reliable indicator of the gestational maturity of a stillborn baby. Since the epiphysal plate is level with the adductor tubercle, the epiphysis is partly extra-articular. Operations here may damage the distal epiphysal cartilage in children and result in subsequent shortening of the leg.

JOINTS

PUBIC SYMPHYSIS

The pubic bones meet in the midline at the pubic symphysis, a secondary cartilaginous joint (see [Fig. 80.4](#)).



Fig. 80.20 A radiograph of the pelvis of a 9-month-old infant showing the centre of ossification of the femoral epiphysis (arrow). (Courtesy of Mr Christopher Edward Bache.)



Fig. 80.21 An ultrasound image of a 4-week-old baby taken in the lateral position and demonstrating the cartilaginous femoral head (yellow arrow) and edge of the acetabulum (red arrow). (Courtesy of Mr Christopher Edward Bache.)

Articulating surfaces The articulating surfaces are the medial (symphyseal) surfaces of the pubic bones, each covered by a thin layer of tightly adherent hyaline cartilage (surface growth cartilage in the young). The junction is not flat but is marked by reciprocal crests and papillae. Theoretically, this would resist shearing. The surfaces of hyaline cartilage are connected by fibrocartilage, varying in thickness and constituting the interpubic disc. The symphysis often contains a cavity, probably due to absorption. It rarely appears before the tenth year and is non-synovial. The cavity, which is better developed in females, is usually posterosuperior but may reach the front or even occupy most of the cartilage.

Ligaments The interpubic disc is strengthened anteriorly by several interlacing collagenous fibrous layers, passing obliquely from bone to bone, decussating with fibres of the external oblique aponeuroses and the medial tendons of the recti abdominis. These layers constitute the anterior pubic ligament. There are less well developed posterior fibres, sometimes named the posterior pubic ligament. The main ligaments of the joint are the superior and arcuate pubic ligaments. The superior pubic ligament connects the bones above, extending to the pubic tubercles. The arcuate pubic ligament, a thick arch of fibres, connects the lower borders of the symphyseal pubic surfaces bounding the pubic arch. It blends superiorly with the interpubic disc and extends laterally attached to the inferior pubic rami. Its inferior edge is separated from the anterior border of the perineal membrane by an opening, which is traversed by the deep dorsal vein of the penis or clitoris.

Vascular supply The pubic symphysis is supplied by pubic branches of the obturator, superficial external pudendal and inferior epigastric arteries.

Innervation The pubic symphysis is innervated by branches from the iliohypogastric, ilioinguinal and pudendal nerves.

Factors maintaining stability The interpubic disc and the superior and arcuate ligaments are the main stabilizing factors of the pubic symphysis.

Movements Angulation, rotation and displacement are possible but slight, and are likely during movement at the sacroiliac and hip joints. The movements at the pubic symphysis range between 0.1 and 2 mm and are higher along the vertical axis than along the sagittal and transverse axes (Walheim et al 1984). Excessive movement may occur following an injury. Some separation occurs late in gestation and during childbirth; on occasion, this is considerable.

Relations Anteriorly, the pubic symphysis is related to subcutaneous tissue and skin. Because of the obliquity of the joint, the proximal ends of the penile or clitoral shafts lie anterior to the lower half of the joint. Inferiorly, the urethra lies about 2.5 cm away in the male, and somewhat closer in the female, as it passes through the perineal membrane. Closer to the joint, the deep dorsal vein of the penis or clitoris passes between the arcuate ligament and the anterior border of the perineal membrane. Posteriorly, the upper part of the joint is separated from the inferolateral surfaces of the urinary bladder by the retropubic fat pad. Inferiorly in the male, the prostatic venous plexus separates the prostate from the lower part of the joint. The region is sometimes termed the retropubic space (of Retzius). These relationships explain why traumatic disruption of the anterior bony pelvis may be associated with serious urogenital injury.

SACROILIAC JOINT

The sacroiliac joint occurs between the sacral and iliac auricular surfaces and, in essence, is a stress-relieving joint (Bogduk 1997) (see Fig. 80.8A). The joint consists of syndesmotic and synovial parts. Fibrous adhesions and gradual obliteration occur in both sexes, earlier in males, and after menopause in females. Radiological evidence of obliteration in normal subjects is occasionally seen before 50 years but is not uncommon thereafter; in old age, the joint may be completely fibrosed and occasionally even ossified.

Articulating surfaces The surfaces are nearly flat in infants, whereas in adults they are irregular, often markedly so, and sometimes undulant. The curvatures and irregularities, greater in males, are reciprocal; they restrict movements and contribute to the considerable strength of the joint in transmitting weight from the vertebral column to the lower limbs. The sacral surface is covered by hyaline cartilage, which is thicker

anteriorly than posteriorly in adults. The thinner cartilage on the iliac surface is also hyaline in type, as confirmed by the presence of type II collagen.

Fibrous capsule The capsule is attached close to both articular margins.

Ligaments The ligaments of the sacroiliac joint are the anterior, interosseous and posterior sacroiliac, iliolumbar, sacrotuberous and sacrospinous ligaments.

Anterior sacroiliac ligament The anterior sacroiliac ligament (see Fig. 80.9A), an anteroinferior capsular thickening, is particularly well developed near the arcuate line and the posterior inferior iliac spine, where it connects the third sacral segment to the lateral side of the preauricular sulcus. It is thin elsewhere.

Interosseous sacroiliac ligament The interosseous sacroiliac ligament is the major bond between the bones, filling the irregular space posterosuperior to the joint. The posterior sacroiliac ligament covers it superficially. Its deeper part has superior and inferior bands passing from depressions posterior to the sacral auricular surface to those on the iliac tuberosity. These bands are covered by, and blend with, a more superficial fibrous sheet connecting the posterosuperior margin of a rough area posterior to the sacral auricular surface to the corresponding margins of the iliac tuberosity. This sheet is often partially divided into superior and inferior parts, the former uniting the superior articular process and lateral crest on the first two sacral segments to the neighbouring ilium as a short posterior iliac ligament (see Fig. 43.68).

Posterior sacroiliac ligament The posterior sacroiliac ligament (see Fig. 80.9B) overlies the interosseous ligament: the dorsal rami of the sacral spinal nerves and vessels intervene. It consists of several weak fasciculi connecting the intermediate and lateral sacral crests to the posterior superior iliac spine and posterior end of the internal lip of the iliac crest. Inferior fibres, from the third and fourth sacral segments, ascend to the posterior superior iliac spine and posterior end of the internal lip of the iliac crest; they may form a separate long posterior sacroiliac ligament. This ligament is continuous laterally with part of the sacrotuberous ligament and medially with the posterior lamina of the thoracolumbar fascia.

Iliolumbar ligament See page 738.

Sacrotuberous ligament The sacrotuberous ligament (see Figs 80.9, 43.68) is attached by its broad base to the posterior superior iliac spine, the posterior sacroiliac ligaments (with which it is partly blended), lateral sacral crest, and the lateral margins of the lower sacrum and upper coccyx. Its oblique fibres descend laterally, converging to form a thick, narrow band that widens again below and is attached to the medial margin of the ischial tuberosity. It then spreads along the ischial ramus as the falciform process, whose concave edge blends with the fascial sheath of the internal pudendal vessels and pudendal nerve. The lowest fibres of gluteus maximus are attached to the posterior surface of the ligament; superficial fibres of the lower part of the ligament continue into the tendon of biceps femoris. The coccygeal branches of the inferior gluteal artery, the perforating cutaneous nerve and filaments of the coccygeal plexus pierce the ligament.

Sacrospinous ligament The thin, triangular sacrospinous ligament (see Fig. 80.9) extends from the ischial spine to the lateral margins of the sacrum and coccyx anterior to the sacrotuberous ligament, with which it blends in part. Its anterior surface is in fact the coccygeus muscle, i.e. muscle and ligament are coextensive. The sacrospinous ligament is often regarded as a degenerate part of coccygeus.

Sciatic foramina The sacrotuberous and sacrospinous ligaments convert the sciatic notches into foramina (see Fig. 80.9).

Greater sciatic foramen The greater sciatic foramen is bounded anterosuperiorly by the greater sciatic notch, posteriorly by the sacrotuberous ligament and inferiorly by the sacrospinous ligament and ischial spine (see Fig. 80.9). It is partly filled by the emerging piriformis, above which the superior gluteal vessels and nerve leave the pelvis. Below it, the inferior gluteal vessels and nerve, internal pudendal vessels and pudendal nerve, sciatic and posterior femoral cutaneous nerves and the nerves to obturator internus and quadratus femoris all leave the pelvis.

Lesser sciatic foramen The lesser sciatic foramen is bounded anteriorly by the ischial body, superiorly by its spine and sacrospinous ligament, and posteriorly by the sacrotuberous ligament (see Fig. 80.9A). It transmits the tendon of obturator internus, the nerve to obturator internus, and the internal pudendal vessels and pudendal nerve.

Vascular supply The arterial supply of the sacroiliac joint is derived from the iliolumbar, superior gluteal and superior lateral sacral arteries, with corresponding venous drainage. Lymphatic drainage follows the arteries, reaching the iliac and lumbar nodes.

Innervation Nerve fibres ramify within the joint capsule and adjoining ligaments, but their source is uncertain. It is thought that the joint probably receives branches from the anterior and posterior rami of the first two sacral spinal nerves, and from the superior gluteal nerve, and that there may also be contributions from the obturator nerve and the lumbosacral trunk.

The sacroiliac joint has sometimes been implicated as the source of pain in the lower back and buttocks. Diagnosing sacroiliac joint-mediated pain is difficult because the presenting complaints are similar to those seen with other causes of back pain. Patients with sacroiliac joint-mediated pain rarely report pain above the level of L5; most localize their pain to the area around the posterior superior iliac spine.

Factors maintaining stability The sacroiliac joint is one of the most stable joints in the body and supports the weight of the trunk. The reciprocal irregularity of the joint surfaces allows very little movement. The tendency of the sacrum to be forced downwards by the trunk is resisted by the extremely strong posterior ligaments, while the iliolumbar ligaments help to resist displacement of the fifth lumbar vertebra over the sacrum. The sacrotuberous and sacrospinous ligaments oppose upward tilting of the lower part of the sacrum when downward thrust is applied at its upper end.

Movements Primary movement of the sacroiliac joint is minimal. All muscles that cross the joint act on the lumbar spine or on the hip. Such movements as do occur are secondarily imposed on the joint as the pelvis moves. Data from living subjects are technically difficult to obtain, and those based on plain radiographs are unreliable. Studies using implanted tantalum spheres and biplanar radiography have shown mean rotational ranges of less than 2°. Even when there is recordable movement, the direction of movement is irregular. Biplanar radiography has also shown that the axes of movement of the sacroiliac joint during hip movement are oblique, and that the axes differ in flexion and extension.

During pregnancy, the pelvic joints and ligaments loosen under the influence of the hormone relaxin. Movements in the joints increase. Relaxation renders the sacroiliac locking mechanism less effective, permitting greater rotation and perhaps allowing alterations in pelvic diameters at childbirth, although the effect is probably small. The impaired locking mechanism diverts the strain of weight-bearing to the ligaments, with frequent sacroiliac strain after pregnancy.

Relations The sacroiliac joints have many important anterior relations. The internal and external iliac veins join to form the common iliac veins immediately anteriorly, separating the joints from the bifurcations of the common iliac arteries and, more anteriorly, the ureters. The lumbosacral trunk and the obturator nerve cross the anterior aspect of the joint behind the vessels. Piriformis partly attaches to the anterior capsule, separating the joint from the upper part of the sacral plexus.

Variants Accessory sacroiliac articulations are not uncommon. They develop behind the articular surface between the lateral sacral crest and posterior superior iliac spine and iliac tuberosity, and are acquired fibrocartilaginous joints resulting from the stresses of weight-bearing. They have a joint capsule, are saddle-shaped, and may be single, double, unilateral or bilateral (Weisl 1954).

MUSCLES

MUSCLES OF THE ILIAC REGION

Although there is no 'iliac region' as such, this heading conveniently describes a group of three muscles that originate from the lumbar vertebral column (psoas major and minor) and the ilium (iliacus). Psoas major and iliacus are attached together on the femur as flexors of the hip joint and are often considered as a functional unit, iliopsoas. Psoas minor only reaches the pubis, and acts on the spine and sacroiliac joint.

Psoas major

Attachments Psoas major is a long muscle that lies on either side of the lumbar vertebral column and the pelvic brim (Fig. 80.22, see Fig. 62.7). Its proximal attachments are complex. They include the anterior surfaces and lower borders of the transverse processes of all the lumbar vertebrae. There are five digitations, each from the bodies of two adjoining vertebrae and their intervertebral disc. The highest of these arises from the lower margin of the body of the twelfth thoracic vertebra, the upper margin of the body of the first lumbar vertebra and the interposed intervertebral disc. The lowest arises from the adjacent margins of the bodies of the fourth and fifth lumbar vertebrae and the interposed disc. A series of tendinous arches extend across the narrow parts of the bodies of the lumbar vertebrae between the digitations already described. The lumbar arteries and veins, and filaments from

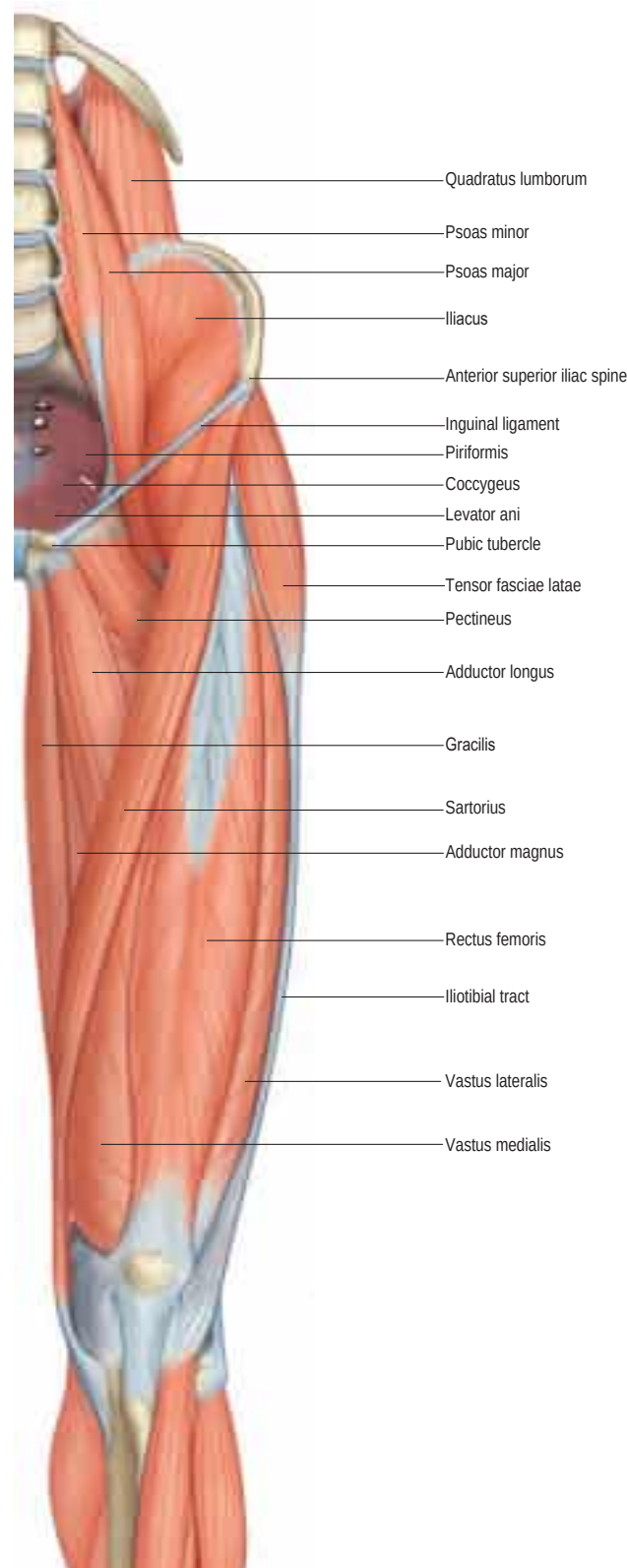


Fig. 80.22 The hip flexors and superficial muscles of the thigh.

the sympathetic trunk, pass medial to these arches. The upper four lumbar intervertebral foramina bear important relations to these attachments of the muscle. The foramina lie anterior to the transverse processes and posterior to the attachments to vertebral bodies, intervertebral discs and tendinous arches. Thus, the roots of the lumbar plexus enter the muscle directly, the plexus is lodged within it, and its branches emerge from its borders and surfaces.

The muscle descends along the pelvic brim, continues posterior to the inguinal ligament and anterior to the capsule of the hip joint, and converges to a tendon that, having received on its lateral side nearly all the fibres of iliacus, becomes attached to the lesser trochanter of the femur. The large subtendinous iliac bursa, which occasionally communicates with the cavity of the hip joint, separates the tendon from the pubis and the capsule of the joint.

The complex vertebral attachments of psoas major sometimes display minor numerical variations.

Relations The upper limit of psoas major is posterior to the diaphragm in the lowest part of the posterior mediastinum. It may be in contact with the posterior extremity of the pleural sac. In the abdomen, its anterolateral surface is related to the medial arcuate ligament (a linear, arched thickening in the psoas fascia), extraperitoneal tissue and peritoneum, the kidney, psoas minor, renal vessels, ureter, testicular or ovarian vessels and the genitofemoral nerve. Anteriorly, the right psoas is overlapped by the inferior vena cava and crossed by the terminal part of the ileum; the colon crosses the left psoas. Its posterior surface is related to the transverse processes of the lumbar vertebrae and the medial edge of quadratus lumborum. The lumbar plexus is embedded posteriorly in the substance of psoas major. Medially, the muscle is related to the bodies of the lumbar vertebrae and lumbar vessels. Along its anteromedial margin, it is in contact with the sympathetic trunk, aortic lymph nodes and, along the pelvic brim, with the external iliac artery. This margin is covered by the inferior vena cava on the right side, and lies posterior and lateral to the abdominal aorta on the left side. In the thigh, iliopsoas is related anteriorly to the fascia lata and the femoral artery, and posteriorly to the capsule of the hip joint, from which it is separated by a bursa. At its medial border, it is related to pectineus and the medial circumflex femoral artery, and to the femoral vein, which may overlap it slightly. At its lateral border, it is related to the femoral nerve and iliacus. The femoral nerve descends at first through the fibres of psoas major, and then in the furrow between it and iliacus.

Branches of the lumbar plexus diverge from the abdominal part of psoas major. Emerging from the lateral border, from above downwards, are the iliohypogastric, ilioinguinal and lateral femoral cutaneous and femoral nerves. Emerging from the anterolateral surface is the genitofemoral nerve. The obturator and accessory obturator nerves (when present) and the upper root of the lumbosacral trunk all emerge from the medial border.

Vascular supply Psoas major is supplied by a rich network of arteries derived from the lumbar, ilio-lumbar, obturator, external iliac and femoral arteries. In general terms, the upper part of the muscle is supplied by the lumbar arteries, the mid part by the anterior branch of the ilio-lumbar artery (the main artery to the muscle) with contributions from the deep circumflex and external iliac arteries, and the distal part by the femoral artery and its branches. The psoas sheath has an arterial supply independent from that of the muscle, though the same vessels contribute.

Innervation Psoas major is innervated by the ventral rami of the lumbar spinal nerves, mainly L1 and 2 with some contribution from L3.

Actions Psoas major acts together with iliacus; the combination is referred to as iliopsoas. See below ('Iliacus').

Testing See below (after 'Iliacus').

Psoas minor

Attachments Psoas minor (see Fig. 80.22) is sometimes absent. When present, it lies anterior to psoas major, entirely within the abdomen. It arises from the sides of the bodies of the twelfth thoracic and first lumbar vertebrae, and from the intervertebral disc between them. It ends in a long, flat tendon that is attached to the pecten pubis, ilio-pubic ramus and, laterally, to the iliac fascia.

Relations Psoas minor lies on psoas major, and its proximal anterior relations are those of the anterior or anteromedial surface of that muscle.

Vascular supply The main arterial supply to psoas minor is from the lumbar arteries, though there may be minor contributions from other vessels of the network that supplies psoas major.

Innervation Psoas minor is innervated by a branch from L1.

Actions Psoas minor is probably a weak flexor of the trunk.

Testing Psoas minor cannot be tested clinically.

Iliacus

Attachments Iliacus (see Figs 80.22, 62.7) is a triangular sheet of muscle that arises from the superior two-thirds of the concavity of the iliac fossa, the inner lip of the iliac crest, the ventral sacroiliac and ilio-lumbar ligaments, and the upper surface of the lateral part of the sacrum (see Fig. 43.49). In front, it reaches as far as the anterior superior and anterior inferior iliac spines, and receives a few fibres from the upper part of the capsule of the hip joint (iliocapsularis, p. 1377). Most of its fibres converge into the lateral side of the strong tendon of psoas major, and the muscles then insert together into the lesser trochanter, but some fibres are attached directly to the femur 2.5 cm below and in front of the lesser trochanter.

Relations In the abdomen, the anterior surface of iliacus is related to its fascia (which separates the muscle from extraperitoneal tissue and peritoneum), the lateral femoral cutaneous nerve, the caecum (on the right) and the iliac part of the descending colon (on the left). On its posterior surface is the iliac fossa, and at its medial border, psoas major and the femoral nerve. In the thigh, its anterior surface is in contact with the fascia lata, rectus femoris, sartorius and the profunda femoris artery, and its posterior surface is in contact with the capsule of the hip joint, from which it is partially separated by a bursa it shares with psoas major.

Vascular supply Iliacus is supplied by the same arterial network as psoas major, and there is mutual overlap of the arterial territories of each muscle. The main supply is from the iliac branches of the ilio-lumbar artery, with contributions from the deep circumflex iliac and obturator arteries, and branches of the femoral artery.

Innervation Iliacus is innervated by branches of the femoral nerve, L2 and 3.

Actions Psoas major, acting from above together with iliacus, flexes the thigh on the pelvis. Electromyographic studies do not support the common view that psoas major acts as a medial rotator of the hip joint, but activity has been described in lateral rotation, particularly in the young. When psoas major and iliacus of both sides act from below, they contract powerfully to bend the trunk and pelvis forwards against resistance, as in raising the trunk from the recumbent to the sitting posture when undertaking a 'sit-up' exercise.

Electromyography does not support the commonly held belief that unilateral action of psoas major causes lateral and forward flexion of the trunk to that side (Joseph 1975).

In symmetrical upright stance, iliopsoas has some action from below to maintain the vertebral column upright. Psoas major is active in balancing the trunk while sitting.

Testing Both psoas and iliacus may be the sites of pathological collections of fluid. An abscess (typically tuberculous) of vertebral origin may track down through psoas and present as a mass in the thigh. Haematoma or infection within the iliacus fascia may present as a mass or as a flexion deformity of the hip.

Iliopsoas may be tested clinically by actively flexing the hip against resistance, in the supine position with the hip and knee flexed.

MUSCLES OF THE GLUTEAL REGION

Tensor fasciae latae

Attachments Tensor fasciae latae (see Fig. 80.22) arises from the anterior 5 cm of the outer lip of the iliac crest, from the lateral surface of the anterior superior iliac spine and part of the border of the notch below it, between gluteus medius and sartorius, and from the deep surface of the fascia lata. Proximal attachments may extend to the aponeurotic fascia superficial to gluteus medius. It descends between, and is attached to, the two layers of the iliotibial tract of the fascia lata and usually ends approximately one-third of the way down the thigh,

although it may occasionally extend as far as the lateral femoral condyle.

Vascular supply The muscle itself is supplied mainly by a large ascending branch of the lateral circumflex femoral artery. The tensor fasciae latae musculocutaneous flap is raised on this pedicle. The superior part of the muscle receives branches from the superior gluteal artery. The fascia surrounding the muscle is supplied on its superficial aspect by the superficial circumflex iliac artery and on its deep surface by the lateral circumflex femoral artery.

Innervation Tensor fasciae latae is innervated by the superior gluteal nerve, L4, 5 and S1.

Actions Tensor fasciae latae, acting through the iliotibial tract, extends the knee with lateral rotation of the leg; it may also assist in abduction and medial rotation of the thigh, though its role as an abductor is debatable. The muscle helps to maintain upright posture while minimizing energy expenditure on muscle activity: when the subject is standing it acts from below to steady the pelvis on the head of the femur and, through the iliotibial tract, to steady the condyles of the femur on the tibial condyles while the knee extensors are relaxed. The muscle aids gluteus medius in postural abduction at the hip. Postural control is its main function.

In the last 20° or so of extension, the pull of the iliotibial tract is anterior to the flexion axis of the knee and so the tensor fasciae latae is a weak extensor. Flexion of greater than 20° leads to the iliotibial tract passing posterior to the axis of flexion so that the muscle becomes a weak flexor.

Testing When the thigh is flexed against gravity and the knee is extended, an angular depression appears immediately below the anterior superior iliac spine; its lateral boundary is tensor fasciae latae. If the thigh is then abducted against resistance, the muscle can be felt and sometimes seen.

Gluteus maximus

Attachments Gluteus maximus (Fig. 80.23) is the largest and most superficial muscle in the gluteal region. It is a broad, thick, quadrilateral mass, which, with its overlying adipose fascia, forms the familiar prominence of the buttock. Gluteus maximus is thicker and more extensive in humans than in any non-human primate, features that presumably correlate with the evolutionary transition to bipedality and a permanently upright posture. The muscle has a coarse fascicular architecture, with large bundles of fibres separated by fibrous septa. It arises from the posterior gluteal line of the ilium and the rough area of bone, including the crest, immediately above and behind it; from the aponeurosis of erector spinae; the dorsal surface of the lower part of the sacrum and the side of the coccyx; the sacrotuberous ligament; and the fascia (gluteal aponeurosis) that covers gluteus medius. There may be additional slips from the lumbar aponeurosis or ischial tuberosity. The muscle may also be bilaminar. The fibres descend laterally; the upper part of the muscle, together with the superficial fibres of the lower part, ends in a thick tendinous lamina that passes lateral to the greater trochanter and is attached to the iliotibial tract of the fascia lata. The deeper fibres of the lower part of the muscle are attached to the gluteal tuberosity between vastus lateralis and adductor magnus.

Relations A thin fascia separates the superficial surface of gluteus maximus from the overlying thick, adipose subcutaneous tissue. The deep surface of the muscle is related to the ilium, sacrum, coccyx, sacrotuberous ligament, part of gluteus medius, piriformis, the gemelli, obturator internus, quadratus femoris, the ischial tuberosity, greater trochanter, and the attachments of biceps femoris, semitendinosus, semimembranosus and adductor magnus to the ischial tuberosity. Three bursae lie deep to gluteus maximus: trochanteric, over the greater trochanter; gluteofemoral, between the tendon of gluteus maximus and that of vastus lateralis; and ischiofemoral, over the gluteal tuberosity, which is less commonly present.

The superficial division of the superior gluteal artery reaches the deep surface of the muscle between piriformis and gluteus medius. The inferior gluteal and internal pudendal vessels, the sciatic, pudendal and posterior femoral cutaneous nerves, muscular branches from the sacral femoral cutaneous nerves, and muscular branches from the sacral plexus all leave the pelvis below piriformis. The first perforating artery and the terminal branches of the medial circumflex femoral artery are also deep to the lower part of gluteus maximus. Its upper border is thin and overlies gluteus medius. Its prominent lower border is free and slopes downwards and laterally. It is crossed by the horizontal gluteal

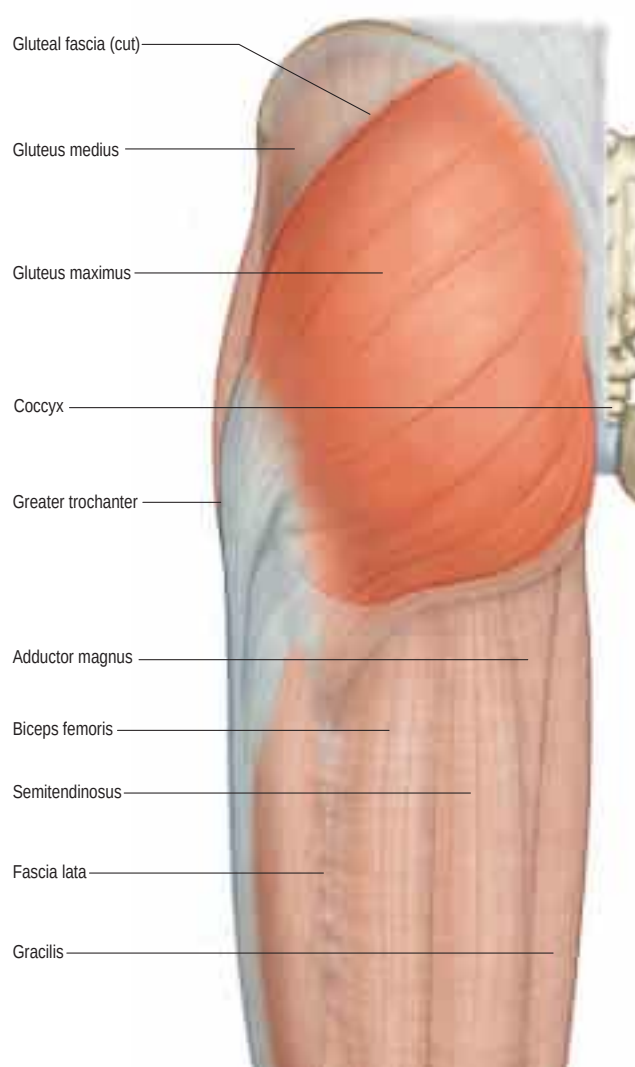


Fig. 80.23 Gluteus maximus, posterior view with the posterior thigh muscles seen through the overlying fascia lata.

fold (the posterior flexure line of the hip joint), which marks the upper limit of the back of the thigh on the surface.

Vascular supply The dominant vascular pedicle is usually that of the inferior gluteal artery, which supplies approximately two-thirds of the muscle. The remainder is supplied mainly by the superior gluteal artery, though this may sometimes be the dominant vessel. The lateral and distal borders of the muscle receive a supply from the first perforator given off from the profunda femoris artery and from the medial circumflex femoral artery. Minor branches may be derived from the lateral circumflex femoral, lateral sacral and internal pudendal arteries.

Gluteus maximus musculocutaneous flaps may be based on either of the gluteal vessels or on the first perforator given off from the profunda femoris artery, depending on the site and size of the defect requiring coverage.

Innervation Gluteus maximus is innervated by the inferior gluteal nerve, L5, S1 and S2.

Actions Acting from the pelvis, gluteus maximus can extend the flexed thigh and bring it into line with the trunk. Acting from its distal attachment, it may prevent the forward momentum of the trunk from producing flexion at the supporting hip during bipedal gait. The muscle is inactive during standing, when swaying forwards at the ankle joints, or when bending forwards at the hip joints to touch the toes. However, it acts with the hamstrings in raising the trunk after stooping, by rotating the pelvis backwards on the head of the femur. It is intermittently active in the walking cycle and in climbing stairs, and continuously active in strong lateral rotation of the thigh. Its upper fibres are active in powerful abduction of the thigh. It is a tensor of the fascia lata, and through the iliotibial tract it stabilizes the femur on the tibia when the extensor muscles of the knee are relaxed.

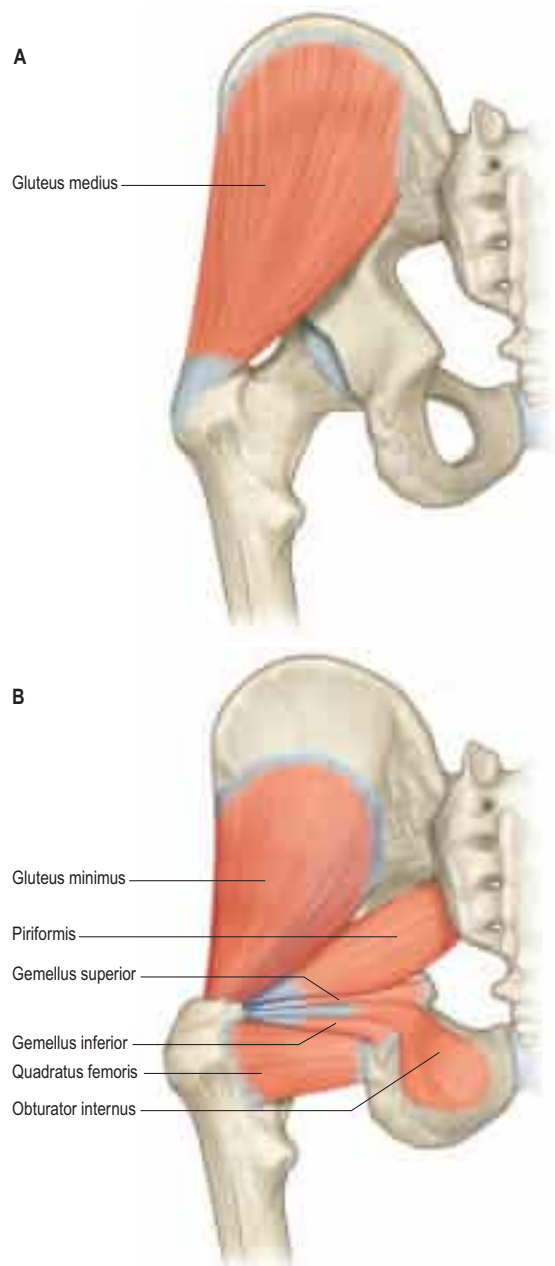


Fig. 80.24 **A**, Gluteus medius, posterior view. **B**, Gluteus minimus and short lateral rotators of the hip, posterior view.

Testing Gluteus maximus may be tested by extension of the hip against resistance, in the supine or prone position.

Gluteus medius

Attachments Gluteus medius is a broad, thick muscle (see Fig. 80.23; Fig. 80.24A). It arises from the outer surface of the ilium between the iliac crest and posterior gluteal line above, and the anterior gluteal line below, and also from the strong fascia superficial to its upper part. The fibres converge to a flat tendon that attaches to a ridge that slants downwards and forwards on the lateral surface of the greater trochanter.

A deep slip of the muscle may be attached to the upper border of the trochanter. The posterior edge of gluteus medius sometimes blends with piriformis.

Relations The posterior third of gluteus medius is covered by gluteus maximus but it is superficial in its anterior two-thirds, where a strong layer of deep fascia covers it. Its deep surface is related to gluteus minimus. Branches of the deep divisions of the superior gluteal nerve and artery run between the medius and minimus muscles, and are vulnerable during anterolateral and lateral approaches to the hip that involve splitting gluteus medius. Where the tendon glides on the anterosuperior part of the lateral surface of the trochanter, a bursa (trochanteric bursa of gluteus medius) separates it from the bone.

Vascular supply The main supply to gluteus medius is from the deep branch of the superior gluteal artery (Fig. 80.25A). The distal part of the muscle is supplied by the trochanteric anastomosis.

Innervation Gluteus medius is innervated by the superior gluteal nerve, L4, 5 and S1.

Actions These are considered below with gluteus minimus.

Testing This is considered below with gluteus minimus.

Gluteus minimus

Attachments Gluteus minimus lies deep to gluteus medius (see Fig. 80.24B). The fan-shaped muscle arises from the outer surface of the ilium between the anterior and inferior gluteal lines and, behind, from the margin of the greater sciatic notch. The fibres converge below to the deep surface of an aponeurosis that ends in a tendon attached to an anterolateral ridge on the greater trochanter and contributes an expansion to the capsule of the hip joint.

The muscle may divide into anterior and posterior parts. Separate slips may pass to piriformis, gemellus superior or vastus lateralis.

Relations Branches of the deep division of the superior gluteal artery and nerve run on the superficial surface of the muscle. The reflected tendon of rectus femoris and the capsule of the hip joint are deep to gluteus minimus. A bursa (trochanteric bursa of gluteus minimus) separates the tendon from the medial part of the anterior surface of the greater trochanter.

Vascular supply Gluteus minimus is supplied from both its surfaces, from the main trunk and the deep branch of the superior gluteal artery, with a contribution at its femoral attachment from the trochanteric anastomosis (see Fig. 80.25).

Innervation Gluteus minimus is innervated by the superior gluteal nerve, L4, 5 and S1.

Actions of gluteus medius and minimus

Both gluteus medius and minimus, acting from the pelvis, abduct the thigh, and their anterior fibres rotate it medially. Acting from the femur, they play an essential part in maintaining the trunk upright when the foot of the opposite side is raised from the ground in walking and running. In this phase, the body weight tends to make the pelvis sag downwards on the unsupported side. This is counteracted by the gluteus medius and minimus of the supporting side, which, acting from below, exert such powerful traction on the hip bone that the pelvis is actually raised a little on the unsupported side. In symmetrical standing with the feet somewhat separated, the abductor muscles are usually 'silent' on electromyography, but with the feet placed parallel and close together they are active.

Testing The supportive effect of the glutei (medius and minimus) on the pelvis when the contralateral foot is raised depends on the following conditions. The two muscles, and their innervation, must be functioning normally. The components of the hip joint, which forms the fulcrum, must be in their usual relation. The neck of the femur must be intact, with its normal angulation to the shaft.

When any one of these conditions is not fulfilled, e.g. in paralysis of the glutei, developmental dysplasia of the hip or coxa vara, the supporting mechanism is upset and the pelvis sinks on the unsupported side when the patient tries to stand on the affected limb. This is known clinically as Trendelenburg's sign. Individuals with paralysis of gluteus medius and minimus have a characteristic lurching gait. Provided that these two muscles are intact, paralysis of other muscles acting on the hip joint produces remarkably little deficit in walking, or even in running.

Gluteus medius and minimus may be tested together by medial rotation of the thigh against resistance, in the supine position with hip and knee flexed. Both muscles may be tested together with tensor fasciae latae by abduction of the lower limb against resistance, in the supine position with the knee extended.

Piriformis

Attachments Piriformis (see Figs 80.24B, 80.25A) occupies a central position in the buttock, where it lies in the same plane as gluteus medius. It arises from the anterior surface of the sacrum by three digitations, which are attached to the portions of bone between the pelvic sacral foramina, and to the grooves leading from the foramina (see Fig. 43.49). It also arises from the gluteal surface of the ilium near the

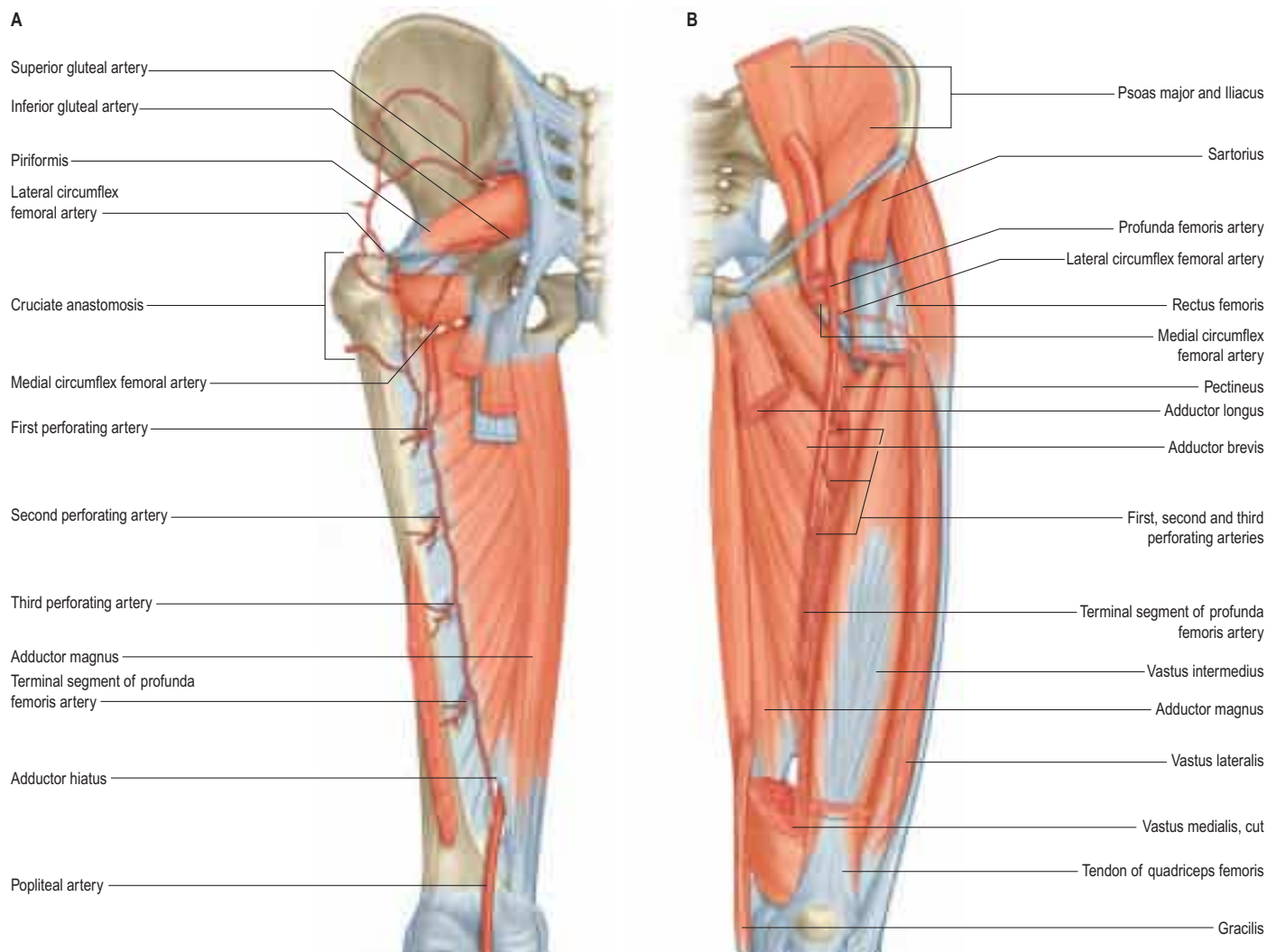


Fig. 80.25 The arteries and muscles of the left gluteal and femoral regions. **A**, Posterior aspect. **B**, Anterior aspect. (With permission from Drake RL, Vogl AW, Mitchell A (eds), *Gray's Anatomy for Students*, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

posterior inferior iliac spine, from the capsule of the adjacent sacroiliac joint, and sometimes from the upper part of the pelvic surface of the sacrotuberous ligament. The muscle passes out of the pelvis through the greater sciatic foramen, which it substantially fills. Here it constitutes an important surgical landmark in the identification of structures that emerge above and below it. It inserts into the medial side of the upper border of the greater trochanter of the femur via a rounded tendon that lies behind and above, but is often partially blended with, the common tendon of obturator internus and the gemelli. The muscle itself may be fused with gluteus medius.

Relations Within the pelvis, the anterior surface of piriformis is related to the rectum (especially on the left), the sacral plexus of nerves and branches of the internal iliac vessels. The posterior surface lies against the sacrum. Outside the pelvis, its anterior surface is in contact with the posterior surface of the ischium and capsule of the hip joint, and its posterior surface with gluteus maximus. Its upper border is in contact with gluteus medius and the superior gluteal vessels and nerve, its lower border with coccygeus and gemellus superior. The inferior gluteal and internal pudendal vessels, the sciatic, posterior femoral cutaneous and pudendal nerves, and muscular branches from the sacral plexus appear in the buttock in the interval between piriformis and gemellus superior. The relationship between piriformis and the sciatic nerve is variable. The undivided nerve may emerge above the muscle or through the muscle. The major divisions of the nerve may lie either side of the muscle, or (the most common variant) one division passes between the heads of a divided muscle and one division either above or below.

Piriformis can occasionally cause entrapment of the sciatic nerve in the buttock, giving rise to the so-called 'piriformis syndrome'.

Vascular supply In the buttock, piriformis is supplied mainly from the superior gluteal artery, with contributions from the gemellar branches of the internal pudendal. There may be a separate branch from

the inferior gluteal artery (see Fig. 80.25A). In the pelvis, the main supply is from the lateral sacral artery, with contributions from both gluteal vessels.

Innervation Piriformis is innervated by branches from S1 and 2 (sometimes only from S2). It may also contain fibres from L5.

Actions Piriformis rotates the extended thigh laterally, but abducts the flexed thigh.

Testing Clinically, it is not possible to test piriformis alone; however, for suspected injury to piriformis the best provocative test is to ask the seated subject to abduct the thighs. Buttock pain suggests piriformis injury.

Obturator internus

Attachments Obturator internus (see Figs 80.24B, 80.27) is situated partly within the true pelvis and partly posterior to the hip joint. It arises from the internal surface of the anterolateral wall of the lesser pelvic cavity. Its attachments, which almost surround the obturator foramen, are to the inferior ramus of the pubis, the ischial ramus, and the pelvic surface of the hip bone below and behind the pelvic brim, to the upper part of the greater sciatic foramen above and behind, to the obturator foramen below and in front. It also arises from the medial part of the pelvic surface of the obturator membrane, from the tendinous arch that completes the obturator canal, and, to a small extent, from the obturator fascia that covers the muscle. The fibres converge towards the lesser sciatic foramen and end in four or five tendinous bands on the deep surface of the muscle. These bands make a lateral right-angled turn around the grooved surface of the ischium between its spine and tuberosity. The grooved surface is covered with a smooth layer of hyaline cartilage and is separated from the tendon by a bursa; ridges on the surface correspond to furrows between the tendinous bands. These bands leave the pelvis through the lesser sciatic

foramen and unite to form a single flattened tendon that passes horizontally across the capsule of the hip joint. The gemelli fuse with this tendon before it inserts on to an anterior impression on the medial surface of the greater trochanter anterosuperior to the trochanteric fossa. A long, narrow bursa is usually interposed between the tendon and the capsule of the hip joint, and occasionally communicates with the bursa between the tendon and the ischium.

Relations Within the pelvis, the anterolateral surface of the muscle is in contact with the obturator membrane and inner surface of the lateral wall of the pelvis. Its posteromedial surface is related to the obturator fascia, the origin of levator ani, and the sheath that surrounds the internal pudendal vessels and pudendal nerve, and forms the lateral wall of the ischio-anal fossa. Outside the pelvis, the muscle is covered by gluteus maximus, is crossed posteriorly by the sciatic nerve and passes behind the hip joint. As the tendon of obturator internus emerges from the lesser sciatic foramen, it is overlapped both above and below by the two gemelli, which form a muscular canal for it. Near its termination, the gemelli pass anterior to the tendon and form a groove in which it lies.

Vascular supply The main arterial supply of the extrapelvic part of obturator internus is from the gemellar branches of the internal pudendal artery. Intrapelvic and extrapelvic parts are supplied by the branches of the obturator artery.

Innervation Obturator internus is innervated by the nerve to obturator internus, L5 and S1.

Actions See below actions of [obturator externus](#).

Obturator externus

Attachments Obturator externus (Figs 80.26–80.29; see Fig. 80.16) is a flat, triangular muscle covering the external surface of the anterior pelvic wall. It arises from the anteromedial two-thirds of the external surface of the obturator foramen, and from the adjacent bone of the pubic and ischial rami, extending for a short distance on to their pelvic surfaces between the margin of the obturator foramen and the obturator membrane. The whole muscle, and the tendon into which its fibres converge, spiral backwards, laterally and upwards, and thus cross the inferior aspect and then the back of the neck of the femur and lower part of the capsule of the hip joint to end in the trochanteric fossa of the femur.

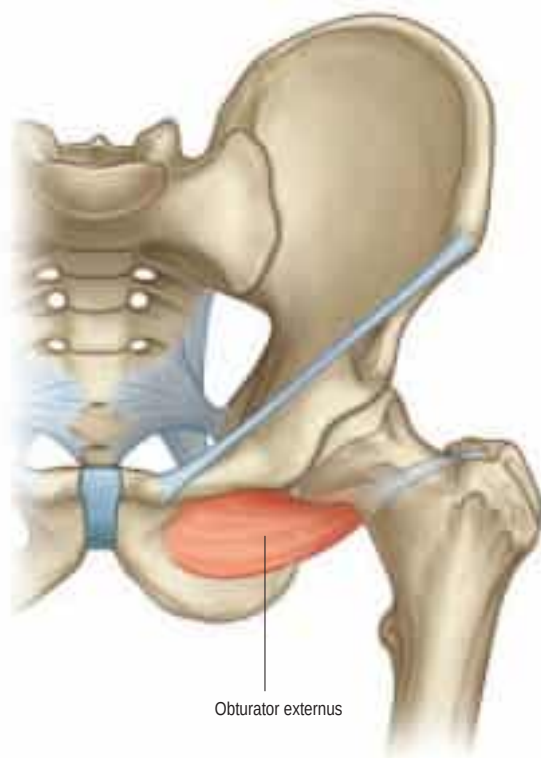


Fig. 80.26 Obturator externus, anterior aspect. (Adapted from Drake RL, Vogl AW, Mitchell A (eds), Gray's Anatomy for Students, 2nd ed, Elsevier, Churchill Livingstone. Copyright 2010.)

Relations A bursa, which communicates with the hip joint, may be interposed between the tendon and the hip joint capsule and femoral neck. The obturator vessels lie between the muscle and the obturator membrane. The anterior branch of the obturator nerve reaches the thigh by passing in front of the muscle, and the posterior branch by piercing it.

Vascular supply Obturator externus receives a variable pattern of supply from the obturator and medial circumflex femoral arteries.

Innervation Obturator externus is innervated by the posterior branch of the obturator nerve, L3 and 4.

Actions It has been suggested that the short muscles around the hip joint (pectineus, piriformis, obturator externus and internus, the gemelli and quadratus femoris) are more important as postural muscles than as prime movers, acting as adjustable ligaments to maintain the stability and integrity of the hip. However, these muscles are largely inaccessible to direct observation, and, because of the hazards presented by their close relationship to important neurovascular structures, there is a lack of electromyographic data in humans. In both bipedal walking and vertical climbing, obturator externus is recruited during the early part of the swing phase; in climbing it effects lateral rotation of the thigh, and in walking it probably counteracts the tendency to medial rotation produced by the anterior adductor muscles at this stage of the cycle. Obturator internus differs from obturator externus in its pattern of use but its role in bipedal walking remains unclear. Its attachments suggest

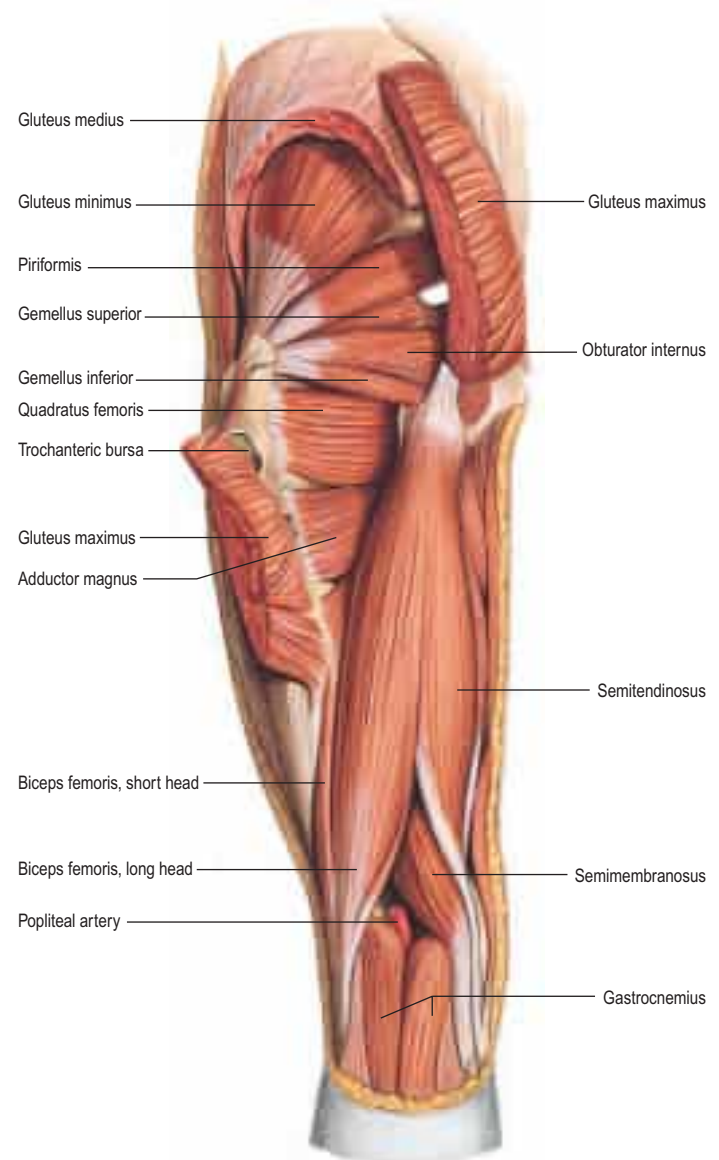


Fig. 80.27 The posterior thigh and gluteal muscles with gluteus maximus and medius partially removed. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

that it, like the gemelli, is a lateral rotator of the extended thigh and an abductor of the flexed thigh. These actions may be used to antagonize unwanted components of movement produced by the primary locomotor muscles.

Testing The short rotators cannot be tested individually, but lateral rotation of the extended hip and abduction of the flexed hip against resistance tests them as a group.

Gemellus inferior and gemellus superior

Attachments Gemellus inferior arises from the upper part of the lateral surface of the ischial tuberosity, immediately below the groove for the tendon of obturator internus (Fig. 80.27). It blends with the lower border of this tendon, and inserts with it into the medial surface of the greater trochanter.

Gemellus superior, the smaller of the two gemelli, arises from the dorsal surface of the ischial spine, blends with the upper border of the tendon of obturator internus, and inserts with it into the medial surface of the greater trochanter. It is sometimes absent.

The two gemelli can be regarded as accessory to obturator internus (see Fig. 80.24B).

Relations The relations of the gemelli are as described for the extrapelvic part of obturator internus.

Vascular supply Gemellus superior is supplied by the internal pudendal artery and its gemellar branches; by the inferior gluteal artery; and sometimes also by the superior gluteal artery. Gemellus inferior is supplied mainly by the medial circumflex femoral artery.

Innervation Gemellus superior is innervated by the nerve to obturator internus, L5, S1 and S2. Gemellus inferior is innervated by the nerve to quadratus femoris, L4, L5 and S1.

Actions The gemelli rotate the extended thigh laterally and abduct the flexed thigh.

Testing See testing of obturator externus above.

Quadratus femoris

Attachments Quadratus femoris (see Fig. 80.24B) is a flat, quadrilateral muscle lying between gemellus inferior and the upper margin of adductor magnus, from which it is separated by the transverse branch of the medial circumflex femoral artery. It arises from the upper part of the external aspect of the ischial tuberosity and inserts on to a small tubercle a little above the middle of the trochanteric crest of the femur and into the bone for a short distance below. It may be absent.

Relations The muscle passes behind the hip joint and the neck of the femur, separated from them by the tendon of obturator externus and the ascending branch of the medial circumflex femoral artery (which may cause troublesome bleeding if injured during the posterior approach to the hip). The sciatic nerve crosses the muscle posteriorly. A bursa is often present between the muscle and the lesser trochanter.

Vascular supply Branches pass to the superficial surface of the muscle from the inferior gluteal artery and from the medial circumflex femoral artery, which also supplies the deep surface.

Innervation Quadratus femoris is innervated by the nerve to quadratus femoris, L5 and S1.

Actions Quadratus femoris produces lateral rotation of the thigh.

Testing See testing of obturator externus.

MUSCLES OF THE THIGH

The presence and position of the femoral neck cause the femoral shaft to lie obliquely; consequently, the anterior (extensor) muscle group, quadriceps femoris, runs obliquely distally and medially and so applies a pull to the patella that is both laterally and proximally directed. The adductor muscles occupy the region between quadriceps femoris and the medial margin of the thigh. They are attached distally to the posterior surface of the femur and lie more posteriorly than quadriceps femoris. The posterior muscle group, the hamstrings, lie behind the adductors. The attachments of these muscles determine the nature and direction of displacement of femoral shaft fractures.

Anterior compartment

The muscles of the anterior compartment (see Figs 80.2, 80.22) include sartorius and rectus femoris, which can act at both the hip and knee joints, and vasti medialis, lateralis and intermedius, which act only at the knee. Articularis genus completes the group: it retracts the synovial capsule of the knee joint. Rectus femoris and the vasti extend the knee joint through a common tendon and hence are collectively termed quadriceps femoris. Adductor longus and pectineus are sometimes considered to be part of both the anterior and the adductor compartments.

Sartorius

Attachments Sartorius is a narrow strap muscle and is the longest muscle in the body (see Fig. 80.31). It arises by tendinous fibres from the anterior superior iliac spine and the upper half of the notch below it. It crosses the thigh obliquely over to the medial side and then descends more vertically to the medial side of the knee. The muscle fibres terminate at this point and a thin, flattened tendon curves obliquely forwards and expands into a broad aponeurosis. The aponeurosis is attached to the proximal part of the medial surface of the tibia in front of gracilis and semitendinosus, together forming the so-called pes anserinus (Fig. 80.28). A slip from its upper margin blends with the capsule of the knee joint, and another from its lower margin merges with the superficial layer of the deep fascia on the medial side of the leg. This sheet of tissue passes inferiorly to lie superficial to and over the distal insertions of gracilis and semitendinosus. It has to be split by sharp dissection to reveal these two tendons if they need to be harvested, e.g. in cruciate ligament surgery.

In some cases, sartorius is absent, while in others, the muscle may possess an extra head; when the latter occurs, the extra head is attached to the pectineal line or to the femoral sheath. Variations in the distal attachment of sartorius, in the vicinity of the knee, have been described.

Relations The relations of sartorius are described in the femoral triangle (see below).

Vascular supply The main arterial supply to sartorius is derived from multiple branches of the femoral system and enters the medial half of the muscle from its deep surface. Vessels supplying the proximal third of sartorius may arise from the femoral artery, the profunda femoris artery, the 'artery of the quadriceps' or the lateral circumflex femoral artery. There may be an additional proximal supply from the superficial circumflex iliac artery. Those vessels supplying the middle third of the

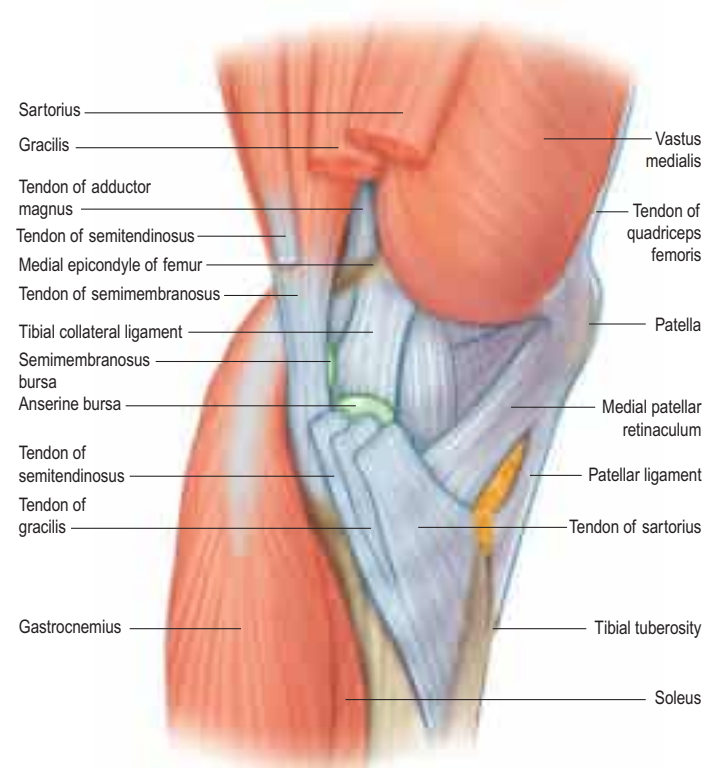


Fig. 80.28 The distal attachment of muscles of the thigh, medial aspect.

muscle arise from the femoral artery. The distal vessels arise from the femoral artery within the adductor canal and from the descending genicular artery.

Innervation Sartorius is innervated by the femoral nerve, L2 and 3.

Actions Sartorius assists in flexing the leg, and the thigh on the pelvis, particularly when these two movements are combined. It also helps to abduct the thigh and to rotate it laterally. Together with inversion of the foot, these movements bring the sole of the foot into direct view. The fact that sartorius represents only 1% of the physiological cross-sectional area of all muscles that cross the hip or knee joint would appear to suggest that its role in walking is a minor one. In a subject ascending steps, electromyographic activity of sartorius increases during lateral rotation of the thigh at the end of the swing phase immediately preceding heel strike, which presumably decelerates the limb. Sartorius may therefore have a significant involvement in climbing.

Testing When it contracts against gravity, as it usually does, sartorius can be both seen and felt in the living subject.

Femoral triangle

The femoral triangle is a depressed, intermuscular space in the antero-medial aspect of the proximal thigh, lying immediately distal to the inguinal ligament. The latter constitutes the base of the femoral triangle's inverted triangular outline. Its lateral boundary is the medial margin of sartorius. Its medial boundary is the medial margin of adductor longus. Its distal extremity, the apex, is where sartorius overlaps adductor longus (see Fig. 80.31). Its floor is provided laterally by iliacus and psoas major, and medially by pectineus and adductor longus. Its roof is the overlying fascia lata. The femoral vessels, passing from mid-base to apex, are in the deepest part of the triangle. Lying lateral to the artery and outside the femoral sheath is the femoral nerve, which, on entering the femoral triangle divides into multiple branches. The triangle also contains fat and lymph nodes.

Quadriceps femoris

Quadriceps femoris (see Figs 80.2, 80.22, 80.31), the great extensor muscle of the leg, covers almost all of the front and sides of the femur. It can be divided into four parts, each named individually. One, rectus femoris, arises from the ilium and travels straight down the middle of the thigh, its shape and path determining its name. The other three arise from the shaft of the femur and surround it (apart from the linea aspera) from the trochanters to the condyles: vastus lateralis is lateral to the femur, vastus medialis is medial to it, and vastus intermedius lies anterior to the femur. Rectus femoris crosses both hip and knee joints, while the three vasti only cross the knee joint.

The tendons of the four components of quadriceps femoris unite in the lower part of the thigh to form a single strong tendon attached to the base of the patella, and some fibres continue over it to blend with the patellar ligament. The patella is a sesamoid bone in the quadriceps femoris tendon, and the patellar ligament, which extends from the patellar apex to the tibial tuberosity, is the continuation of the main tendon. The medial and lateral patellar retinacula are expansions from its borders. The suprapatellar bursa (a synovial extension of the knee joint) lies between the femur and the suprapatellar part of the quadriceps femoris tendon. The deep infrapatellar bursa lies between the patellar ligament and the proximal end of the tibia.

The arterial supply to the quadriceps femoris has been traditionally ascribed to a single branch of either the profunda femoris artery or the lateral circumflex femoral artery, the 'artery of the quadriceps'. This vessel, which may be large, occasionally arises directly from the femoral artery. However, according to Taylor and Razaboni (1994), this artery does not supply all four components of quadriceps femoris; vastus medialis is supplied directly from the femoral artery. The supply of the individual muscle components is described below; their sheaths may have an additional and variable supply.

Quadriceps femoris as a group shows little anatomical variation.

Rectus femoris

Attachments Rectus femoris is fusiform. Its superficial fibres are bipennate, the deep fibres parallel. It has a double origin on the ilium: a straight tendon arises from the anterior inferior iliac spine, and a thinner, flatter reflected tendon from a groove above the acetabulum and from the fibrous capsule of the hip joint. The two unite at an acute angle and spread into an aponeurosis that is prolonged downwards on the anterior surface of the muscle; the muscular fibres arise from this aponeurosis. The fibres end in a broad, thick aponeurosis that forms over the lower two-thirds of its posterior surface and gradually narrows

into the thick, flat tendon by which it is attached to the base of the patella. This constitutes the superficial central part of the quadriceps femoris tendon. A third head of rectus femoris may exist; it attaches deeply to the iliofemoral ligament and superficially to the tendon of gluteus minimus as it attaches on to the femur (Tubbs et al 2006). Rectus femoris may also arise from the anterior superior iliac spine, and its reflected head may be absent.

Relations Proximally, the muscle is covered by tensor fasciae latae, iliacus and sartorius. The reflected head lies beneath gluteus minimus. The capsule of the hip joint, vastus intermedius, the anterior borders of vasti lateralis and medius, the lateral circumflex femoral artery and some branches of the femoral nerve all lie deep to rectus femoris.

Vascular supply There are two main vascular pedicles: a superior and an inferior. The superior enters rectus femoris at the junction of its upper and middle thirds, and arises from the 'artery of the quadriceps'. The inferior and larger branch arises from the 'artery of the quadriceps' at about mid-thigh level and enters the muscle a few centimetres more distally. There are additional contributions from the lateral circumflex femoral artery and, less often, from the superficial circumflex iliac artery.

Vastus medialis and vastus medialis obliquus

Attachments Vastus medialis arises from the lower part of the intertrochanteric line, spiral line, medial lip of the linea aspera, proximal part of the medial supracondylar line, the tendons of adductor longus and magnus, and the medial intermuscular septum. Its fibres pass downwards and forwards at an average angle of 15° to the long axis of the femur, most of them into an aponeurosis on the deep surface of the muscle that is attached to the medial border of the patella and to the quadriceps femoris tendon. An expansion from this aponeurosis reinforces the capsule of the knee joint and is attached below to the medial condyle of the tibia.

The lowest fibres are nearly horizontal and form a bulge in the living subject, medial to the upper half of the patella. Some authors distinguish this part of the muscle as the vastus medialis obliquus, with fibres that originate largely from the tendon of adductor magnus and insert into the medial border of the patella. It plays an important role in the function of patellofemoral joint.

Relations Vastus medialis is partly covered by rectus femoris and sartorius. In the central part of the thigh it forms the lateral wall of the adductor (Hunter's) canal.

Vascular supply Vastus medialis is supplied by three branches of the femoral artery. The superior and middle branches arise, sometimes from a common trunk, proximal to the adductor canal, while the inferior arises within the canal. There may also be minor branches from the profunda femoris artery and descending genicular arteries.

Vastus lateralis

Attachments Vastus lateralis is the largest component of quadriceps femoris. It arises by a broad aponeurosis from the upper part of the intertrochanteric line, the anterior and inferior borders of the greater trochanter, the lateral lip of the gluteal tuberosity, and the proximal half of the lateral lip of the linea aspera. This aponeurosis covers the proximal three-quarters of the muscle; many additional fibres arise from its deep surface. A few fibres also arise from the tendon of gluteus maximus and the lateral intermuscular septum between vastus lateralis and the short head of biceps femoris. The muscular mass thus formed is attached to a strong aponeurosis on the deep surface of the lower part of the muscle. This narrows to a flat tendon, which is attached to the base and lateral border of the patella and blends into the compound quadriceps femoris tendon. It contributes an expansion to the capsule of the knee joint that descends to the lateral condyle of the tibia and blends with the iliotibial tract.

Relations Vastus lateralis is covered laterally by the fascia lata and the aponeurotic insertions of tensor fasciae latae and gluteus maximus. It is separated from vastus intermedius by branches of the femoral nerve and the lateral circumflex femoral artery. Posteriorly, it is separated from biceps femoris by the lateral intermuscular septum.

Vascular supply There are three main arteries of supply: the superior medial artery arises directly from the lateral circumflex femoral artery; the inferior medial (the largest of the three) from the 'artery of the quadriceps'; and the lateral from the first perforating branch given off from the profunda femoris artery.

Vastus intermedius

Attachments Vastus intermedius arises from the anterior and lateral surfaces of the upper two-thirds of the femoral shaft, and from the lower part of the lateral intermuscular septum. Its fibres end on the anterior surface of the muscle in an aponeurosis that forms the deep part of the quadriceps femoris tendon and is attached to the lateral border of the patella and the lateral condyle of the tibia.

Vastus intermedius appears to be inseparable from vastus medialis. However, when rectus femoris is reflected, a narrow cleft can be seen extending upwards from the medial border of the patella between the two muscles, sometimes as far as the lower part of the intertrochanteric line; proximal to this, the two muscles are frequently fused.

Relations Vastus intermedius is almost completely covered by the other components of quadriceps femoris, except laterally, where a small part becomes superficial behind vastus lateralis.

Vascular supply Vastus intermedius receives a lateral artery of supply, which arises from the 'artery of the quadriceps', and a medial artery that arises directly from the profunda femoris artery. Arteries to the other components of quadriceps femoris may also contribute. There is an anastomotic network deep to the muscle that supplies the suprapatellar bursa and articularis genus, and may form a collateral pathway when the femoral artery is blocked.

Articularis genus

Articularis genus is a small muscle, usually distinct from vastus intermedius but occasionally blending with it. It consists of several muscular bundles that arise from the anterior surface of the lower part of the femoral shaft and are attached to a proximal reflection of the synovial membrane of the knee joint. It is visible during knee arthroscopy when viewed looking superiorly from within the suprapatellar bursa (see [Commentary 9.3](#)).

Innervation of quadriceps femoris

Quadriceps femoris and articularis genus are supplied by the femoral nerve, L2, 3 and 4.

Actions of quadriceps femoris

Quadriceps femoris extends the knee. Rectus femoris helps to flex the thigh on the pelvis; if the thigh is fixed, it helps to flex the pelvis on the thigh. Rectus femoris can flex the hip and extend the knee simultaneously. Electromyographic studies indicate that the three vasti are not equally active in different phases of extension or rotation. There is little or no activity in quadriceps femoris during standing. Rectus femoris pulls the patella along the line of the mechanical axis of the lower limb (i.e. the line connecting the centres of the hip, knee and ankle) because its attachment is anterior to the hip joint. The remaining muscles in the group are attached to the shaft of the femur and their pull is lateral as well as proximal. An extremely important dynamic function of vastus medialis obliquus is to counter this lateral vector on the patella during knee motion. Other static factors that help are the depth of the femoral patellar surface, the buttressing effect of the lateral femoral condyle in the groove, and the restraining action of the medial patellofemoral ligament. Inadequacy of vastus medialis obliquus is a factor in producing patellar instability and pain. Strengthening of this muscle to improve patellar 'tracking' on the femur is an important part of physiotherapy regimens for patellofemoral problems.

Articularis genus retracts the synovial suprapatellar bursa proximally during extension of the leg, presumably to prevent interposition of redundant synovial folds between patella and femur.

Testing The quadriceps femoris is tested by extending the knee against resistance, in the supine position with the hip flexed.

Adductor compartment

The muscles of the adductor compartment – gracilis, pectineus, adductor longus, adductor brevis, and adductor magnus ([Fig. 80.29](#)) – have evolved, as their nerve supply suggests, from both flexor and extensor columns. All five muscles cross the hip joint but only gracilis reaches beyond the knee. They are known collectively as the adductors of the thigh, although their actions are more complex than this, e.g. acting from below they have important roles in balancing the trunk on the lower limb during walking.

Their blood supply is derived from the profunda femoris artery, medial circumflex femoral, femoral and obturator arteries. The profunda femoris artery (or sometimes the first perforator) supplies a large branch, the 'artery of the adductor muscles'.

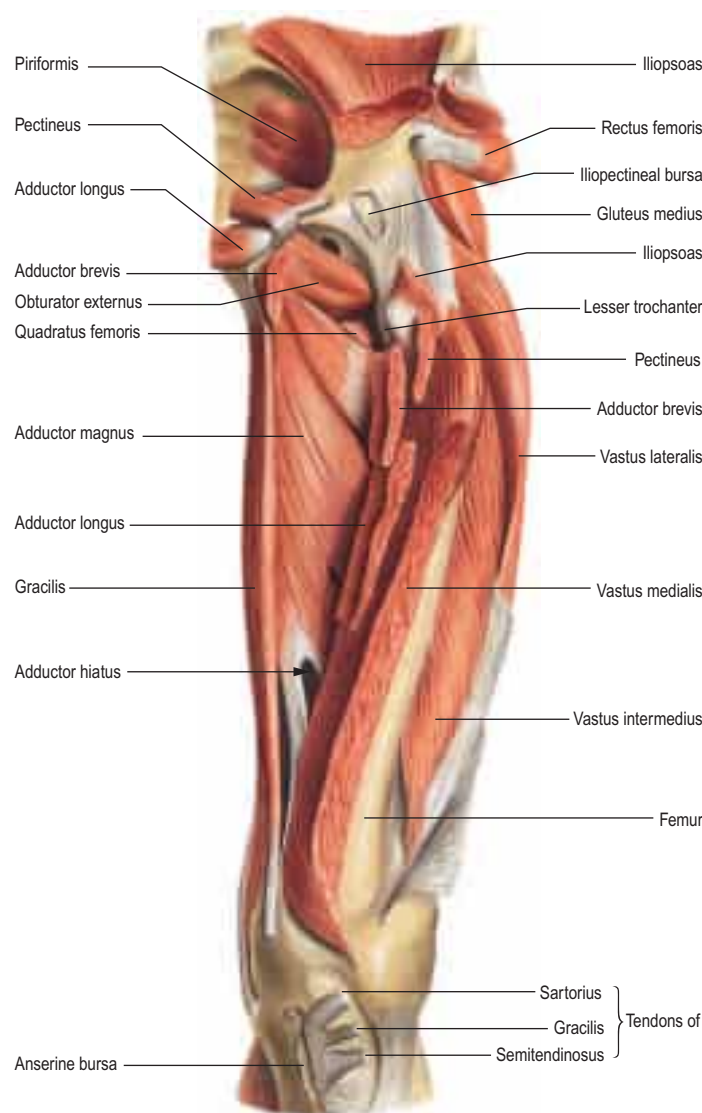


Fig. 80.29 The adductor muscles of the thigh, anterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

Gracilis

Attachments Gracilis (see [Figs 80.2, 80.29](#)) is the most superficial of the adductor group. It is thin and flat, broad above, and narrow and tapering below. It arises by a thin aponeurosis from the medial margins of the lower half of the body of the pubis, the whole of the inferior pubic ramus, and the adjoining part of the ischial ramus. The fibres descend vertically into a rounded tendon, often harvested as a knee ligament graft, which passes across the medial condyle of the femur posterior to the tendon of sartorius. It then curves around the medial condyle of the tibia, where it fans out and is attached to the upper part of the medial surface of the tibia, just below the condyle, forming part of the pes anserinus (see [Fig. 80.28](#)). A few fibres from the lower part of the tendon continue into the deep fascia of the lower leg. Often there is a slip that blends with the tendon of the medial head of gastrocnemius. Unless divided, this can lead to problems during surgical harvesting of the gracilis tendon.

Relations The muscle is covered medially by the fascia lata throughout most of its length. Deep to gracilis lie adductor brevis and adductor magnus. The tibial attachment is immediately proximal to that of semitendinosus, and the tendon of sartorius, with which it is partly blended, overlaps its upper edge. It is separated from the tibial collateral ligament of the knee joint by the anserine bursa.

Vascular supply The arterial supply to gracilis enters via its lateral surface. The main pedicle arises from the 'artery to the adductors', a branch of the profunda femoris artery, and enters at the junction of the upper and middle thirds of the muscle. The gracilis musculocutaneous flap is based on this pedicle. A less important artery enters the distal third of the muscle from the femoral artery. There is a minor proximal supply from the medial circumflex femoral artery.

Innervation Gracilis is innervated by the obturator nerve, L2 and 3.

Actions Gracilis flexes the leg and rotates it medially. It may also act as an adductor of the thigh. When the foot is fixed, gracilis rotates the femur and pelvis laterally on the tibia, and helps balance the trunk during walking.

Testing While palpating over the tendon posteromedially at the flexed knee, the knee is actively further flexed and the leg actively rotated medially. Gracilis can also be tested with the other adductors of the hip.

Adductor longus

Attachments Adductor longus (see Figs 80.2, 80.22), the most anterior of the three adductors, is a large, fan-shaped muscle that lies in the same plane as pectineus. It arises by a narrow tendon with a flattened (sometimes C-shaped) cross-section, which is attached to the front of the pubis in the angle between the crest and the symphysis. It expands into a broad fleshy belly that descends posterolaterally and inserts by an aponeurosis into the linea aspera in the middle third of the femur, between vastus medialis and adductors magnus and brevis, usually blending with all of them. Its proximal attachment is vulnerable to overload from sporting activity: this is one cause of sport-related groin pain.

Adductor longus is occasionally double.

Relations Anterior to adductor longus are the spermatic cord, fascia lata (which separates it from the long saphenous vein) and, near its attachment, the femoral artery and vein and sartorius. Posterior to it are adductor brevis and adductor magnus, the anterior branch of the obturator nerve and, near its attachment, the deep artery and vein of the thigh. Lateral and medial to adductor longus are, respectively, pectineus and gracilis.

Vascular supply The main supply to adductor longus is to the central part of the muscle from the 'artery to the adductors', a branch of the profunda femoris artery. There is an additional proximal supply from the medial circumflex femoral artery, and a more distal supply from the femoral artery and sometimes the descending genicular artery.

Innervation Adductor longus is innervated by the anterior division of the obturator nerve, L2, 3 and 4.

Actions The actions of the named adductors as a group are discussed below, after each muscle has been described.

Testing The adductors are tested as a group by adduction of the thigh against resistance, in the supine position and with the knee extended.

Adductor brevis

Attachments Adductor brevis (see Fig. 80.2A; Fig. 80.25B) lies posterior to pectineus and adductor longus. It arises by a narrow attachment from the external aspect of the body and inferior ramus of the pubis, between gracilis and obturator externus. Like adductor longus, it is somewhat triangular, and expands as it descends posterolaterally to insert via an aponeurosis into the femur, along a line from the lesser trochanter to the linea aspera, and on the upper part of the linea immediately behind pectineus and the upper part of adductor longus.

Adductor brevis often has two or three separate parts, or may be integrated into adductor magnus.

Relations Anteriorly lie pectineus, adductor longus, the profunda femoris artery and the anterior branch of the obturator nerve; posteriorly are adductor magnus and the posterior branch of the obturator nerve. The upper border of adductor brevis is related to the medial circumflex femoral artery, obturator externus, and the conjoint tendon of psoas major and iliacus. Its lower border is related to gracilis and adductor magnus. The second perforating artery, or first and second perforating arteries, pierces it near its femoral attachment.

Vascular supply The vascular supply to adductor brevis is variable. Usually the main supply is directly from the profunda femoris artery distally and from the 'artery to the adductors' more proximally. There is an additional proximal supply from the medial circumflex femoral artery. The deep surface receives branches from the obturator artery.

Innervation Adductor brevis is innervated by the obturator nerve, L2 and 3.

Actions The actions of the named adductors as a group are discussed below, after each muscle has been described.

Testing Adductor brevis is tested in the same way as adductor longus.

Adductor magnus

Attachments Adductor magnus (see Figs 80.2, 80.25A, 80.29, 80.30), a massive triangular muscle, arises from a small part of the inferior ramus of the pubis, from the conjoint ischial ramus, and from the inferolateral aspect of the ischial tuberosity. The short, horizontal fibres from the pubic ramus are inserted into the medial margin of the gluteal tuberosity of the femur, medial to gluteus maximus; this part of the muscle, in a plane anterior to the rest, is sometimes called adductor minimus. The fibres from the ischial ramus fan out downwards and laterally, to insert via a broad aponeurosis into the linea aspera and the proximal part of the medial supracondylar line. The medial part of the muscle, composed mainly of fibres from the ischial tuberosity, is a thick mass that descends almost vertically, and ends in the lower third of the thigh in a rounded tendon, which can be palpated proximal to its attachment to the adductor tubercle on the medial condyle of the femur. The tendon is connected by a fibrous expansion to the medial supracondylar line.

The long, linear attachment of the muscle is interrupted by a series of osseo-aponeurotic openings, bridged by tendinous arches attached to the bone. The upper four are small and transmit the perforating branches and the termination of the profunda femoris artery. The lowest is large and allows the femoral vessels to cross to the popliteal fossa.

The vertical, ischiocondylar part of the muscle varies in its degree of separation from the rest. The upper border of adductor magnus may fuse with quadratus femoris.

Relations Anteriorly lie pectineus, adductor brevis and adductor longus, the femoral and deep femoral vessels, and the posterior branch of the obturator nerve. A bursa separates the proximal part of the muscle from the lesser trochanter of the femur. Posteriorly are the sciatic nerve, gluteus maximus, biceps femoris, semitendinosus and semimembranosus. The superior border is parallel with quadratus femoris, and the transverse branch of the medial circumflex femoral artery passes between the muscles. The medial border is related to gracilis, sartorius and the fascia lata.

Vascular supply As expected from its position, adductor magnus is supplied from both its anterior and its posterior aspects. The contribution from the anterior compartment is the more important. The profunda femoris artery, obturator artery and femoral artery all contribute; the main supply comes directly from the distal part of the profunda femoris artery. Distally, there may be contributions from the femoral and descending genicular arteries. Posteriorly, there are branches from the medial circumflex femoral, the first and second perforating and the popliteal vessels.

Innervation Adductor magnus is composite and is doubly innervated by the obturator nerve and by the tibial division of the sciatic nerve (L2, 3 and 4), which supplies the ischiocondylar part. Both nerves are derived from anterior divisions in the lumbosacral plexus, indicating a primitive flexor origin for both parts of the muscle.

Actions of the adductors

Extensive or forcible adduction of the femur is not often required. Although the adductors can act in this way, they more commonly act as synergists in the complex patterns of gait activity, and to some degree as controllers of posture. They are active during flexion and extension of the knee. Adductors magnus and longus are probably medial rotators of the thigh. The adductors are inactive during adduction of the abducted thigh in the erect posture (when gravity assists), but active in other postures, such as the supine position, or during adduction of the flexed thigh when standing. They are also active during flexion (longus) and extension (magnus) of the thigh at the hip joint. In symmetrical easy standing their activity is minimal.

Testing of the adductors The adductors are usually tested as a group by adduction of the thigh against resistance, in the supine position with the knee extended. The tendon of adductor magnus can be felt just proximal to the adductor tubercle on the medial condyle of the femur. Clinical testing of the other actions mentioned above is not feasible for the individual muscles.

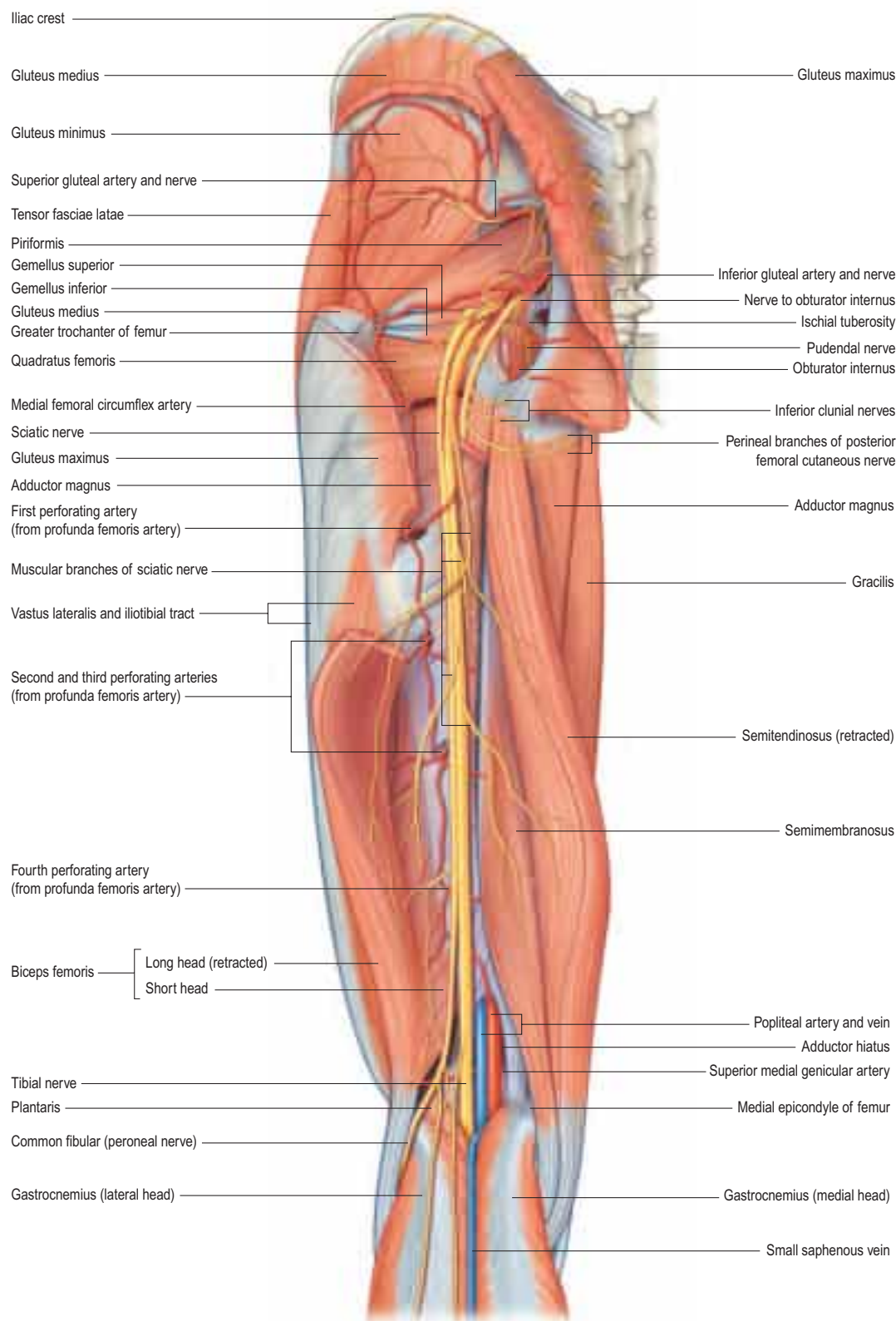


Fig. 80.30 The posterior thigh and gluteal regions with associated neurovascular structures. (With permission from Netter, FH, Atlas of Human Anatomy, 5th ed, Elsevier, Saunders. Copyright 2011.)

Pectineus

Attachments Pectineus (see Fig. 80.22) is a flat, quadrangular muscle in the femoral triangle. It may also be considered as part of the anterior compartment of the thigh. It arises from the pecten pubis, from the bone anterior to it between the iliopubic ramus and the pubic tubercle, and from the fascia on its own anterior surface. The fibres descend, initially posteromedially and then posterolaterally, to be attached along a line from the lesser trochanter to the linea aspera. Proximally, it may be partially or wholly attached to the capsule of the hip joint.

Relations Pectineus is related anteriorly to the deep lamina of the fascia lata, which separates it from the femoral vessels and long saphenous vein; posteriorly, to the capsule of the hip joint, adductor brevis, obturator externus and the anterior branch of the obturator nerve;

laterally, to psoas major and the medial circumflex femoral vessels; and medially, to the lateral margin of adductor longus.

Vascular supply The main arterial supply to pectineus is derived from the medial circumflex femoral artery, which enters the superficial surface of the muscle. There may be a branch from the femoral artery more proximally, and a deep supply from the obturator artery.

Innervation Pectineus is innervated by the femoral nerve, L2 and 3, and the accessory obturator nerve, L3 (when present). Occasionally, it receives a branch from the obturator nerve. The muscle may be incompletely divided into dorsal and ventral layers, which are supplied by the obturator and femoral (or rarely, accessory obturator) nerves, respectively.

Actions Pectineus adducts the thigh and flexes it on the pelvis.

Testing The action of pectineus cannot be tested clinically in isolation.

Adductor canal

The adductor canal (Hunter's canal; subsartorial canal) is a trough-shaped intermuscular tunnel occupying the distal two-thirds of the medial aspect of the thigh (see Fig. 80.2B). It starts at the apex of the femoral triangle and extends distally as far as the distal attachment of the tendon of adductor magnus. It is triangular in section and is bounded anterolaterally by vastus medialis; posteromedially by adductor longus; and distal to adductor longus, by adductor magnus. Its anteromedial boundary (often referred to as the roof) is a strong and dense fascia that extends from the medial surface of vastus medialis to the medial edge of the adductors longus and magnus, overlapping in its stride the femoral vessels in the adductor canal. This fascia, on account of being overlain by sartorius, is termed the subsartorial fascia.

The adductor canal contains the femoral artery and vein, the descending genicular and muscular branches of the femoral artery and their corresponding venous tributaries, the saphenous nerve, and the nerve to vastus medialis (until it enters its muscle). The femoral vessels pass from the adductor canal into the popliteal fossa via the adductor hiatus, an opening in the tendon of adductor magnus adjacent to the femoral shaft, two-thirds of the way down the adductor canal.

Posterior compartment

The posterior thigh muscles, biceps femoris, semitendinosus and semimembranosus, are colloquially termed the 'hamstrings'. They cross both hip and knee joints, and integrate extension at the hip with flexion at the knee. As the muscles span the back of the knee, they form the proximal lateral and medial margins of the popliteal fossa. The actions of these muscles and their clinical anatomy will be considered as a group after they have been described individually.

The muscles of the posterior compartment (see Figs 80.2, 80.25A, 80.27) receive their blood supply from the perforating branches of the profunda femoris artery, most importantly through the first perforator given off from this artery. This vessel has important anastomoses with the inferior gluteal artery (on or within semitendinosus) and with the medial circumflex femoral artery, while the third perforator anastomoses with the superior medial genicular artery within the short head of biceps femoris. This anastomotic chain forms an important collateral arterial pathway when the femoral artery is blocked.

Semitendinosus

Attachments Semitendinosus (see Figs 80.2B, 80.30), notable for the length of its tendon, lies posteromedial in the thigh. It arises from an inferomedial impression on the upper area of the ischial tuberosity (see Fig. 80.5), by a tendon it shares with the long head of biceps femoris, and from an aponeurosis connecting the adjacent surfaces of the two muscles for 7.5 cm from their common origin. The belly is fusiform and ends a little below mid-thigh in a long, rounded tendon that runs on the posterior surface of semimembranosus. The tendon curves around the medial condyle of the tibia, passes over the tibial collateral ligament (from which it is separated by the anserine bursa), and inserts into the upper part of the medial surface of the tibia behind the attachment of sartorius and distal to that of gracilis (see Fig. 80.28). At its termination, it is united with the tendon of gracilis and gives off a prolongation to the deep fascia of the leg and to the medial head of gastrocnemius. A tendinous intersection is usually present near the midpoint of the muscle, which may also receive a muscular slip from the long head of biceps femoris. These connections with the medial head of gastrocnemius and biceps femoris can cause difficulty when harvesting the tendon surgically for a graft.

Relations Semitendinosus lies on semimembranosus throughout its length. The relations of the distal part of the muscle are described above and with the pes anserinus.

Vascular supply The two main arteries of supply to semitendinosus are superior and inferior. The superior is derived from either the medial circumflex femoral artery or the first perforating artery. The inferior and larger branch arises from the first perforator distal to the origin of the superior branch. An accessory supply at the ischial attachment is derived from the inferior gluteal artery, and at the tibial attachment from the inferior medial genicular artery.

Innervation Semitendinosus is innervated by the sciatic nerve, L5, S1 and 2, through its tibial division (see Fig. 80.30).

Semimembranosus

Attachments Semimembranosus (see Figs 80.2, 80.30), so named because of the flattened form of its upper attachment, is posteromedial in the thigh. It arises by a long, flat tendon from a superolateral impression on the ischial tuberosity (see Fig. 80.5). Inferomedially, the tendinous fibres intermingle to some extent with those of biceps femoris and semitendinosus. The tendon receives, from the ischial tuberosity and ramus, two fibrous expansions that flank adductor magnus. It then broadens and descends deep to semitendinosus and the long head of biceps femoris. Muscle fibres arise from the tendon at about mid-thigh and converge to a second aponeurosis on the posterior aspect of the lower part of the muscle, which tapers to the heavy, rounded tendon of the distal attachment. The tendon divides at the level of the knee into five components. The main one is attached to a tubercle (sometimes called the tuberculum tendinis) on the posterior aspect of the medial tibial condyle. The others are: a series of slips to the medial margin of the tibia, immediately behind the tibial collateral ligament; a thin fibrous expansion to the fascia over popliteus; a cord-like tendon to the inferior lip and adjacent part of the groove on the back of the medial tibial condyle, deep to the tibial collateral ligament; and a strong expansion that passes obliquely upwards to the femoral intercondylar line and lateral femoral condyle, and forms much of the oblique popliteal ligament of the knee joint.

Semimembranosus varies considerably in size and may be absent. It may be double, and arise mainly from the sacrotuberous ligament. Slips to the femur or to adductor magnus may occur.

Relations Semimembranosus overlaps the popliteal vessels and is itself partly overlapped by semitendinosus throughout its extent (see Fig. 80.2). Its deep surface lies on adductor magnus. The sciatic nerve lies laterally and surprisingly close to the surface. The distal end of the muscle partially overlies the medial head of gastrocnemius before passing anteromedially to it. An important bursa lies between the semimembranosus tendon and gastrocnemius, and often communicates with the knee joint and with a smaller bursa between the tendon and the posterior tibial margin.

Vascular supply Semimembranosus is supplied from the perforating arteries, usually from all, though predominantly from the first. Sometimes the predominant artery arises from the fourth perforator. A branch of the femoral or popliteal artery supplies the distal part of the muscle, and there may be a contribution from the inferior gluteal artery at the proximal attachment.

Innervation Semimembranosus is innervated by the sciatic nerve, L5, S1 and 2, through its tibial division.

Biceps femoris

Attachments Biceps femoris (see Figs 80.2, 80.30) occupies a posterolateral position in the thigh. It has two proximal attachments. One, the long head, arises from an inferomedial impression on the upper area of the ischial tuberosity (see Fig. 80.5), via a tendon that it shares with semitendinosus, and from the lower part of the sacrotuberous ligament. The other, the short head, arises from the lateral lip of the linea aspera, between adductor magnus and vastus lateralis. This attachment extends proximally almost to gluteus maximus and distally along the lateral supracondylar line to within 5 cm of the lateral femoral condyle, and from the lateral intermuscular septum. The long head forms a fusiform belly that descends laterally across the sciatic nerve. The fibres end in an aponeurosis that covers the posterior surface of the muscle. This aponeurosis receives on its deep surface the fibres of the short head, and gradually narrows to a tendon (the lateral hamstring). The main part of the tendon splits round the fibular collateral ligament and is attached to the head of the fibula. The remainder splits into three laminae. The intermediate lamina fuses with the fibular collateral ligament, while the others pass superficial and deep to the ligament to attach to the lateral condyle of the tibia.

The short head may be absent. Additional slips may arise from the ischial tuberosity, linea aspera or medial supracondylar line.

Relations Proximally, biceps femoris is covered by gluteus maximus, but elsewhere it lies relatively superficially. Deep to it lie semimembranosus proximally, and the sciatic nerve, adductor magnus and the lateral head of gastrocnemius more distally. Semitendinosus and semimembranosus lie medially. A bursa may lie between the tendon and the fibular collateral ligament. The common fibular nerve descends along the medial border of the tendon of biceps femoris, separating it distally from the lateral head of gastrocnemius.

As a guide during surgery, it is safest to find a nerve proximally and dissect it free distally. The common fibular nerve is found emerging posterior to the biceps femoris tendon, which is a useful guide to locating the nerve and avoiding iatrogenic injury. The nerve is adherent to the tendon; if part of the fibular head or the attachment of the tendon of biceps femoris, usually with the insertion of the fibular collateral ligament, is avulsed, then the tendon will exert proximal traction on the nerve, which may contribute to common fibular nerve traction injury. When the lateral meniscus is sutured arthroscopically, care must be taken that all needle passages are anterior to biceps femoris. A proximal fibular fracture may result in the nerve being trapped in the fracture line.

Vascular supply The long head of biceps femoris is supplied by the first and second perforating arteries. There are accessory supplies at the ischial attachment from the inferior gluteal and medial circumflex femoral arteries, and in the distal quarter from the superior lateral genicular artery. The short head is supplied superiorly by the second or third perforating artery and inferiorly by the superior lateral genicular artery.

Innervation Biceps femoris is innervated by the sciatic nerve, L5, S1 and 2: the long head through the tibial division and the short head through the common fibular division, a distribution that reflects the composite derivation from flexor and extensor musculature.

Actions of posterior thigh muscles

Acting from above, the posterior thigh muscles flex the knee. Acting from below, they extend the hip joint, pulling the trunk upright from a stooping posture against the influence of gravity, biceps femoris being the main agent. When the knee is semi-flexed, biceps femoris can act as a lateral rotator and semimembranosus and semitendinosus as medial rotators of the lower leg on the thigh at the knee. When the hip is extended, biceps femoris is a lateral rotator and semimembranosus and semitendinosus are medial rotators of the thigh. As is the case with quadriceps femoris, the adductors and gluteus maximus, the hamstrings are quiescent in easy symmetrical standing. However, any action that takes the centre of gravity in front of a transverse axis through the hip joints, e.g. forward reaching, forward sway at the ankle joints or forward bending at the hips, is immediately accompanied by strong contraction of the hamstrings. This is in marked contrast to gluteus maximus, which contracts only when there is a call for powerful extension at the hip joint.

When the knee is flexed against resistance, the tendon of biceps femoris can be felt lateral to the popliteal fossa. Medial to the fossa, the tendons of gracilis (which is the more medial) and semitendinosus stand out sharply. The semimembranosus tendon is just palpable in the interval between them (and also by deep pressure from a 'pincer' grip beyond their margins). There is some evidence that semimembranosus, semitendinosus and biceps femoris, although they cross both hip and knee joints, may produce movement at one of these joints without resisting antagonists at the other. Usually, however, each of these muscles contracts as a whole, and whether or not movement takes place at hip or knee is determined by other muscles that act as fixators of these joints.

Testing The posterior thigh muscles are tested clinically by active knee flexion against resistance in the supine or prone position. In the prone position, the individual hamstring tendons can be identified more easily.

When relaxed, these muscles show considerable variation in length; in some individuals, the muscles are so short as to impose a serious limitation on flexion of the trunk at the hip joints when the knees are kept extended. Movements such as stooping must then be accomplished by flexing the vertebral column or squatting. Perhaps the need for more stress on the lumbar spine accounts for the occurrence of hamstring tightness in adolescents who develop spondylolysis. These muscles are prone to tearing, which may be related to the relative complexity of muscles that cross two joints, or to the fact that they are subject to large forces including eccentric contractions.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Iliac vessels and external iliac arteries and veins

The iliac vessels are described on [page 1224](#), and the external iliac arteries and veins on [pages 1227](#) and [1228](#), respectively.

ARTERIES

Gluteal vessels

The superior and inferior gluteal arteries (see [Fig. 80.30](#)) are described on [pages 1226](#) and [1227](#), respectively.

Arteria comitans nervi ischiadici (artery to sciatic nerve)

The arteria comitans nervi ischiadici is a direct or indirect branch of the internal iliac artery, and runs on the surface of, or within, the sciatic nerve. It represents the primitive axial artery of the lower limb. The artery is usually a very small vessel; occasionally, it persists as a large vessel, in which case the femoral artery is correspondingly reduced in size. The artery may participate in collateral circulatory pathways.

Femoral artery

The femoral artery (see [Figs 78.4, 80.2–80.3; Figs 80.31–80.32](#)) is a continuation of the external iliac artery ([Benninger 2014](#)). It begins behind the inguinal ligament, midway between the anterior superior

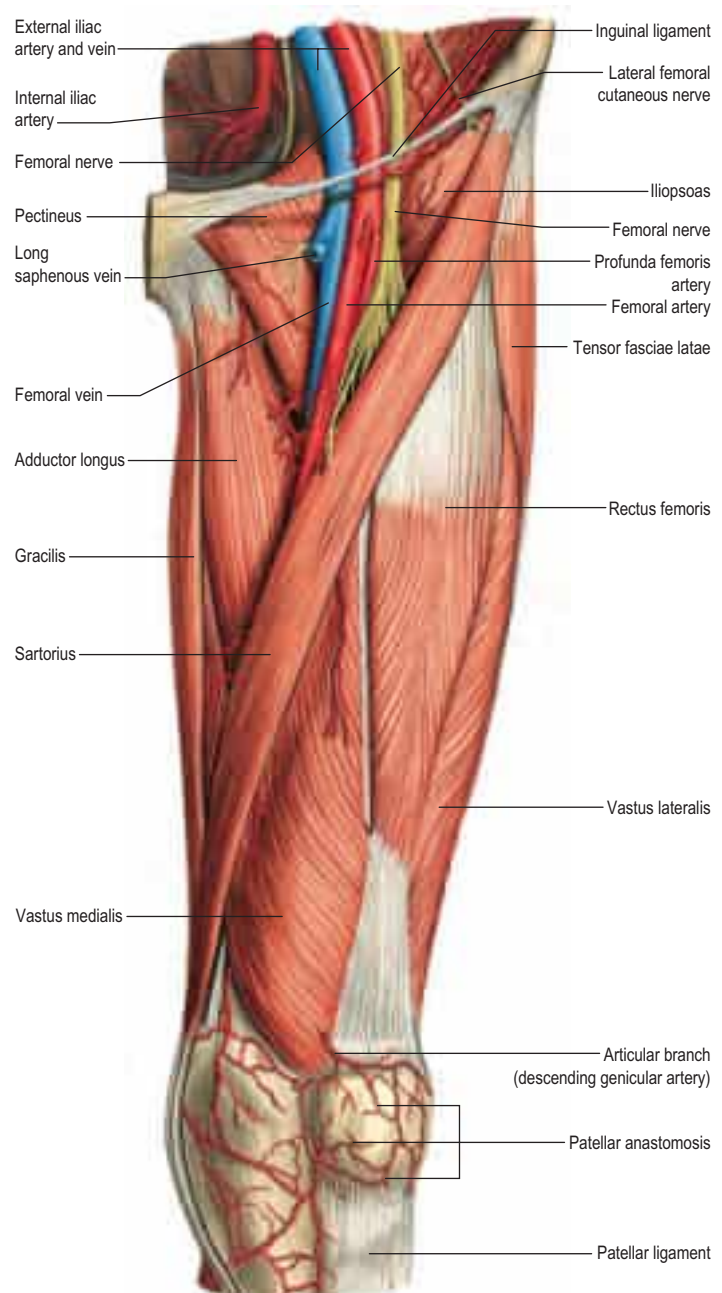


Fig. 80.31 The left femoral triangle. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

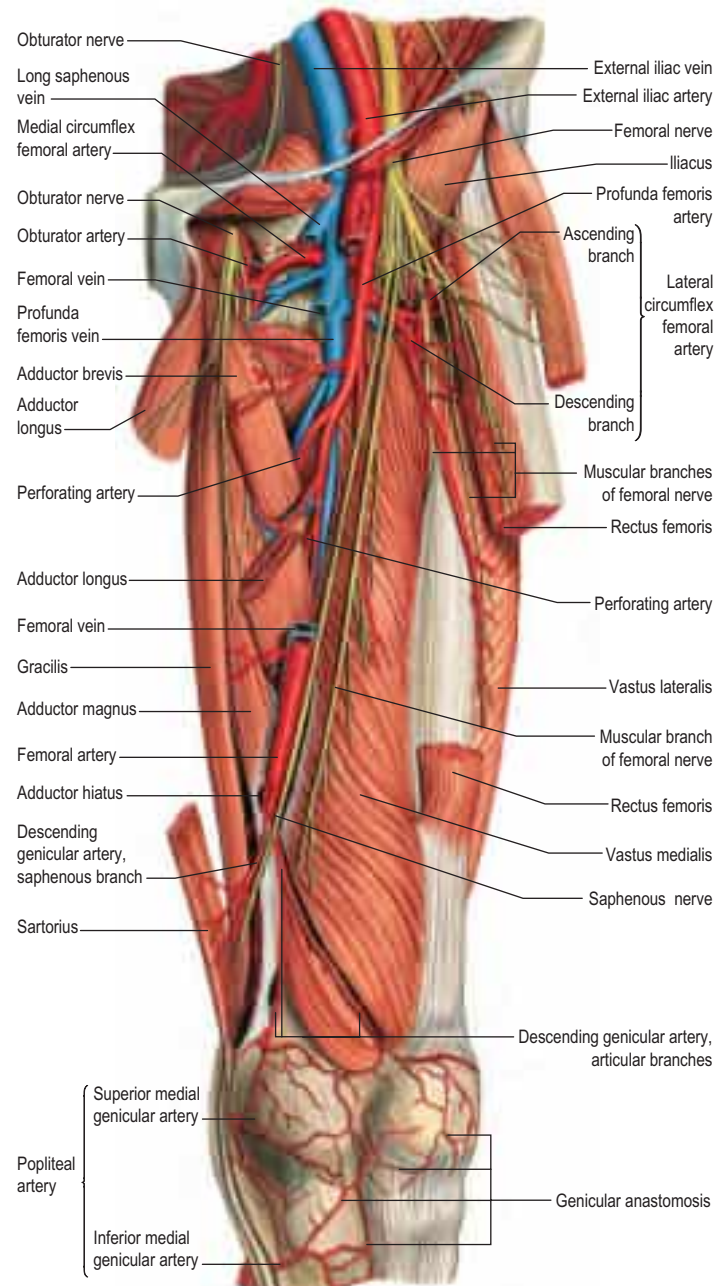


Fig. 80.33 The neurovascular structures of the anterior thigh. This is a deeper dissection than is shown in [Figure 80.32](#). (With permission from Waschke J, Paulsen F (eds), *Sobotta Atlas of Human Anatomy*, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

to the medial side. Posteriorly, the posterior wall of the femoral sheath separates the artery from the tendons of psoas, pectineus and adductor longus. The artery is separated from the hip joint by the tendon of psoas major, from pectineus by the femoral vein and deep femoral vessels, and from adductor longus by the femoral vein. Proximally, the nerve to pectineus passes medially behind the artery. The femoral nerve lies lateral to the artery, outside the femoral sheath. The femoral vein is medial to the artery in the proximal part of the triangle and becomes posterior distally at the apex.

Within the adductor canal, the artery is covered by skin, subcutaneous tissue, deep fascia, sartorius and the fibrous roof (subsartorial fascia) of the canal. The saphenous nerve lies at first lateral, then anterior and finally medial to the artery. Adductor longus and adductor magnus are posterior, vastus medialis and its nerve are anterolateral, and the femoral vein is also posterior proximally but it becomes lateral distally (see Figs 80.31–80.32; Fig. 80.33).

Compression of the femoral artery is most effective just distal to the inguinal ligament, where it is superficial and separated from the bone (iliopubic ramus) only by the psoas tendon. For this reason, arterial injury proximal to the inguinal ligament, such as laceration by a knife, cannot be controlled simply by compression. Gaining proximal control of bleeding involves major exposure of the more proximal arteries as a life-saving manoeuvre.

Anterior to the artery in the femoral triangle are the skin, subcutaneous tissue, superficial inguinal lymph nodes, fascia lata, femoral sheath, superficial circumflex iliac vein (crossing in the subcutaneous tissue) and the femoral branch of the genitofemoral nerve (which is at first lateral and then anterior). Near the apex of the triangle, the medial femoral cutaneous nerve crosses in front of the artery from the lateral

Branches

The branches of the femoral artery include the superficial epigastric, superficial circumflex iliac, superficial external pudendal, deep external pudendal arteries, muscular branches, the profunda femoris artery and the descending genicular artery (see Fig. 78.4).

Superficial epigastric artery

The superficial epigastric artery arises anteriorly from the femoral artery approximately 1 cm distal to the inguinal ligament, and traverses the cribriform fascia to ascend anterior to the ligament and run in the abdominal subcutaneous tissue almost to the umbilicus. It supplies the superficial inguinal lymph nodes, subcutaneous tissue and skin, and anastomoses with branches of the inferior epigastric artery and its contralateral fellow.

Superficial circumflex iliac artery

The superficial circumflex iliac artery is the smallest superficial branch of the femoral artery and arises near or with the superficial epigastric artery. It usually emerges through the fascia lata, lateral to the saphenous opening, and turns laterally distal to the inguinal ligament towards the anterior superior iliac spine. It supplies the skin, subcutaneous tissue and superficial inguinal lymph nodes, anastomosing with the deep circumflex iliac, superior gluteal and lateral circumflex femoral arteries.

The superficial circumflex iliac artery is the basis for the important axial-pattern pedicled groin skin flap. Free flaps based on the vessel may also be raised.

Superficial external pudendal artery

The superficial external pudendal artery arises medially from the femoral artery, close to the preceding branches. Emerging from the cribriform fascia, it passes medially, usually deep to the long saphenous vein, across the spermatic cord (or round ligament) to supply the lower abdominal, penile, scrotal or labial skin, and anastomoses with branches of the internal pudendal artery.

Deep external pudendal artery

The deep external pudendal artery passes medially across pectineus and anterior or posterior to adductor longus, covered by fascia lata, which it pierces to supply the skin of the perineum and scrotum or labium majus. Its branches anastomose with the posterior scrotal or labial branches of the internal pudendal artery.

Muscular branches

These branches supply sartorius, vastus medialis and the adductors.

Profunda femoris artery (deep artery of the thigh)

The profunda femoris artery (see Figs 80.2B, 80.25, 80.32–80.33) is a large branch that arises posterolaterally from the femoral artery about 3.5 cm distal to the inguinal ligament. At first lateral to the femoral artery, it spirals posterior to this and the femoral vein to reach the medial side of the femur. It passes between pectineus and adductor longus, then between the latter and adductor brevis, before it descends between adductor longus and adductor magnus. It pierces adductor magnus and anastomoses with the upper muscular branches of the popliteal artery. This terminal part is sometimes named the fourth perforating artery.

The profunda femoris artery is the main supply to the adductor, extensor and flexor muscles, and also anastomoses with the internal and external iliac arteries above and with the popliteal artery below. It is a major collateral artery of the lower limb in patients with femoropopliteal arterial occlusive disease (see Fig. 80.33) (Shakeri et al 2010).

The profunda femoris artery sometimes arises from the posterior aspect of the femoral artery, and more rarely from the medial aspect. If it arises posteriorly, it may cross anterior to the femoral vein and then pass backwards around its medial side.

Relations

Posteriorly, in proximodistal order, lie iliacus, pectineus, adductor brevis and adductor magnus. Anteriorly, the femoral and deep femoral veins, and distally, adductor longus, separate the profunda femoris artery from the femoral artery. Laterally, vastus medialis separates the proximal part of the artery from the femur.

Branches

The profunda femoris artery gives off the lateral and medial circumflex arteries in the proximal thigh, and perforating and muscular branches more distally.

Lateral circumflex femoral artery The lateral circumflex femoral artery (see Figs 80.32–80.33) is a laterally running branch given off near the root of the profunda femoris artery. It passes between the divisions of the femoral nerve, posterior to sartorius and rectus femoris, and divides into ascending, transverse and descending branches. The ascending branch ascends along the intertrochanteric line, under tensor fasciae latae, lateral to the hip joint. It anastomoses with the superior gluteal and deep circumflex iliac arteries, supplying the greater trochanter, and, with branches of the medial circumflex femoral, forms an anastomotic ring round the femoral neck, from which the femoral neck and head are supplied. The ascending branch forms the pedicle of the tensor fasciae latae musculocutaneous flap. The descending branch descends posterior to rectus femoris, along the anterior border of vastus lateralis, which it supplies; a long branch descends in vastus lateralis to the knee, anastomosing with the superior lateral genicular branch of the popliteal artery, accompanied by the nerve to vastus lateralis. The fasciocutaneous perforators that arise from the descending branch supply the antero-lateral thigh fasciocutaneous flap. The transverse branch, the smallest, passes laterally anterior to vastus intermedius and pierces vastus lateralis to wind round the femur, just distal to the greater trochanter. It anastomoses with the medial circumflex, inferior gluteal and first perforating arteries, forming a cruciate anastomosis.

The lateral circumflex femoral artery may arise from the femoral artery or as a common trunk with the profunda femoris artery and the medial circumflex femoral artery. The ascending and descending branches may occasionally arise separately (Siddharth et al 1985).

Medial circumflex femoral artery The medial circumflex femoral artery (see Figs 80.32–80.33) usually originates from the posteromedial aspect of the profunda femoris artery, but sometimes originates from the femoral artery itself or as a common trunk with either the profunda femoris artery or the lateral circumflex femoral artery or both (Siddharth et al 1985). It supplies the adductor muscles and curves medially round the femur between pectineus and psoas major and then obturator externus and adductor brevis, finally appearing between quadratus femoris and the upper border of adductor magnus, dividing into transverse and ascending branches. The transverse branch takes part in the cruciate anastomosis. The ascending branch ascends on the tendon of obturator externus, anterior to quadratus femoris, to the trochanteric fossa, where it anastomoses with branches of the gluteal and lateral circumflex femoral arteries. An acetabular branch at the proximal edge of adductor brevis enters the hip joint under the transverse acetabular ligament with one from the obturator artery. It supplies the fat in the fossa and reaches the femoral head along its ligament. For details of the blood supply of the proximal end of the femur, consult Crock (1996).

Perforating arteries The perforating arteries (see Fig. 80.25) perforate the femoral attachment of adductor magnus to reach the flexor aspect of the thigh. There are three perforating branches, and the terminal part of the profunda femoris artery itself becomes the fourth perforator. They pass close to the linea aspera under small tendinous arches and give off muscular, cutaneous and anastomotic branches. Diminished, they pass deep to the short head of biceps femoris (the first usually through the attachment of gluteus maximus), traverse the lateral intermuscular septum and enter vastus lateralis. The first arises proximal, the second anterior, and the third distal to adductor brevis. The first perforating artery passes back between pectineus and adductor brevis (sometimes through the latter), pierces adductor magnus near the linea aspera to supply adductor brevis, adductor magnus, biceps femoris and gluteus maximus, and anastomoses with the inferior gluteal, medial and lateral circumflex femoral and second perforating arteries. The larger second perforating artery, often arising with the first, pierces the attachments of adductor brevis and magnus, divides into the ascending and descending branches supplying the posterior thigh muscles and anastomoses with the first and third perforating arteries. The femoral nutrient artery usually arises from it. When two nutrient arteries exist, they usually arise from the first and third perforators. The third perforating artery starts distal to adductor brevis, pierces the attachment of adductor magnus and divides into branches to the posterior thigh muscles. It anastomoses proximally with the perforating arteries, and distally with the end of the profunda femoris artery and muscular branches of the popliteal. The femoral nutrient artery may arise from it. Side branches of the diaphysal nutrient and other branches of the profunda femoris artery provide subsidiary cortical arteries.

The perforating arteries form a double chain of anastomoses, initially in the adductor muscles and subsequently near the linea aspera.

Muscular branches Numerous muscular branches arise from the profunda femoris artery. Some end in the adductors, and others pierce adductor magnus, supply the flexors and anastomose with the medial circumflex femoral artery and superior muscular branches of the popliteal artery. The profunda femoris artery is thus the main supply to the thigh muscles.

Descending genicular artery

The descending genicular artery (see Figs 82.1, 80.32–80.33), the distal branch of the femoral artery, arises just proximal to the adductor opening and immediately supplies a saphenous branch. It then descends in vastus medialis, anterior to the tendon of adductor magnus, to the medial side of the knee, where it anastomoses with the superior medial genicular artery. Muscular branches supply vastus medialis and adductor magnus, and give off articular branches that anastomose round the knee joint. One articular branch crosses above the femoral patellar surface, forming an arch with the superior lateral genicular artery and supplying the knee joint. The saphenous branch (saphenous artery) emerges distally through the roof of the adductor canal to accompany the saphenous nerve to the medial side of the knee. It passes between sartorius and gracilis, and supplies the skin of the proximomedial area of the leg; it anastomoses with the inferior medial genicular artery.

Arterial anastomoses around the hip

In the fetus, a peri-acetabular vascular circle formed by the superior and inferior gluteal, internal pudendal and obturator arteries reduces the risk of bony necrosis of this region in children. However, there is a zone at the anterior portion of the acetabulum where the blood supply is less abundant (Damsin et al 1992).

Anastomoses on the back of the thigh

An important chain of anastomoses extends on the back of the thigh from the gluteal region to the popliteal fossa. It is formed in proximodistal order by anastomoses between the gluteal arteries and terminal branches of the medial circumflex femoral artery; the circumflex femoral arteries and the first perforating artery; the perforating arteries themselves; and between the fourth perforating artery and the superior muscular branches of the popliteal artery (see Fig. 80.18). The trochanteric and cruciate anastomoses are the proximal elements of this chain.

Trochanteric anastomosis

The so-called trochanteric anastomosis lies near the trochanteric fossa of the femur and is an anastomosis between the ascending branch of the medial circumflex femoral artery and descending branches of the superior and inferior gluteal arteries (see Fig. 80.18). The lateral circumflex femoral artery and the first perforating artery from the profunda femoris artery may also contribute, creating an extracapsular 'arterial ring of the femoral neck' (Crock 1996). Branches from this ring, the retinacular vessels, pierce the capsule and ascend along the femoral neck to give the main blood supply to the head of the femur.

Cruciate anastomosis

The cruciate anastomosis lies at the level of the lesser trochanter, near the lower edge of the femoral attachment of quadratus femoris, and is an anastomosis between the transverse branches of the medial and lateral circumflex femoral arteries, a descending branch of the inferior gluteal artery and an ascending branch from the first perforating artery.

Collateral circulation in proximal femoral artery occlusion

After occlusion of the femoral artery proximal to the origin of the profunda femoris artery, five main anastomotic channels are available. These are between branches of the superior and inferior gluteal arteries, the medial and lateral circumflex femoral arteries and the first perforating branch of the profunda femoris artery; the obturator branch of the internal iliac artery and the medial circumflex femoral artery; the internal pudendal branch of the internal iliac artery and the superficial and deep external pudendal branches of the femoral artery; a deep circumflex iliac branch of the external iliac artery, the lateral circumflex femoral branch of the deep artery of the thigh and the superficial circumflex iliac branch of the femoral artery; and the inferior gluteal branch of the internal iliac artery and perforating branches of the profunda femoris artery.

Perforator flaps in the hip and thigh

Gluteal region

The gluteal region of the lower limb has an average of 21 perforators, which arise from three main source arteries: namely, the superior and inferior gluteal and the internal pudendal arteries (see Fig. 78.7). The flaps based on these perforators are used as free flaps for breast reconstruction and as local flaps for covering defects in the sacral and perineal region.

Hip and thigh region

The hip and thigh region has six source arteries and an average of 50 arterial perforators. The thigh can be divided into four areas: antero-medial; anterolateral and trochanteric; posteromedial; and posterolateral. The perforators that supply the anteromedial thigh are derived from the femoral artery, and those for the anterolateral thigh are derived from branches of the lateral circumflex femoral artery. Perforators that supply the skin over the posteromedial and posterolateral thigh regions are derived from the profunda femoris artery and the popliteal arteries. Skin flaps based on the superficial circumflex iliac, superficial external pudendal and superficial inferior epigastric arteries have been used as local flaps, tube pedicles and free tissue transfers. Other popular skin flaps are TFL (tensor fasciae latae) perforator flaps, ALT (anterolateral thigh) and AMT (anteromedial thigh) flaps, gracilis perforator flap and posterior thigh flaps.

VEINS

Femoral vein

The femoral vein accompanies its artery, beginning at the adductor opening as the continuation of the popliteal vein, and ending posterior to the inguinal ligament as the external iliac vein (see Figs 78.8, 80.1–80.3, 80.31). The vein is posterolateral to the femoral artery in the distal adductor canal. More proximally in the canal, and in the distal femoral triangle (i.e. at its apex), the vein lies posterior to the artery and proximally, at the base of the triangle, the vein lies medial to the artery. The vein occupies the middle compartment of the femoral sheath, between the femoral artery and femoral canal; fat in the canal permits expansion of the vein.

The femoral vein has numerous muscular tributaries. The profunda femoris vein (deep vein of the thigh, deep femoral vein) joins the femoral vein posteriorly 4–12 cm distal to the inguinal ligament, and the long saphenous vein then enters anteriorly. Veins accompanying the superficial epigastric, superficial circumflex iliac and external pudendal arteries join the long saphenous vein before it enters the saphenous opening. Lateral and medial circumflex femoral veins are usually tributaries of the femoral vein. There are usually four or five valves in the femoral vein; the two most constant are just distal to the entry of profunda femoris and near the inguinal ligament.

Profunda femoris vein (Deep vein of the thigh)

Profunda femoris lies anterior to its artery and receives tributaries corresponding to the branches of the artery. Through these tributaries it connects distally with the popliteal vein and proximally with the inferior gluteal veins. It sometimes drains the medial and lateral circumflex femoral veins and has a valve just before it empties into the femoral vein.

Long saphenous vein

The long saphenous vein (great saphenous vein) (see Fig. 78.9A), the longest vein in the body, starts distally as a continuation of the medial marginal vein of the foot, and ends in the femoral vein a short distance distal to the inguinal ligament. It ascends immediately anterior to the tibial malleolus, crosses the distal third of the medial surface of the tibia obliquely in an anteroposterior direction to reach its medial border, and then ascends a little behind the border to the knee. Proximally, it is posteromedial to the medial tibial and femoral condyles (lying the breadth of the subject's hand posterior to the medial edge of the patella), and then ascends the medial aspect of the thigh. It passes through the saphenous opening and finally opens into the femoral vein. The 'centre' of the opening is often said to be 2.5–3.5 cm inferolateral to the pubic tubercle, and the vein is then represented by a line drawn from this point to the adductor tubercle. However, the saphenous opening varies greatly in size and disposition so that this 'centre' is not a reliable surface marking for the saphenofemoral junction.

In its course through the thigh, the long saphenous vein is accompanied by the medial branches of the anterior cutaneous branches of the femoral nerve. At the knee, the saphenous branch of the descending genicular artery (the saphenous artery) and, in the leg and foot, the saphenous nerve all lie anterior to the vein. The vein is often duplicated, especially distal to the knee. It has 10–20 valves, which are more numerous in the leg than in the thigh. One is present just before the vein pierces the cribriform fascia, another at its junction with the femoral vein. In almost its entire extent the vein lies in subcutaneous tissue, but it has many connections with the deep veins, especially in the leg.

Tributaries

At the ankle, the long saphenous vein drains the sole by medial marginal veins. In the leg, it often connects with the short saphenous vein and with deep veins via perforating veins. Just distal to the knee, it usually receives three large tributaries from the front of the leg, the tibial malleolar region (connecting with some of the 'perforating' veins) and the calf (communicating with the short saphenous vein). The tributary draining the tibial malleolar region is formed distally from a fine network or 'corona' of delicate veins over the medial malleolus, and then ascends the medial aspect of the calf as the posterior arch vein (Dodd and Cockett 1976). This vein was first illustrated by Leonardo da Vinci, whose name is sometimes given to it. It connects with posterior tibial venae comitantes by a series of perforating (communicating) veins. There are usually three, equally spaced between the medial malleolus and the mid-calf. More than three such perforators are uncommon, and an arch vein perforator above mid-calf is only very rarely found.

Above the posterior crural arch vein, perforating veins join the long saphenous vein or one of its main tributaries at two main sites. The first is at a level in the upper calf indicated by its name, the tibial tubercle perforator; the second is in the lower/intermediate third of the thigh, where it perforates the deep fascia roof of the subsartorial canal to join the femoral vein.

In the thigh, the long saphenous vein receives many tributaries. Some open independently, while others converge to form large named channels that frequently pass towards the basal half of the femoral triangle before joining the long saphenous near its termination. These may be grouped as follows: one or more large posteromedial tributaries, one or more large anterolateral tributaries, and four or more peringuinal veins. The posteromedial vein of the thigh, large and sometimes double, drains a large superficial region indicated by its name; it has (as have the other tributaries) radiological and surgical significance. One of its lower radicles is often continuous with the short saphenous vein. The posteromedial vein is sometimes named the accessory saphenous vein, though some restrict the term accessory to a lower (more distal) posteromedial tributary when two (or more) are present. Another large vessel, the anterolateral vein of the thigh (anterior femoral cutaneous vein), usually commences from an anterior network of veins in the distal thigh and crosses the apex and distal half of the femoral triangle to reach the long saphenous vein. As the latter traverses the saphenous opening, it is joined by the superficial epigastric, superficial circumflex iliac and superficial external pudendal veins. Their mode of union varies. Superficial epigastric and circumflex iliac veins drain the inferior abdominal wall, the latter also receiving tributaries from the proximolateral region of the thigh. The superficial epigastric or the femoral vein may connect with the lateral thoracic veins by means of a thoracoepigastric vein that runs superficially on the anterolateral aspect of the trunk. This vein connects the inferior and superior caval areas of drainage and may be dilated and visible in cases of inferior caval obstruction. Superficial external pudendal veins drain part of the scrotum/labia; one is joined by the superficial dorsal vein of the penis/clitoris. The deep external pudendal veins join the long saphenous vein at the saphenous opening.

The long saphenous vein is often harvested for grafts used in both peripheral and coronary arterial surgery.

LYMPHATIC DRAINAGE

Superficial inguinal nodes

The superficial inguinal nodes form proximal and distal groups. The proximal group usually consists of five or six nodes just distal to the inguinal ligament. Its lateral members receive afferent vessels from the gluteal region and the adjoining infra-umbilical anterior abdominal wall. Medial members receive superficial vessels from the external genitalia (including the inferior vagina), inferior anal canal and perianal region, adjoining abdominal wall, umbilicus and vessels accompanying the round ligament. The distal group usually consists of four or five

nodes along the termination of the long saphenous vein. They receive all the superficial vessels of the lower limb, except those from the posterolateral calf. All superficial inguinal nodes drain to the external iliac nodes, some via the femoral canal and others anterior or lateral to the femoral vessels. Numerous vessels interconnect individual nodes.

Superficial inguinal nodes are frequently enlarged in disease or injury in their region of drainage. Thus the proximal inguinal nodes are almost invariably affected in malignant or infective disease of the prepuce, penis, labia majora, scrotum, perineum, anus and lower vagina, or in diseases affecting the skin and superficial structures in these regions, in the infra-umbilical part of the abdominal wall or in the gluteal region. The distal group is implicated only in disease or injury of the limb.

Deep inguinal nodes

The deep inguinal nodes vary from one to three, and are situated medial to the femoral vein. One lies just distal to the saphenofemoral junction, another in the femoral canal, and the most proximal node lies laterally in the femoral ring. The middle node is the most inconstant and the proximal node is often absent. All receive deep lymphatics that accompany the femoral vessels, lymph vessels from the glans penis or clitoris and a few efferents from the superficial inguinal nodes. Their own efferents traverse the femoral canal to the external iliac nodes.

INNERVATION

LUMBAR PLEXUS AND BRANCHES

The lumbar plexus is formed by the first three and most of the fourth lumbar ventral rami (see Fig. 62.15). The iliohypogastric, ilioinguinal and genitofemoral nerves are described on pages 1094–1096.

Lateral femoral cutaneous nerve

The lateral femoral cutaneous nerve (lateral cutaneous nerve of the thigh) arises from the dorsal branches of the second and third lumbar ventral rami and emerges from the lateral border of psoas major, crossing iliacus obliquely towards the anterior superior iliac spine. It supplies the parietal peritoneum in the iliac fossa. The right nerve passes posterolateral to the caecum, separated from it by the fascia iliaca and peritoneum; the left passes behind the lower part of the descending colon. Both pass behind or through the inguinal ligament, variably medial to the anterior superior iliac spine (commonly 1 cm) and anterior to or through sartorius into the thigh, where they divide into anterior and posterior branches. The anterior branch becomes superficial approximately 10 cm distal to the anterior superior iliac spine and supplies the skin of the anterior and lateral thigh as far distally as the knee. It connects terminally with the cutaneous branches of the anterior division of the femoral nerve and the infrapatellar branch of the saphenous nerve, forming the peripatellar plexus. The posterior branch pierces the fascia lata higher than the anterior, and divides to supply the skin on the lateral surface from the greater trochanter to about mid-thigh. It may also supply skin of the gluteal region.

Lesions of the lateral femoral cutaneous nerve

The lateral femoral cutaneous nerve may become entrapped at three sites along its course: close to the vertebral column; within the abdominal cavity as the nerve travels across the pelvis; and as it leaves the pelvis. This last site is where the nerve is most commonly damaged, and entrapment is thought to occur as the nerve passes through or deep to the inguinal ligament just medial to the anterior superior iliac spine. The angulation of the nerve as it crosses the iliac crest means that it is vulnerable to compression during movement, e.g. hip extension may increase angulation and tension on the nerve.

Injury produces an area of impaired sensation, often with pain and paraesthesia on the anterolateral aspect of the thigh (meralgia paraesthetica) in the distal cutaneous distribution of the nerve. This area does not extend across the midline anteriorly, it does not extend below the knee and it does not extend behind the hamstring tendons laterally. Symptoms may be exacerbated when wearing tight belts or tight-waisted clothes or as a result of a recent weight gain or pregnancy. Exceptionally, the posterior branch of the lateral femoral cutaneous nerve, which supplies a thin strip from the greater trochanter of the femur down about two-thirds of the way to the knee, may be affected separately. This branch leaves the main trunk of the nerve, usually distal to the inguinal ligament, and it then turns laterally to pierce tensor fasciae latae, where it may become entrapped.

Femoral nerve

The femoral nerve is the largest branch of the lumbar plexus and arises from the dorsal branches (posterior divisions) of the second to fourth lumbar ventral rami (see Figs 78.11A, 80.31–80.33). It descends through psoas major, emerging low on its lateral border, and then passes between psoas and iliacus, deep to the iliac fascia. Passing behind the inguinal ligament into the thigh, it is split into anterior and posterior divisions by the lateral circumflex femoral artery. Behind the inguinal ligament, it is separated from the femoral artery by part of psoas major. In the abdomen, the nerve supplies small branches to iliacus and pectineus and a branch to the proximal part of the femoral artery; the latter branch sometimes arises in the thigh.

Nerve to pectineus

The nerve to pectineus branches from the medial side of the femoral nerve near the inguinal ligament. It passes behind the femoral sheath and enters the anterior aspect of the muscle.

Anterior division of the femoral nerve

The anterior division of the femoral nerve supplies intermediate and medial femoral cutaneous nerves and branches to sartorius.

Medial femoral cutaneous nerve of the thigh The medial femoral cutaneous nerve (medial cutaneous nerve of the thigh) is at first lateral to the femoral artery. It crosses anterior to the artery at the apex of the femoral triangle and divides into anterior and posterior branches. Before doing so, it sends a few rami through the fascia lata to supply the skin of the medial side of the thigh, near the long saphenous vein; one ramus emerges via the saphenous opening, while another becomes subcutaneous about mid-thigh. The anterior branch descends on sartorius, perforates the fascia lata beyond mid-thigh, and divides into a branch that supplies the skin as low as the medial side of the knee, and another that runs lateral to the former and connects with the infrapatellar branch of the saphenous nerve. The posterior branch descends along the posterior border of sartorius to the knee, pierces the fascia lata, connects with the saphenous nerve and gives off several cutaneous rami, some as far as the medial side of the leg. The nerve contributes to the subsartorial plexus (see below).

Intermediate cutaneous nerve of the thigh The intermediate femoral cutaneous nerve (intermediate cutaneous nerve of the thigh) usually pierces the fascia lata some 8 cm below the inguinal ligament, either as two branches or as one trunk that quickly divides into two. These descend on the front of the thigh, supplying the skin as far as the knee and ending in the peripatellar plexus. The lateral branch of the intermediate cutaneous nerve communicates with the femoral branch of the genitofemoral nerve, frequently piercing sartorius and sometimes supplying it.

Nerve to sartorius The main nerve to sartorius arises from the femoral nerve in common with the intermediate cutaneous nerve of the thigh.

Posterior division of the femoral nerve

The branches of the posterior division of the femoral nerve are the saphenous nerve and branches to quadriceps femoris and the knee joint.

Saphenous nerve See page 1399.

Muscular branches The muscular branches of the posterior division of the femoral nerve supply quadriceps femoris. A branch to rectus femoris enters its proximal posterior surface and also supplies the hip joint. A larger branch to vastus lateralis forms a neurovascular bundle with the descending branch of the lateral circumflex femoral artery in its distal part and also supplies the knee joint. A branch to vastus medialis descends through the proximal part of the adductor canal, lateral to the saphenous nerve and femoral vessels, and enters the muscle at about its midpoint, sending a long articular filament distally along the muscle to the knee. Two or three branches to vastus intermedius enter its anterior surface about mid-thigh; a small branch from one of these descends through the muscle to supply articularis genu and the knee joint.

Vascular branches

Vascular branches of the femoral nerve supply the femoral artery and its branches.

Lesions of the femoral nerve

The main trunk of the femoral nerve is not subject to an entrapment neuropathy, but it may be compressed by retroperitoneal tumours or retroperitoneal haemorrhage in patients on anticoagulants or with a bleeding diathesis. A localized lesion of the femoral nerve may occur in diabetes mellitus (one of the forms of diabetic amyotrophy). The most striking feature of femoral neuropathy is wasting and weakness of quadriceps femoris, which results in considerable difficulty in walking and a tendency for the leg to collapse. Pain and paraesthesia may occur on the anterior and medial aspect of the thigh, extending down the medial aspect of the leg in the distribution of the saphenous branch of the femoral nerve.

Obturator nerve

The obturator nerve arises from the ventral branches of the second to fourth lumbar ventral rami. The branch from the third is the largest while that from the second is often very small. The nerve descends in psoas major, emerging from its medial border at the pelvic brim to pass behind the common iliac vessels and lateral to the internal iliac vessels. It then descends forwards along the lateral wall of the lesser pelvis on obturator internus, anterosuperior to the obturator vessels, to the obturator foramen, entering the thigh by its upper part. Near the foramen, it divides into anterior and posterior branches, separated at first by part of obturator externus and lower down by adductor brevis.

Anterior branch

The anterior branch (see Fig. 80.33) leaves the pelvis anterior to obturator externus, descending in front of adductor brevis, behind pectineus and adductor longus. At the lower border of adductor longus it communicates with the medial cutaneous and saphenous branches of the femoral nerve, to form a subsartorial plexus that supplies the skin on the medial side of the thigh (see below). It descends on, and supplies, the femoral artery. Near the obturator foramen, the anterior branch supplies the hip joint. Behind pectineus it supplies adductor longus, gracilis, usually adductor brevis and often pectineus, and connects with the accessory obturator nerve (when present). Occasionally, the communicating branch to the femoral medial cutaneous and saphenous branches continues as a cutaneous branch to the thigh and leg. When this occurs, the nerve emerges from behind the distal border of adductor longus to descend along the posterior margin of sartorius to the knee, where it pierces the deep fascia and connects with the saphenous nerve to supply the skin halfway down the medial side of the leg.

Subsartorial nerve plexus The medial cutaneous nerve of the thigh forms a subsartorial plexus with branches of the saphenous and obturator nerves, deep to the fascia lata, at the lower border of adductor longus. When the communicating branch of the obturator nerve is large and reaches the leg, the posterior branch of the medial cutaneous nerve is small, and ends in the plexus from which it gives rise to a few cutaneous filaments.

Posterior branch

The posterior branch pierces obturator externus anteriorly, supplies it and passes behind adductor brevis to the front of adductor magnus, dividing into branches to this and adductor brevis when the latter is not supplied by the anterior division. It usually sends an articular filament to the knee joint, which either perforates adductor magnus distally or traverses its opening with the femoral artery to enter the popliteal fossa. Within the fossa, the nerve descends on the popliteal artery to the back of the knee, pierces its oblique posterior ligament and supplies the articular capsule. It gives filaments to the popliteal artery.

Lesions of the obturator nerve

Isolated lesions of the obturator nerve are extremely rare, but may occasionally occur as a result of direct trauma (sometimes during parturition) or in anterior dislocations of the hip. The nerve may also be damaged by an obturator hernia, or be involved together with the femoral nerve in retroperitoneal lesions that occur close to the origins of the lumbar plexus. Compression of the nerve by herniated bowel loops at the obturator foramen can result in pain referred to the hip, medial thigh and knee, the so-called Howship–Romberg sign. A more distal nerve entrapment syndrome causing chronic medial thigh pain has been described in athletes with large adductor muscles.

Accessory obturator nerve

Occasionally present, the accessory obturator nerve is small and arises from the ventral branches of the third and fourth lumbar ventral rami.

Table 80.1 The branches of the sacral plexus

	Ventral divisions	Dorsal divisions
Nerve to quadratus femoris and gemellus inferior	L4, 5, S1	
Nerve to obturator internus and gemellus superior	L5, S1, 2	
Nerve to piriformis		(L5), S1, 2
Superior gluteal		L4, 5, S1
Inferior gluteal		L5, S1, 2
Posterior femoral cutaneous	S2, 3	S1, 2
Tibial (sciatic)	L4, 5, S1, 2, 3	
Common fibular (sciatic)		L4, 5, S1, 2
Perforating cutaneous		S2, 3
Pudendal	S2, 3, 4	
Branches to levator ani, coccygeus and external anal sphincter	S4	

It descends along the medial border of psoas major, crosses the superior pubic ramus behind pectineus and divides into branches; one branch enters the deep surface of pectineus, another supplies the hip joint, and a third connects with the anterior branch of the obturator nerve. Sometimes the accessory obturator nerve is very small and only supplies pectineus. Any branch may be absent and others may occur; an additional branch sometimes supplies adductor longus.

SACRAL PLEXUS AND BRANCHES

The branches of the sacral plexus (see Fig. 78.11B) are shown in Table 80.1.

Sciatic nerve

The sciatic nerve is 2 cm wide at its origin and is the thickest nerve in the body. It leaves the pelvis via the greater sciatic foramen below piriformis and descends between the greater trochanter and ischial tuberosity, along the back of the thigh, dividing into the tibial and common fibular nerves at a varying level proximal to the knee (Fig. 80.34; see Fig. 78.11B). Superiorly, it lies deep to gluteus maximus, resting first on the posterior ischial surface with the nerve to quadratus femoris between them. It then crosses posterior to obturator internus, the gemelli and quadratus femoris, separated by the latter from obturator externus and the hip joint. It is accompanied medially by the posterior femoral cutaneous nerve and the inferior gluteal artery. More distally, it lies behind adductor magnus and is crossed posteriorly by the long head of biceps femoris. Its course corresponds to a line drawn from just medial to the midpoint between the ischial tuberosity and greater trochanter to the apex of the popliteal fossa.

Articular branches arise proximally to supply the hip joint through its posterior capsule; these are sometimes derived directly from the sacral plexus. Muscular branches are distributed to biceps femoris, semitendinosus, semimembranosus and the ischial part of adductor magnus.

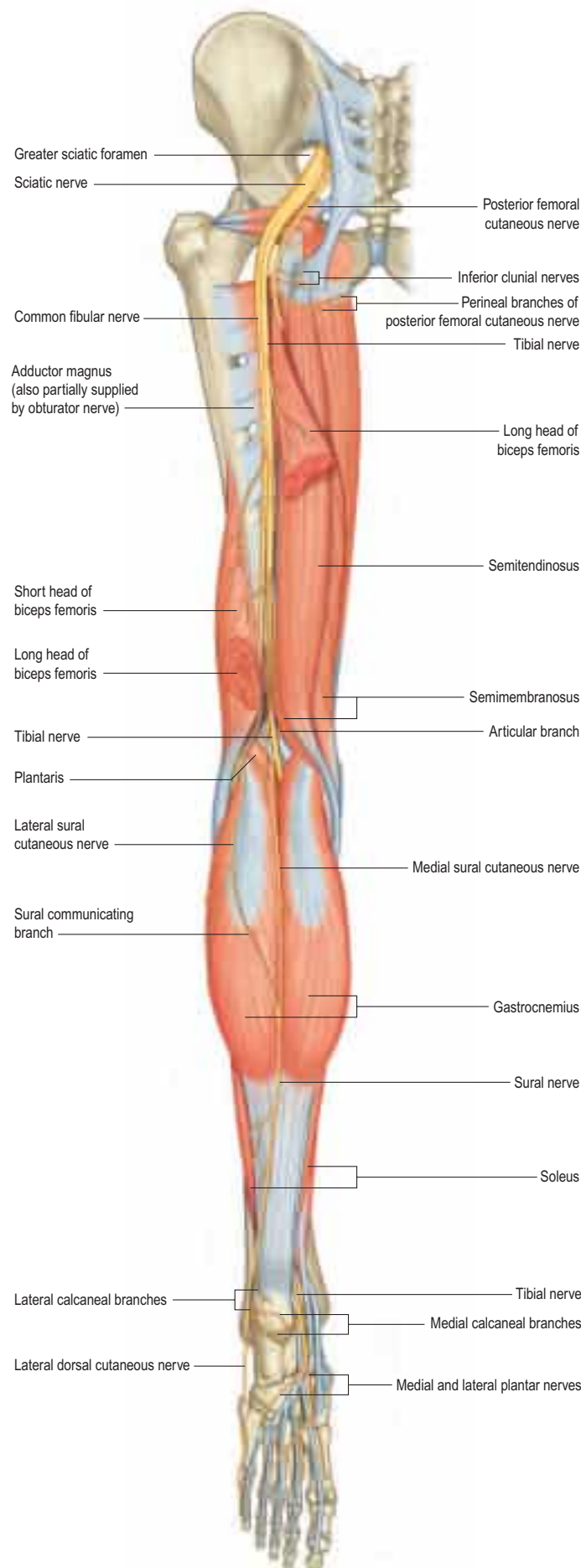
The point of division of the sciatic nerve into its tibial and common fibular components is very variable. The common site is at the junction of the middle and lower thirds of the thigh, near the apex of the popliteal fossa, but the division may occur at any level above this point and, rarely, may occur below it.

Blood supply to the sciatic nerve

In the gluteal region, the sciatic nerve is supplied by the inferior gluteal artery and cruciate anastomosis (the medial and lateral circumflex femoral arteries, inferior gluteal artery and the first perforating branch of the profunda femoris artery) (see Fig. 80.30). On rare occasions, it is supplied by branches from the superior gluteal or internal pudendal arteries, which reach the nerve on its medial side. Lower in the thigh, arterial branches derived from the perforating branches of the profunda femoris artery or the anastomotic chain between them or, occasionally, from the popliteal artery, enter the nerve on its lateral or anterolateral side (Sunderland 1945). The numerous arterial branches to the sciatic nerve anastomose with each other to form extraneural and intraneural arterial chains (Sunderland 1945, Ugrenovic et al 2013).

Lesions of the sciatic nerve

The sciatic nerve supplies the knee flexors and all the muscles below the knee, which means that a complete palsy of the sciatic nerve results in a flail foot and severe difficulty in walking. Fortunately, complete

**Fig. 80.34** The sciatic nerve and its branches, posterior view.

sciatic nerve palsy is very rare. The nerve is vulnerable in posterior dislocation of the hip. As it leaves the pelvis, it passes either behind piriformis or sometimes through the muscle, and at that point it may very rarely become entrapped or tethered; the piriformis syndrome is a controversial condition in which an anomalous relationship between piriformis and the sciatic nerve is thought to cause pain in the buttocks and along the course of the sciatic nerve. External compression over the buttock, e.g. in patients who lie immobile on a hard surface for a considerable length of time, can injure the nerve. However, the most common cause of serious sciatic nerve injury (and of the resulting major medicolegal claims) is iatrogenic (Dillow et al 2013). The nerve may be damaged in misplaced therapeutic injections into gluteus maximus. Sciatic nerve palsy occurs after total hip replacement or similar surgery in 1% of cases, and may be caused by sharp injury, burning from bone cement, traction from instruments, manipulation of the hip, inadvertent lengthening of the femur, or haematoma surrounding the nerve. For some reason, possibly anatomical, the common fibular component of the sciatic nerve is more usually affected; the patient has a foot drop and a high-stepping gait.

Inferior gluteal nerve

The inferior gluteal nerve arises from the dorsal branches of the fifth lumbar and first and second sacral ventral rami. It leaves the pelvis via the greater sciatic foramen below piriformis, and divides into branches that enter the deep surface of gluteus maximus (see Fig. 80.30).

Superior gluteal nerve

The superior gluteal nerve arises from the dorsal branches of the fourth and fifth lumbar and first sacral ventral rami. Accompanied by the superior gluteal vessels, the nerve leaves the pelvis via the greater sciatic foramen above piriformis, and divides into superior and inferior branches (see Fig. 80.30). The superior branch accompanies the upper branch of the deep division of the superior gluteal artery to supply gluteus medius and occasionally gluteus minimus. The inferior branch runs with the lower ramus of the deep division of the superior gluteal artery across gluteus minimus, supplies the glutei medius and minimus, and ends in tensor fasciae latae.

Pudendal nerve

See Figure 62.15 and pages 1230 and 1236.

Perforating cutaneous nerve

The perforating cutaneous nerve usually arises from the posterior aspects of the second and third sacral ventral spinal rami. It pierces the sacrotuberous ligament, curves round the inferior border of gluteus maximus and supplies the skin over the inferomedial aspect of this muscle. The nerve may arise from the pudendal nerve or it may be absent, in which case it may be replaced either by a branch from the posterior femoral cutaneous nerve or from the third and fourth, or fourth and fifth, sacral ventral rami.

Nerve to quadratus femoris

The nerve to quadratus femoris arises from the ventral branches of the fourth lumbar to the first sacral ventral rami. It leaves the pelvis via the greater sciatic foramen below piriformis, descends on the ischium deep to the sciatic nerve, the gemelli and the tendon of obturator internus, and supplies gemellus inferior, quadratus femoris and the hip joint.

Nerve to obturator internus

The nerve to obturator internus arises from the ventral branches of the fifth lumbar and first and second sacral ventral rami. It leaves the pelvis via the greater sciatic foramen below piriformis, supplies a branch to the upper posterior surface of gemellus superior, crosses the ischial spine lateral to the internal pudendal vessels, re-enters the pelvis via the lesser sciatic foramen, and enters the pelvic surface of obturator internus.

Posterior femoral cutaneous nerve (posterior cutaneous nerve of the thigh)

The posterior femoral cutaneous nerve (posterior cutaneous nerve of the thigh) arises from the dorsal branches of the first and second, and the ventral branches of the second and third sacral rami. It leaves the pelvis via the greater sciatic foramen below piriformis and descends under gluteus maximus with the inferior gluteal vessels, lying posterior or medial to the sciatic nerve (see Fig. 80.30). It descends in the back of the thigh superficial to the long head of biceps femoris, deep to the fascia lata. It pierces the deep fascia behind the knee and accompanies the short saphenous vein to mid-calf, its terminal twigs connecting with the sural nerve. Its branches are cutaneous and are distributed to the gluteal region, perineum, posterior thigh and proximal posterior leg.

Three or four gluteal branches (inferior clunial nerves) curl round the lower border of gluteus maximus to supply the skin over the inferolateral portion of the muscle. The perineal branch supplies the superomedial skin in the thigh, curves forwards across the hamstrings below the ischial tuberosity, pierces the fascia lata and then runs in the superficial perineal fascia to the scrotal or labial skin. It communicates with the inferior rectal and posterior scrotal or labial branches of the perineal nerve, and gives numerous branches to the skin of the back and medial side of the thigh, the popliteal fossa and the proximal part of the back of the leg. One study found the perineal branch of the posterior cutaneous nerve of the thigh in about one-half of specimens (Tubbs et al 2009).

Nerve to piriformis

The nerve to piriformis usually arises from the dorsal branches of the first and second sacral ventral rami (sometimes only the second) and enters the anterior surface of piriformis.

Visceral and pelvic muscular branches of sacral plexus

Visceral branches of the sacral plexus are described on page 1229. Pelvic muscular branches are described on page 1221.



Bonus e-book images

Fig. 80.20 An ultrasound image of a 4-week-old baby taken in the lateral position and demonstrating the cartilaginous femoral head and edge of the acetabulum.

Fig. 80.21 A radiograph of the pelvis of a 9-month-old infant showing the centre of ossification of the femoral epiphysis.

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